

Sector × Quintile Return Book

Average monthly returns of each factor's quintile portfolios, by GICS sector. 272 factors, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

■ Q1 ■ Q2 ■ Q3 ■ Q4 ■ Q5

Each cluster of five bars is one sector; bars run Q1 (highest factor score) to Q5 (lowest), dark to light.

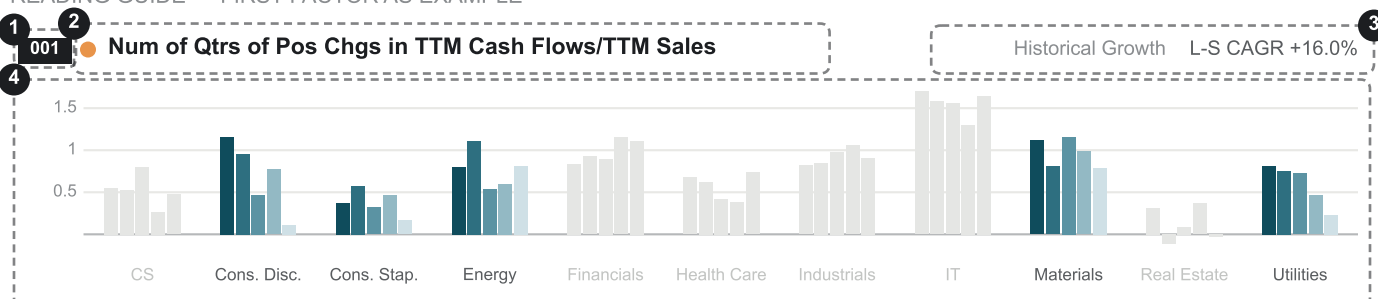
■ Grey clusters mark sectors the live 48-month rules exclude from the factor.

FACTOR STYLES

Valuation	42	Price Momentum	54
Earnings Quality	55	Size	3
Analyst Expectations	24	Historical Growth	56
Capital Efficiency	28	Volatility	10

001 In current model portfolio (50 factors)

READING GUIDE — FIRST FACTOR AS EXAMPLE



1 Rank - ordered by ③ L-S CAGR, descending.
Filled dark badge = in the current Top-50 portfolio.

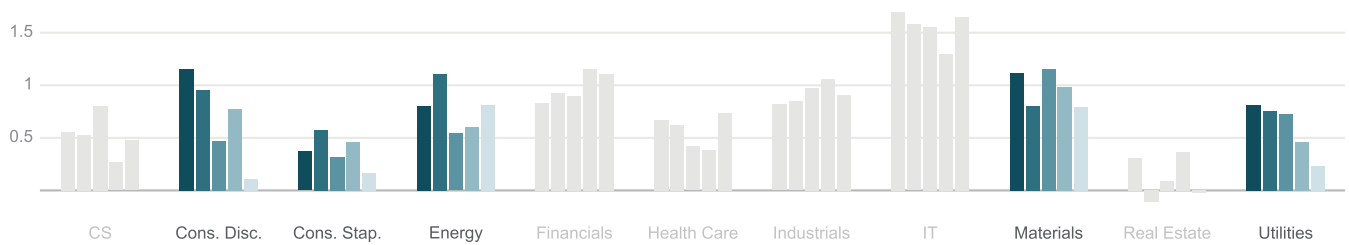
2 Factor name - colour dot = style (legend above).

3 Style name / L-S CAGR - the LAST 48 MONTHS only (rolling IS), net of 10 bp cost. The live selection / sort basis. Grey exclusions in the chart are these same live 48M rules.

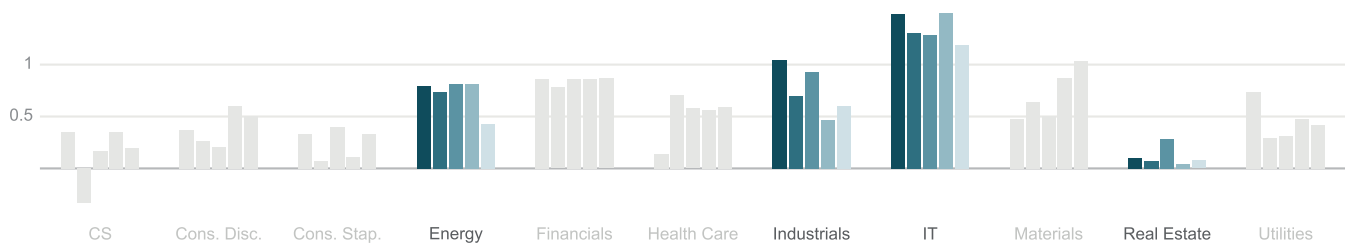
4 Chart - FULL HISTORY (Jul 2015 - Aug 2026): avg monthly return of each sector x quintile portfolio, BEFORE costs. Only the return window differs from ③ - the sector exclusions applied are the live rules.

001 Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

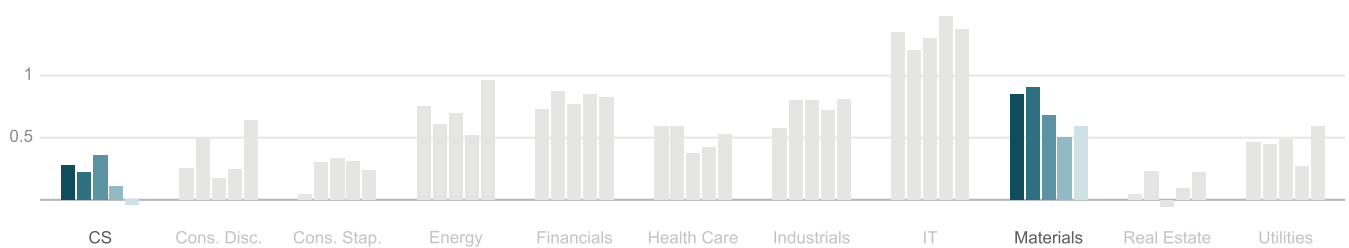
Historical Growth L-S CAGR +16.0%

**002 Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

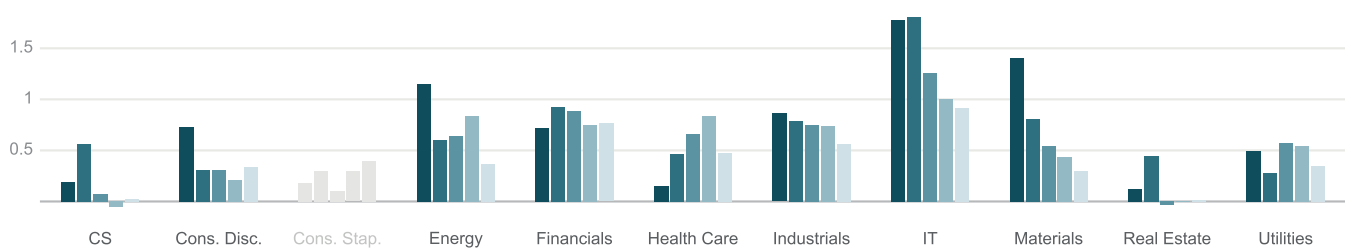
Historical Growth L-S CAGR +13.7%

**003 Long Term Debt to Equity Ratio**

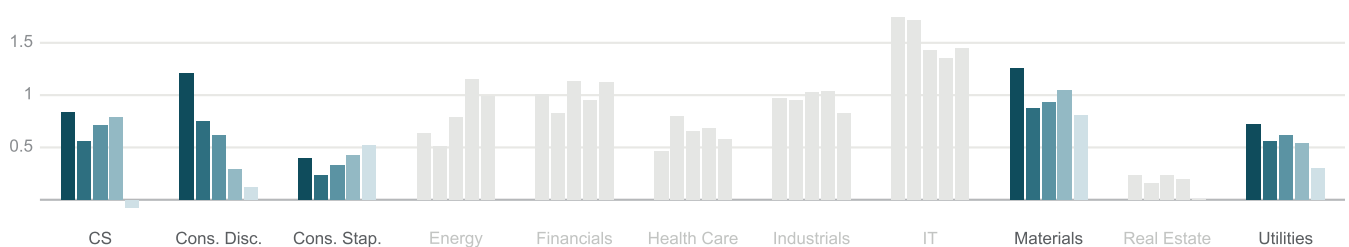
Capital Efficiency L-S CAGR +12.1%

**004 12M Realized Price Volatility**

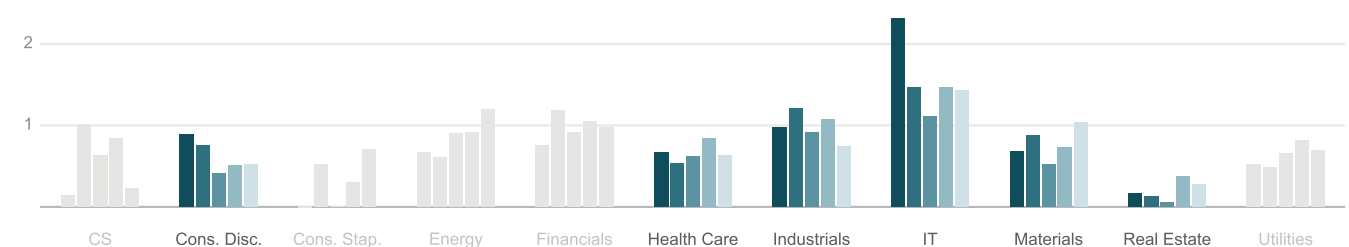
Volatility L-S CAGR +11.9%

**005 Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

Historical Growth L-S CAGR +11.7%

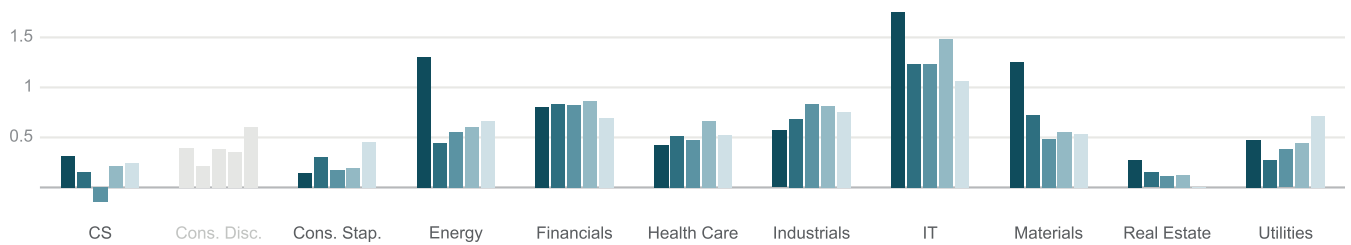
**006 Change of Asset Turnover**

Historical Growth L-S CAGR +11.2%



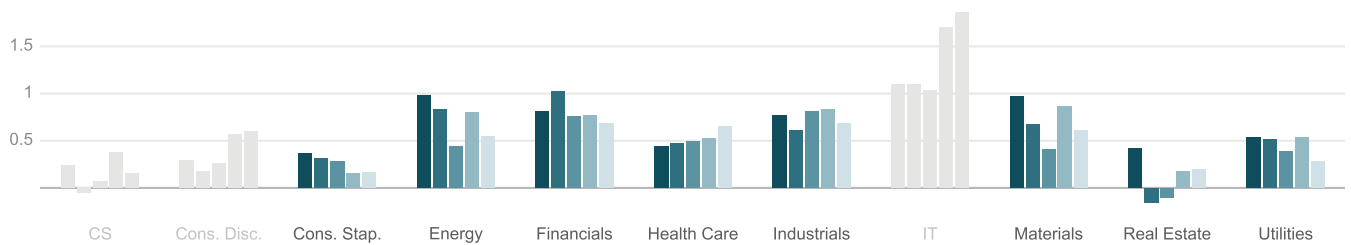
007 ● Analyst Dispersion for FY2 EPS

Analyst Expectations L-S CAGR +11.2%



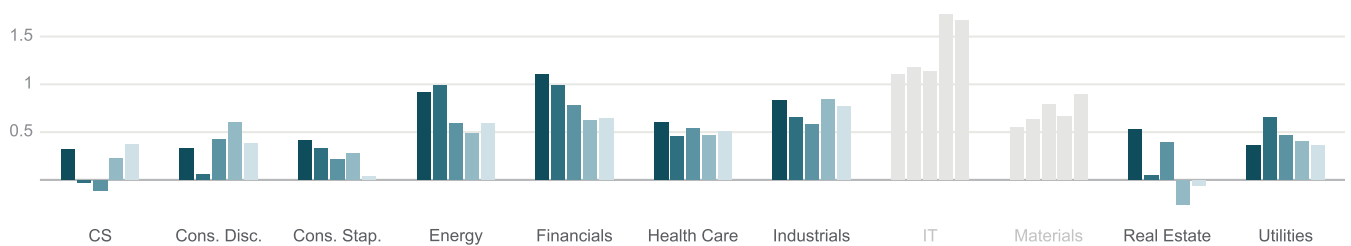
008 ● Assets to Price Ratio

Valuation L-S CAGR +10.9%



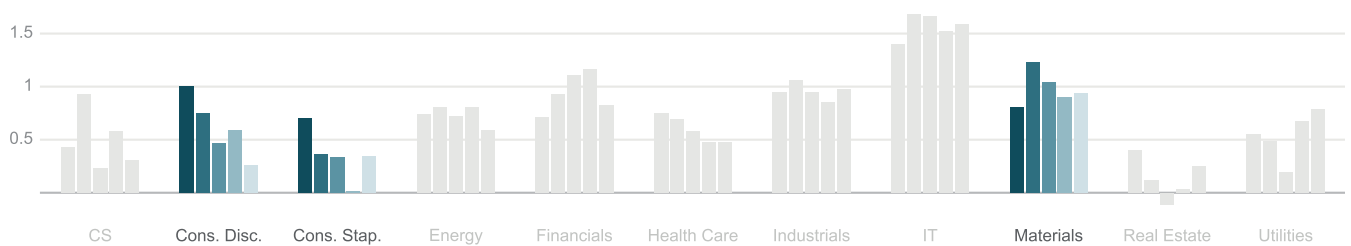
009 ● Current Liability to Price

Valuation L-S CAGR +10.9%



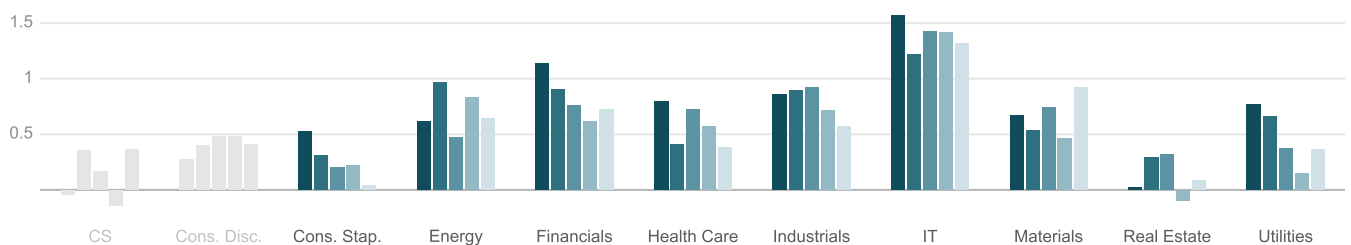
010 ● Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings

Historical Growth L-S CAGR +10.6%



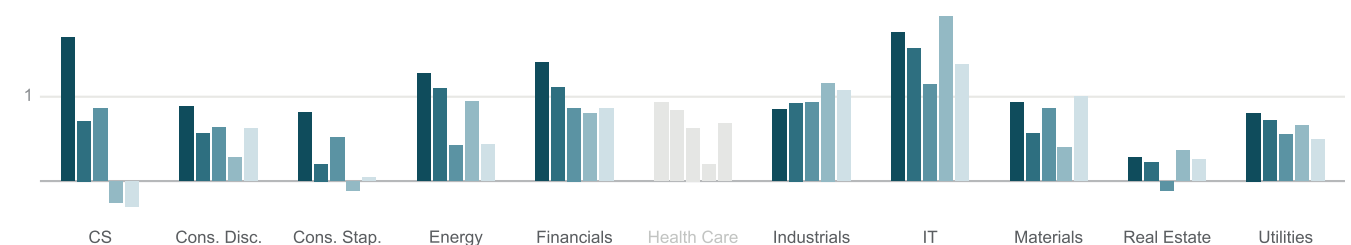
011 ● Inverse P/E Ratio Adj for Growth and Yield

Valuation L-S CAGR +10.5%



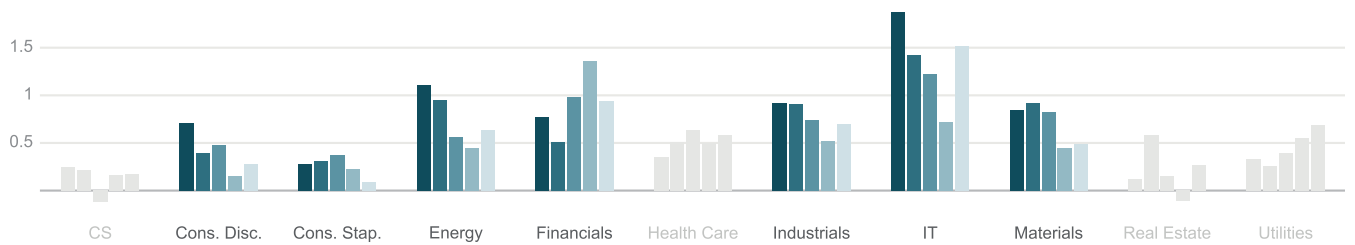
012 ● Cash to Adjusted Enterprise Value

Valuation L-S CAGR +10.4%

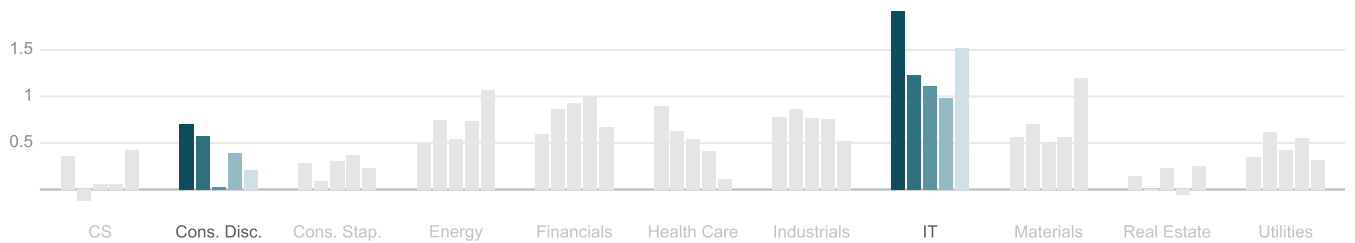


013 Unexpected Inventory Change

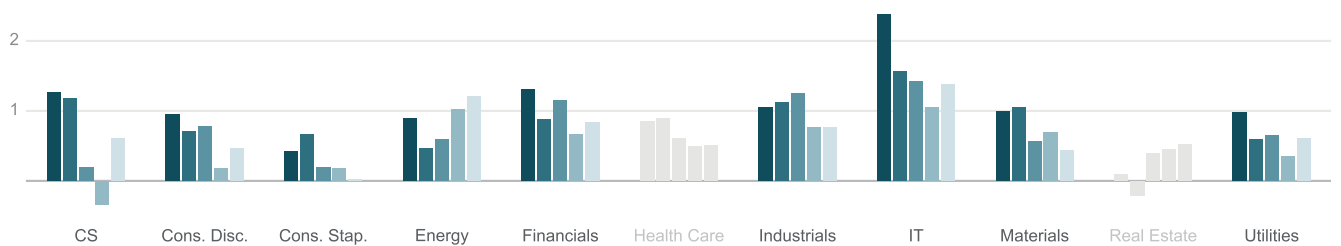
Earnings Quality L-S CAGR +10.4%

**014 Operating Earnings to Assets Ratio**

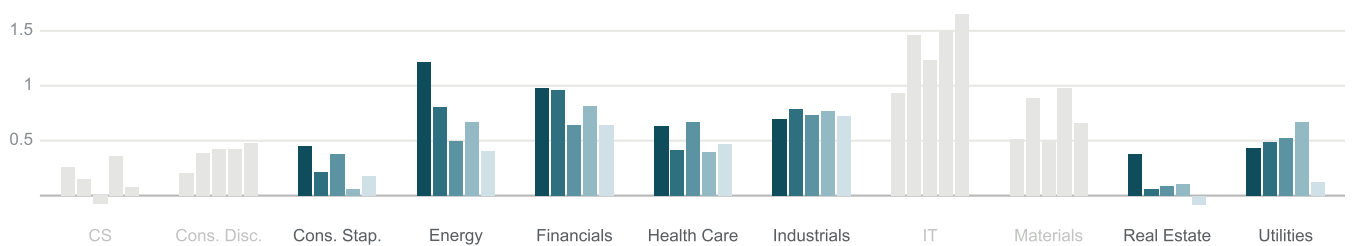
Earnings Quality L-S CAGR +10.4%

**015 Adjusted 3M Revision in FY1 EPS Estimate**

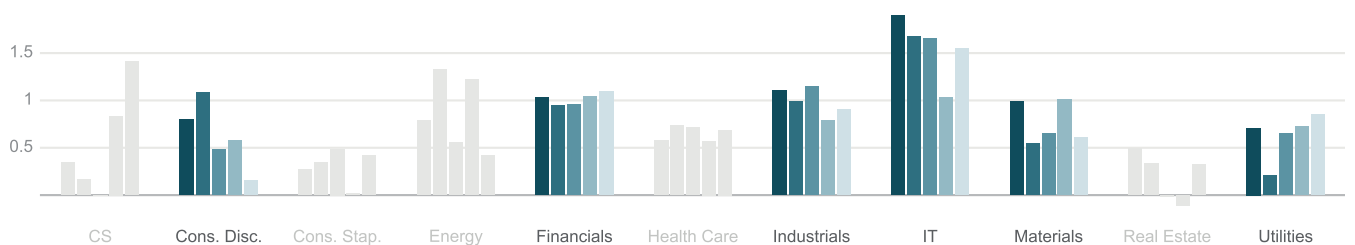
Analyst Expectations L-S CAGR +10.1%

**016 Sales to Price Ratio**

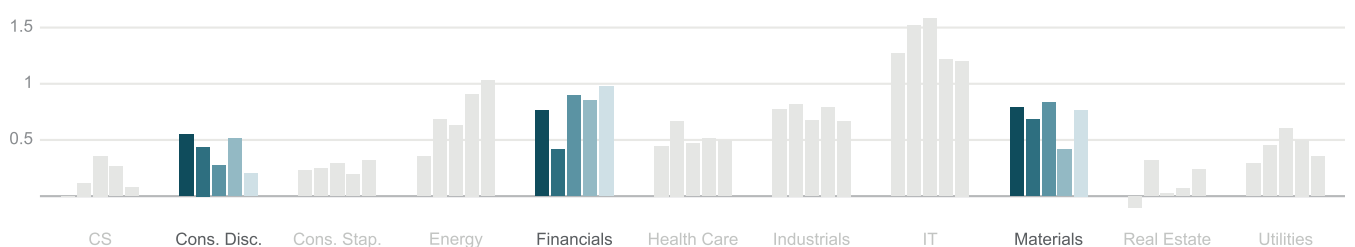
Valuation L-S CAGR +10.1%

**017 Adj 1Y Chg in Asset Adjusted FCF**

Historical Growth L-S CAGR +9.9%

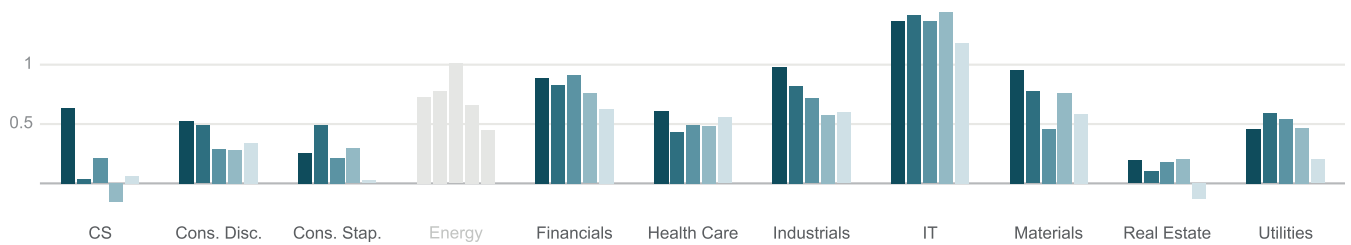
**018 Gross Profit Margin**

Earnings Quality L-S CAGR +9.9%

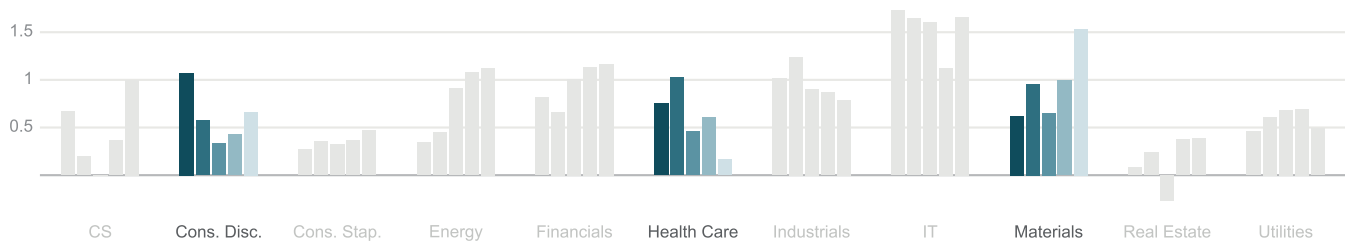


019 Log of Total Last Quarter Assets

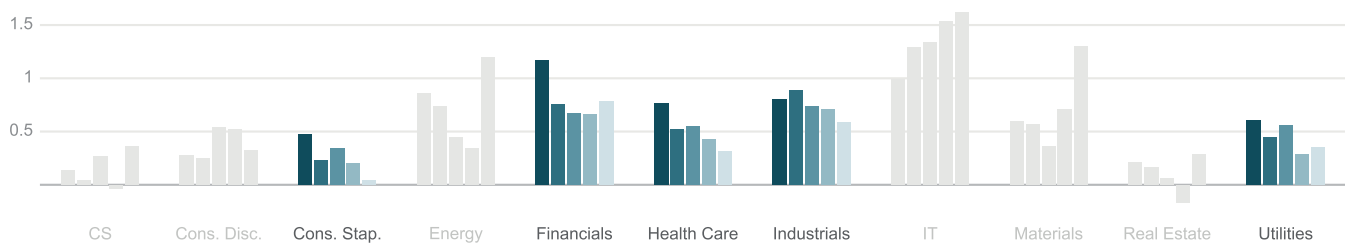
Size L-S CAGR +9.7%

**020 Trailing 12-Month Earnings/Trailing 12-Month Sales**

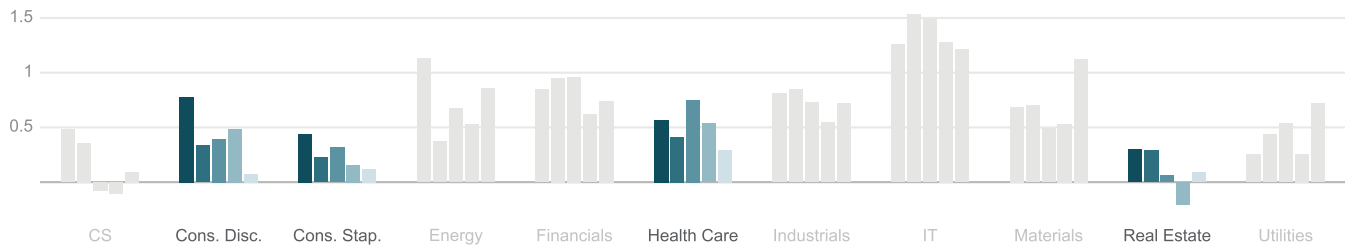
Earnings Quality L-S CAGR +9.4%

**021 Operating Earnings to Price Ratio**

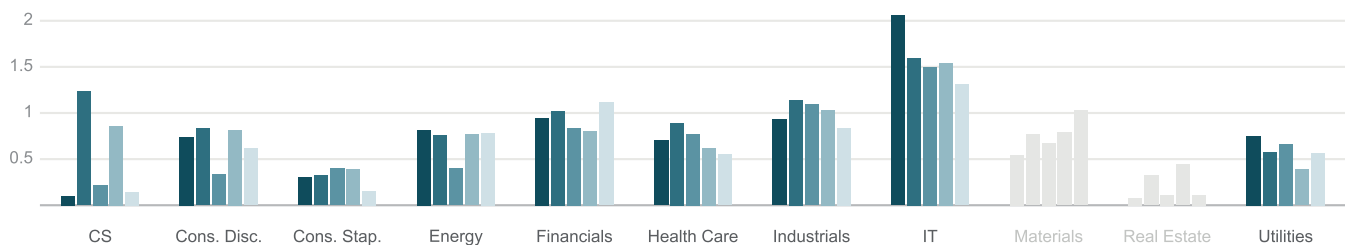
Valuation L-S CAGR +9.2%

**022 3Y Chg in Operating Cash Flow to Price**

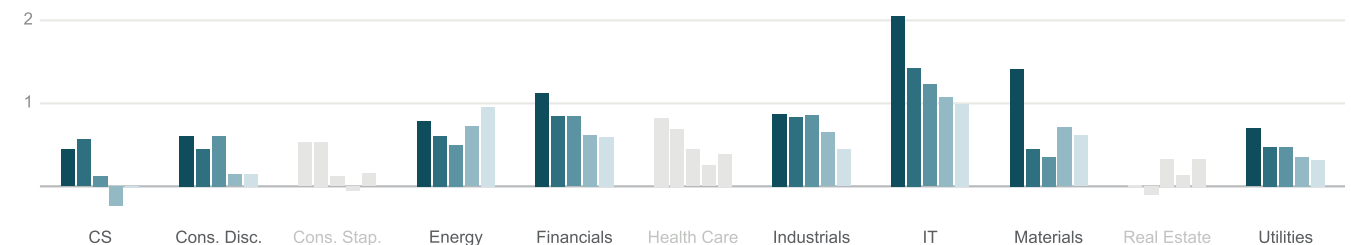
Historical Growth L-S CAGR +9.1%

**023 Adjusted Inverse PEG**

Analyst Expectations L-S CAGR +9.1%

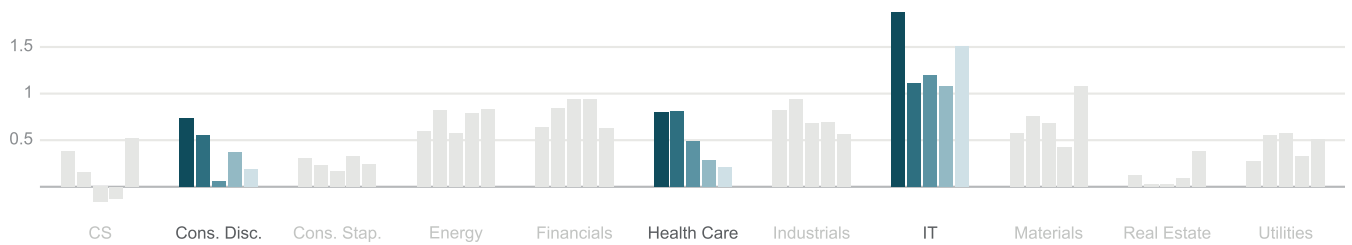
**024 3M Revision in FY1 EPS Estimate**

Analyst Expectations L-S CAGR +8.8%



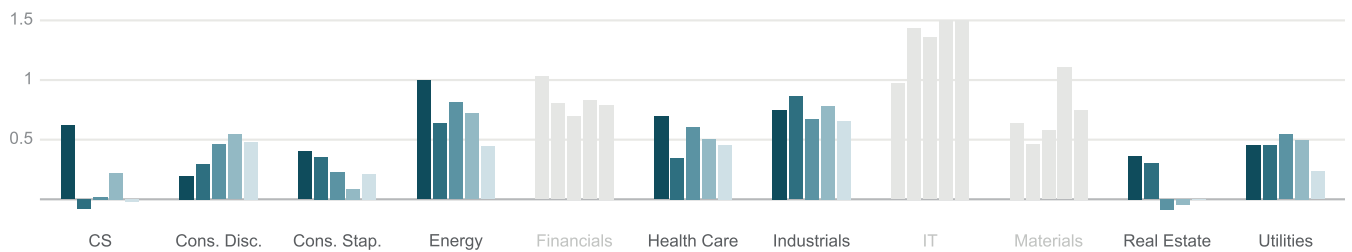
025 ● Return on Assets

Capital Efficiency L-S CAGR +8.8%



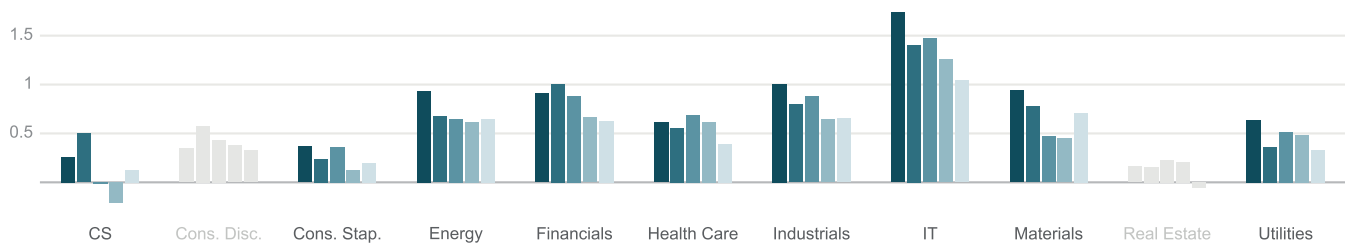
026 ● Sales to EV Ratio

Valuation L-S CAGR +8.8%



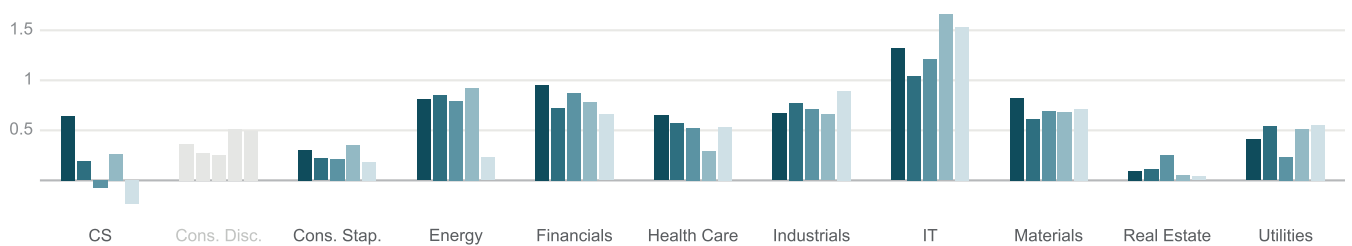
027 ● Inverse PEG

Analyst Expectations L-S CAGR +8.7%



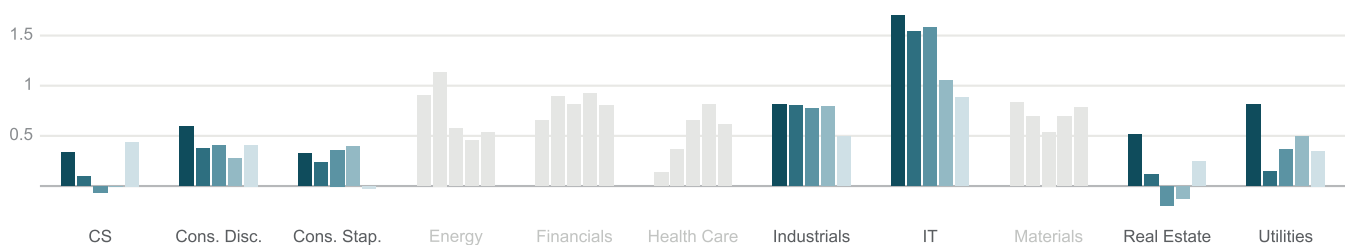
028 ● Cash to Price

Valuation L-S CAGR +8.7%



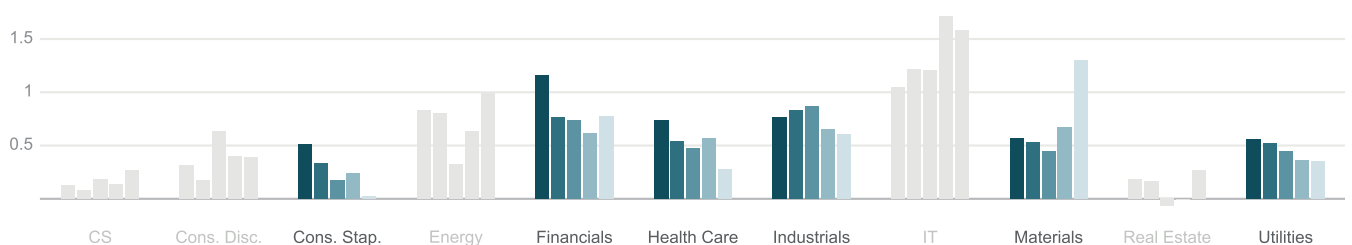
029 ● 60M CAPM Beta

Volatility L-S CAGR +8.7%



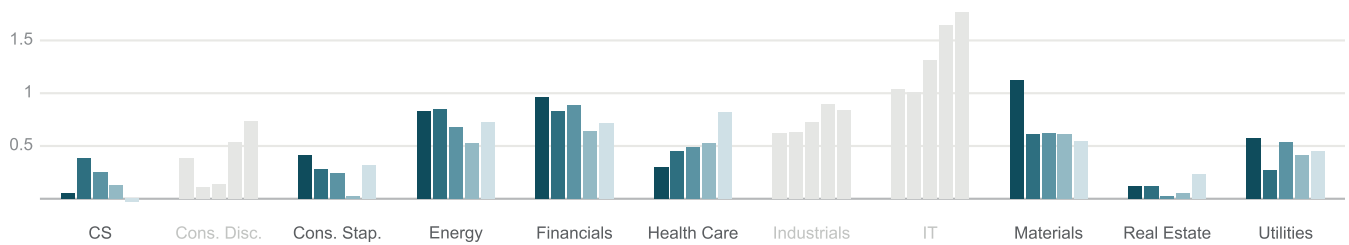
030 ● Earnings to Price

Valuation L-S CAGR +8.7%

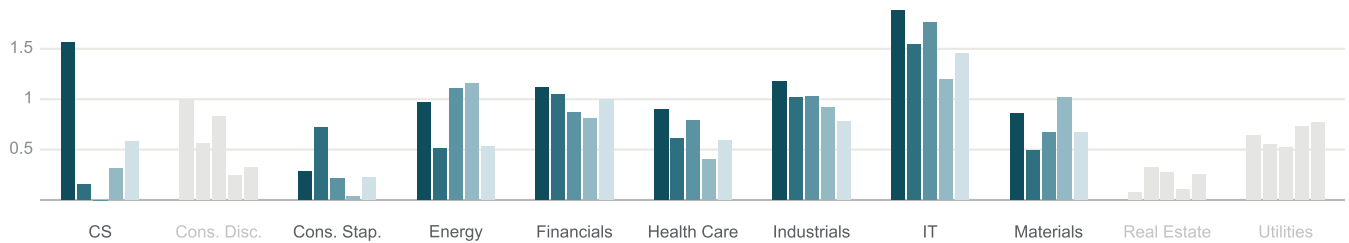


031 ● **Book to Price**

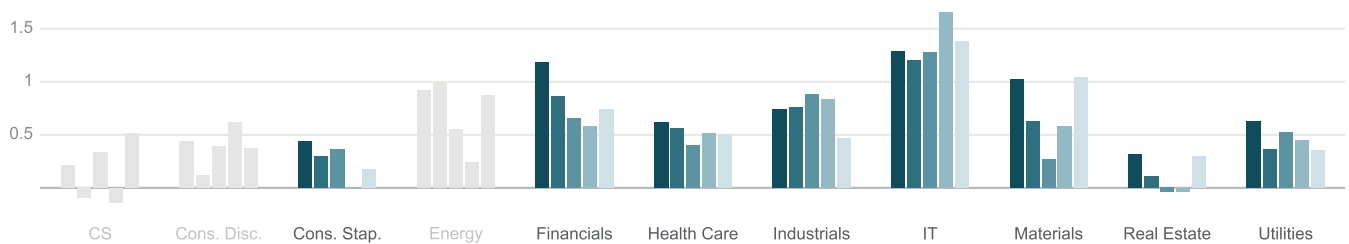
Valuation L-S CAGR +8.7%

032 ● **Short Interest Trading Volume**

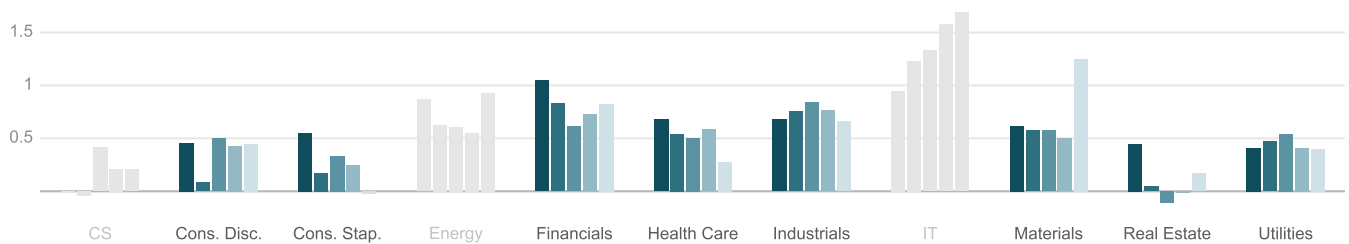
Price Momentum L-S CAGR +8.7%

033 ● **Forward Earnings to Price**

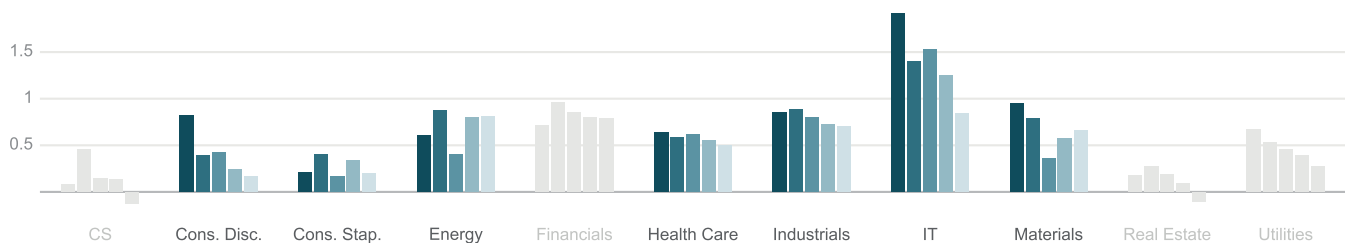
Valuation L-S CAGR +8.6%

034 ● **Pretax Income Per Share to Price**

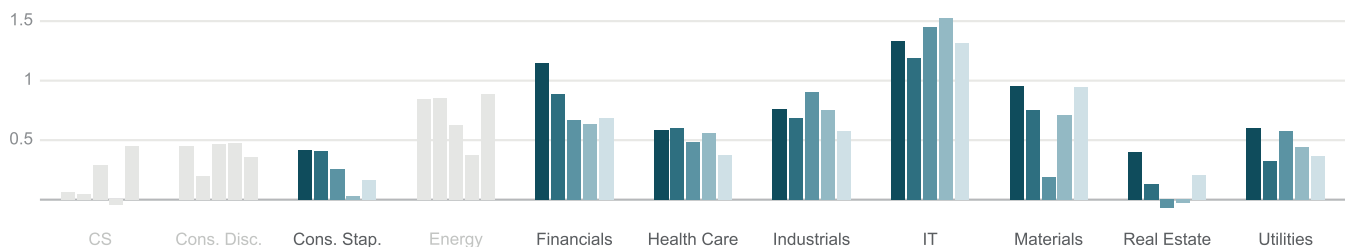
Valuation L-S CAGR +8.4%

035 ● **Expected LTG**

Analyst Expectations L-S CAGR +8.3%

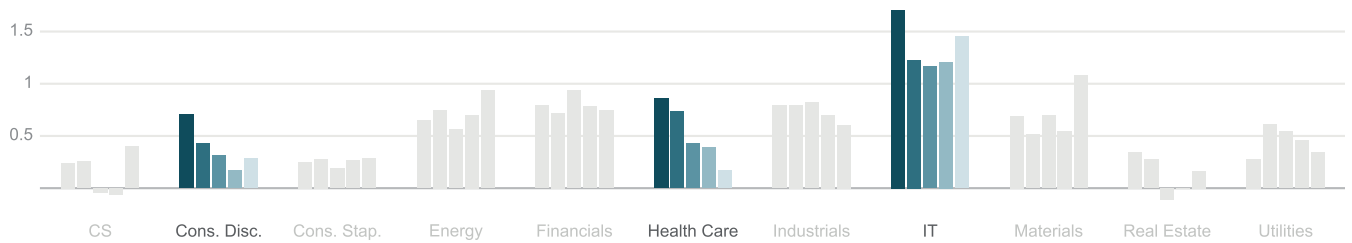
036 ● **Adjusted Forward Earnings to Price**

Valuation L-S CAGR +8.3%



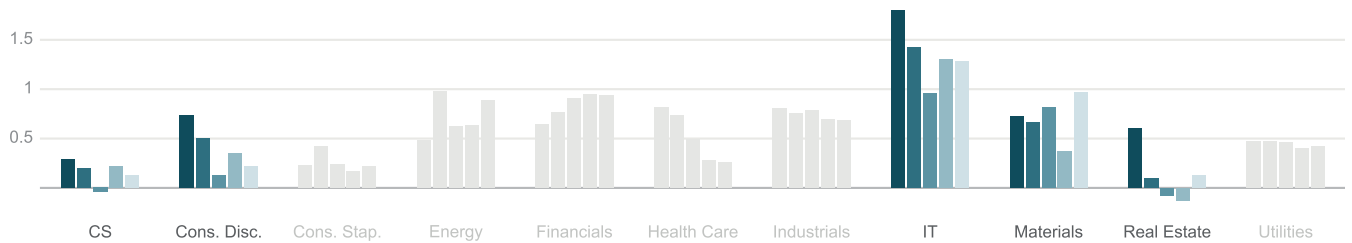
037 ● Return on Invested Capital

Capital Efficiency L-S CAGR +8.1%



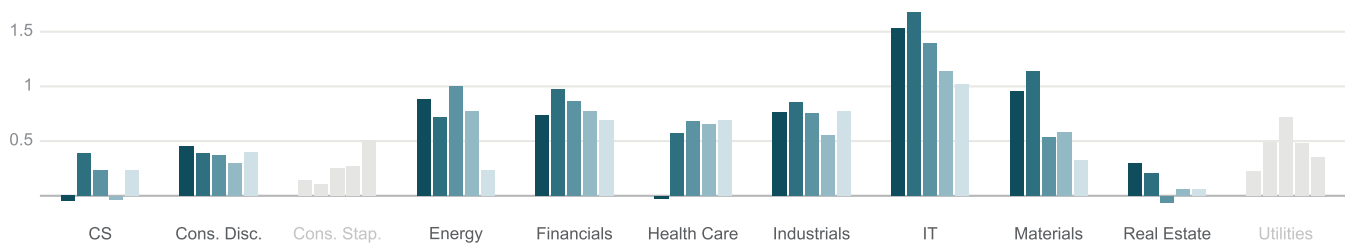
038 ● EBITDA to Assets

Capital Efficiency L-S CAGR +7.9%



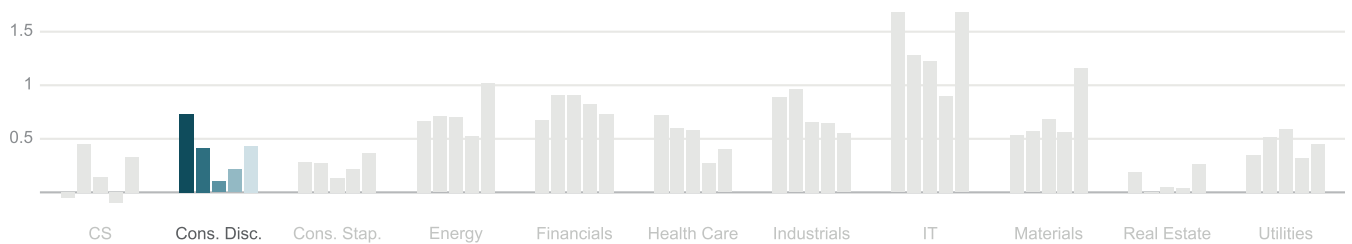
039 ● 1M Realized Price Volatility

Volatility L-S CAGR +7.9%



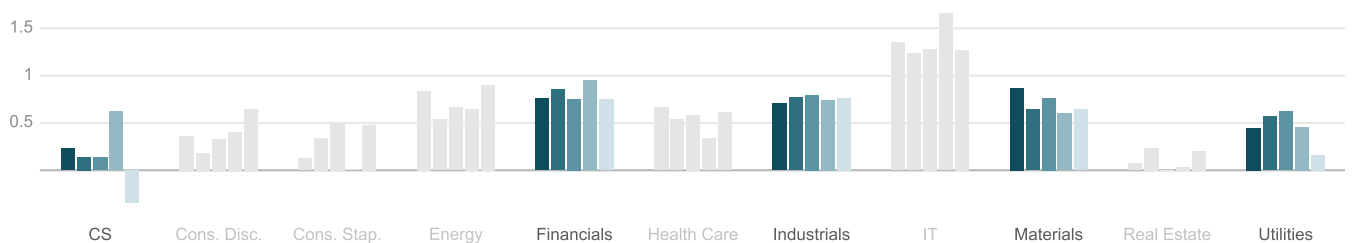
040 ● Return on Equity

Capital Efficiency L-S CAGR +7.9%



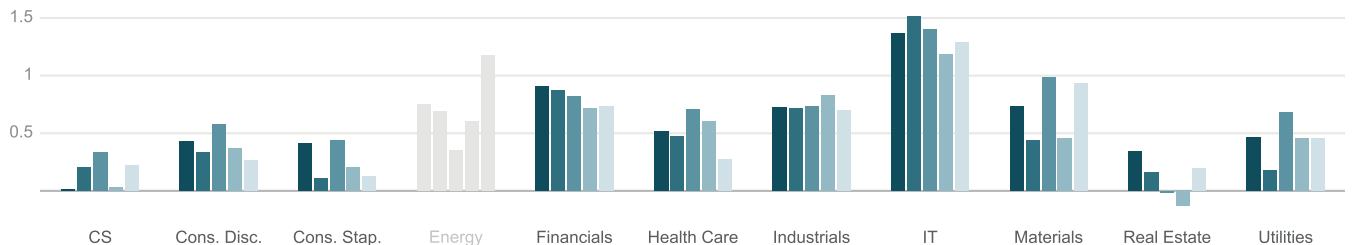
041 ● Long Term Debt to Assets Ratio

Capital Efficiency L-S CAGR +7.8%



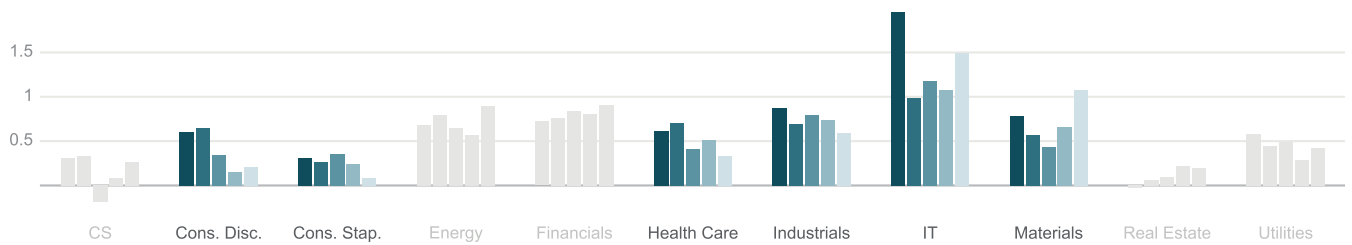
042 ● 1Y Chg in Sales to Price

Historical Growth L-S CAGR +7.7%

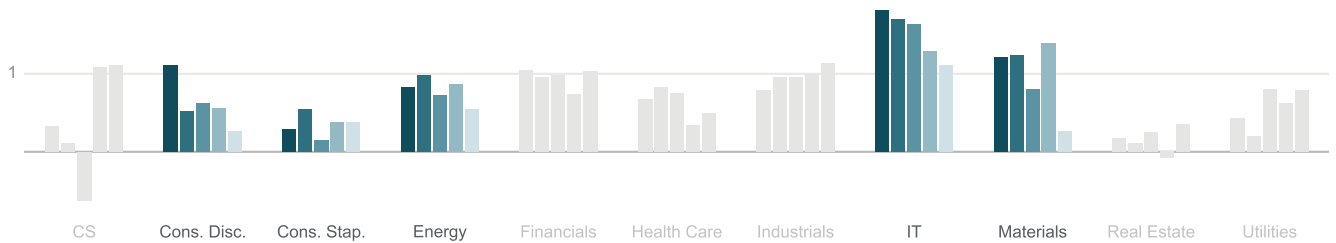


043 1Y Chg in Asset Adjusted EPS

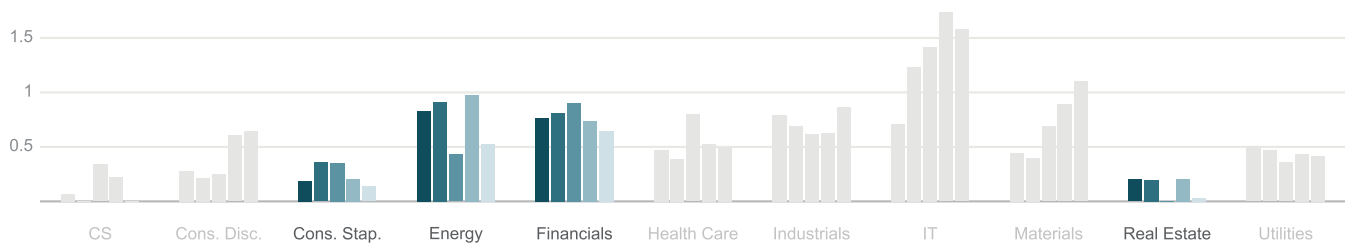
Historical Growth L-S CAGR +7.7%

**044 Consecutive Quarters of Declines in TTM Overhead/Sales**

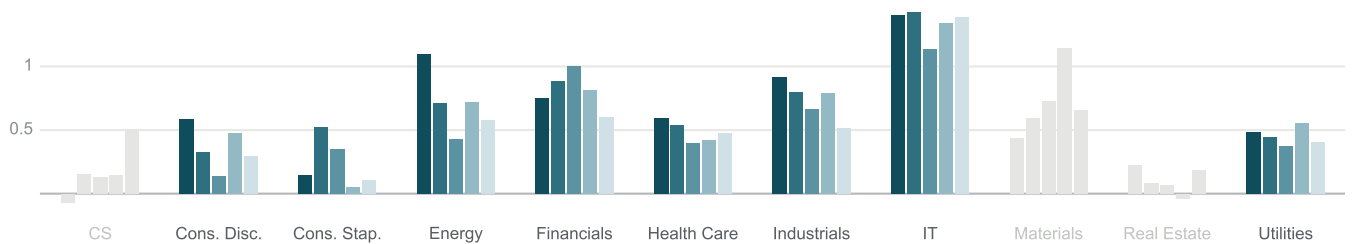
Historical Growth L-S CAGR +7.7%

**045 ROA 20 Qtr Standard Deviation**

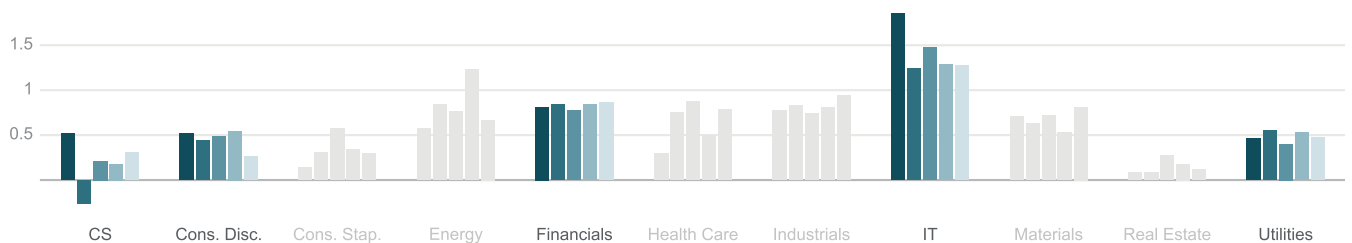
Earnings Quality L-S CAGR +7.7%

**046 Book Leverage**

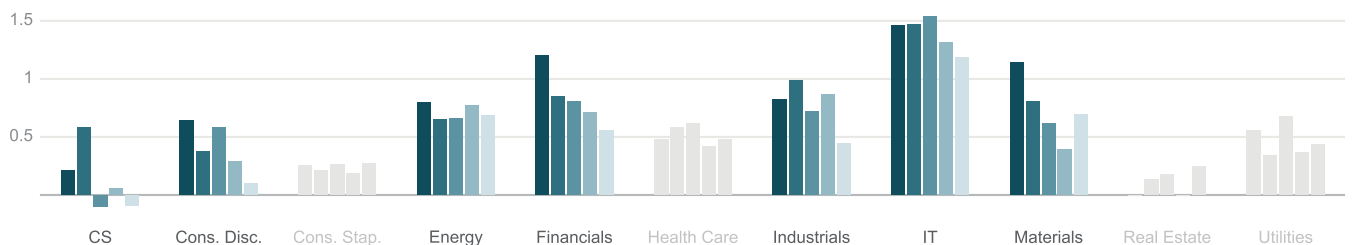
Volatility L-S CAGR +7.6%

**047 1Y Chg in Stock Buybacks**

Capital Efficiency L-S CAGR +7.6%

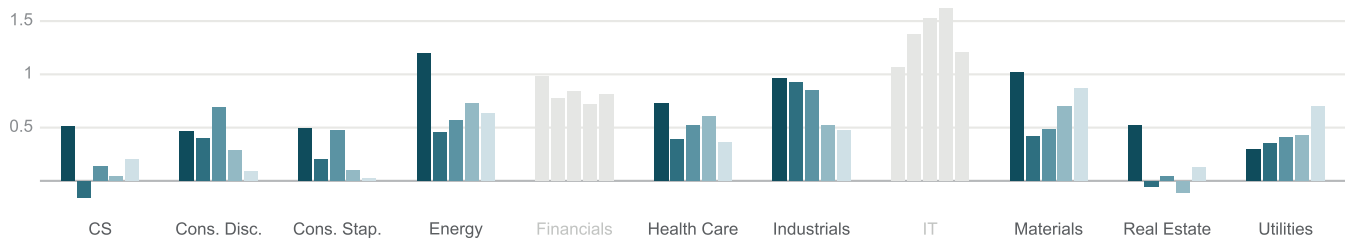
**048 Earnings Surprise**

Analyst Expectations L-S CAGR +7.5%

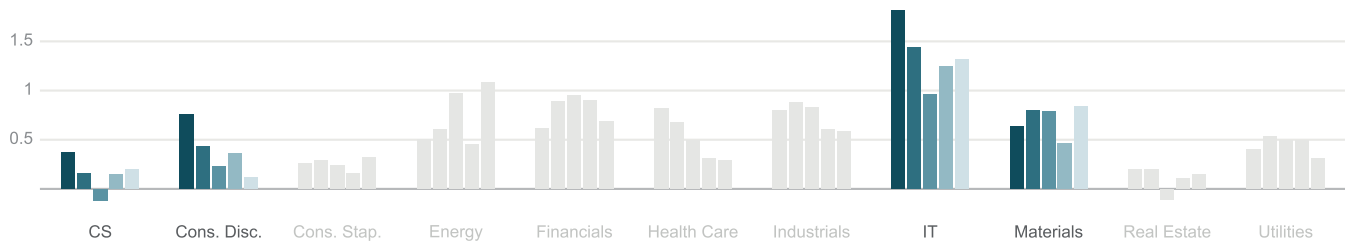


049 ● Free Cash Flow to Enterprise Value

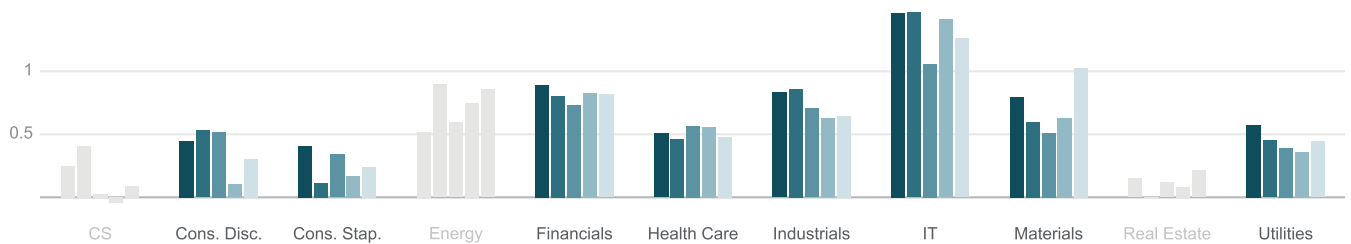
Valuation L-S CAGR +7.5%

**050 ● Cash Flow to Total Asset**

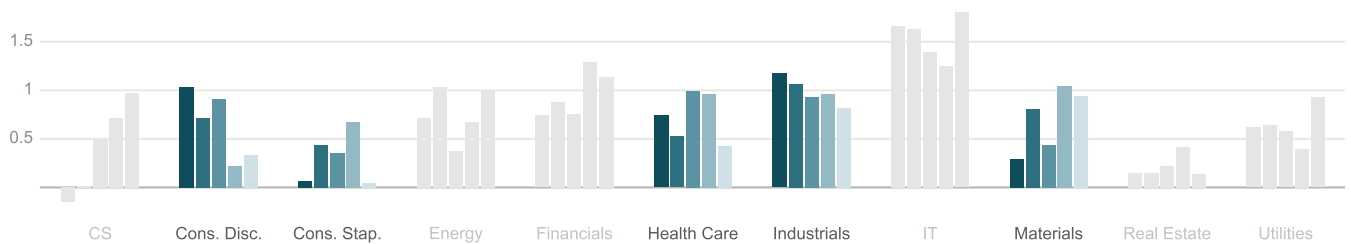
Earnings Quality L-S CAGR +7.5%

**051 ● 1Y Chg in Cash Flow to Price**

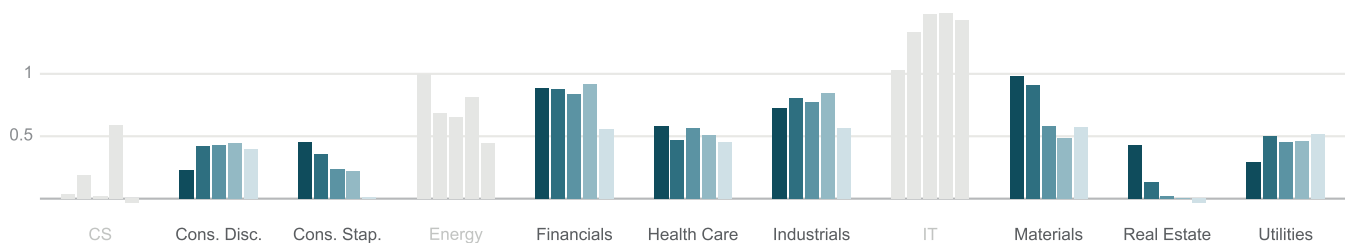
Historical Growth L-S CAGR +7.4%

**052 ● Adj 1Y Chg in Earnings per Share**

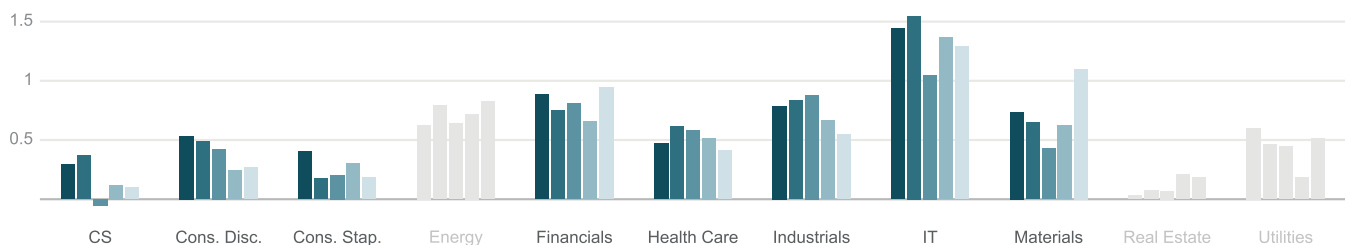
Historical Growth L-S CAGR +7.4%

**053 ● Operating Cash Flow to Price**

Valuation L-S CAGR +7.3%

**054 ● 1Y Chg in Earnings to Price**

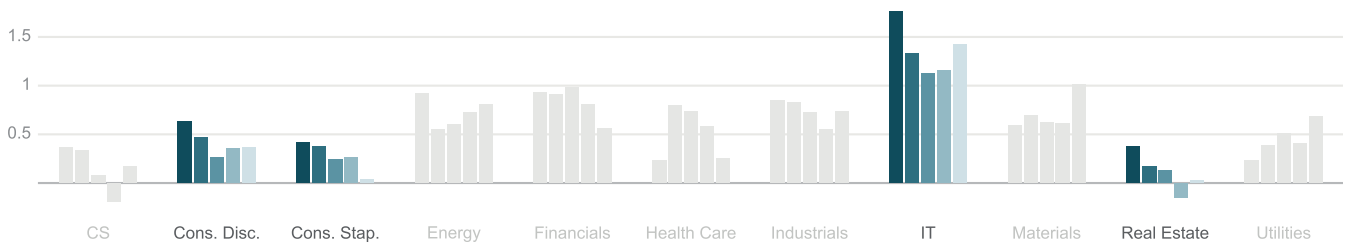
Historical Growth L-S CAGR +7.3%



055 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth

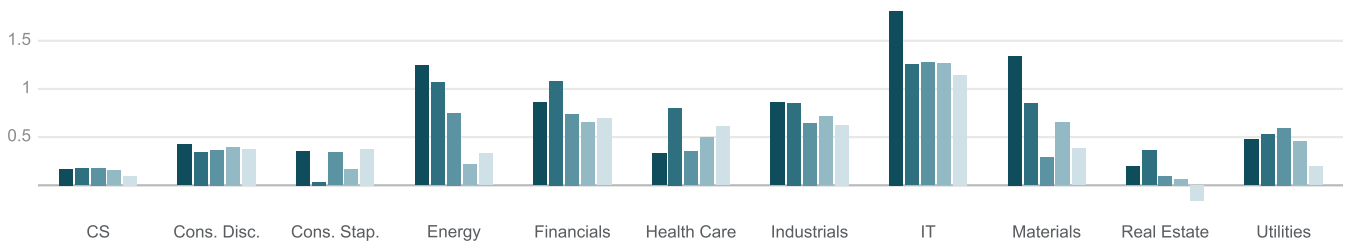
L-S CAGR +7.3%



056 ● 90 Day Coefficient of Variation

Volatility

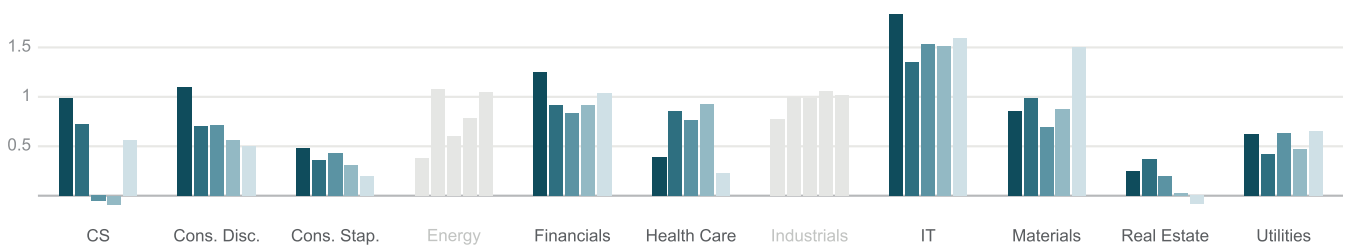
L-S CAGR +7.3%



057 ● 12-Quarter Trendline in Trailing 12-Month Cash Flows

Historical Growth

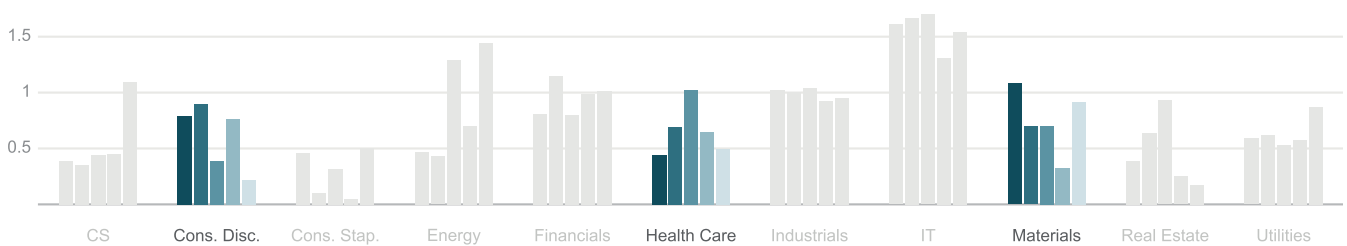
L-S CAGR +7.3%



058 ● EBITDA Margin

Earnings Quality

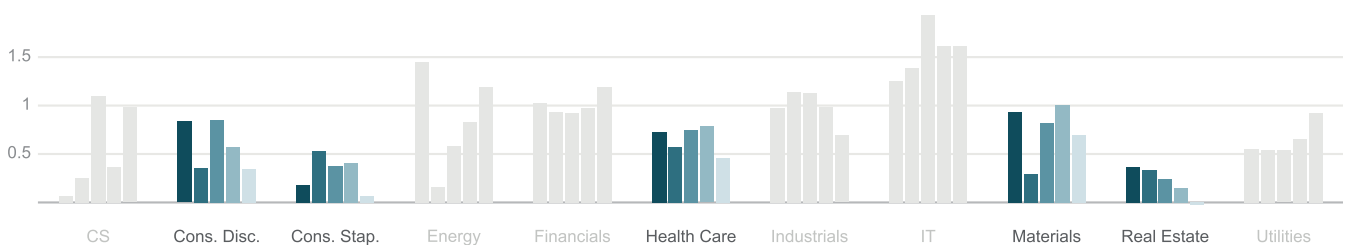
L-S CAGR +7.3%



059 ● Adj Free Cash Flow to Price

Valuation

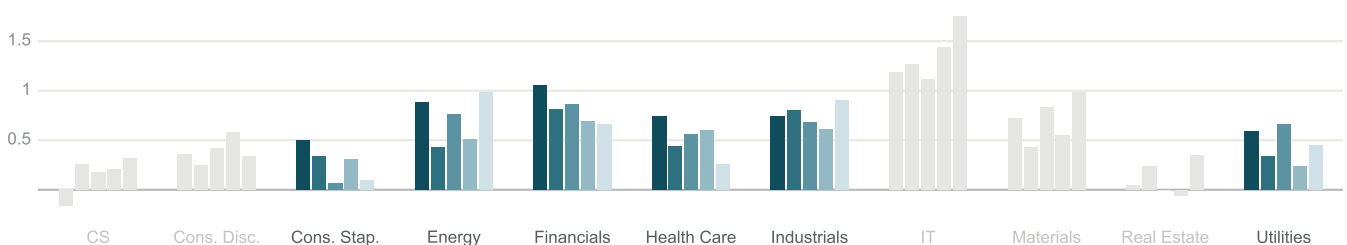
L-S CAGR +7.2%



060 ● Normalized EP

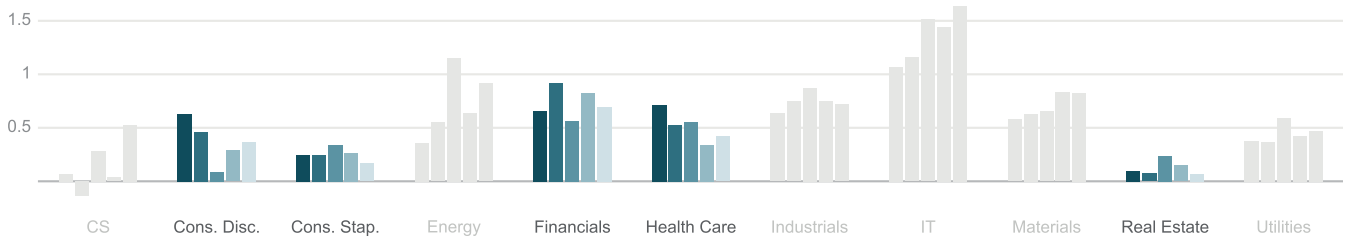
Valuation

L-S CAGR +7.2%



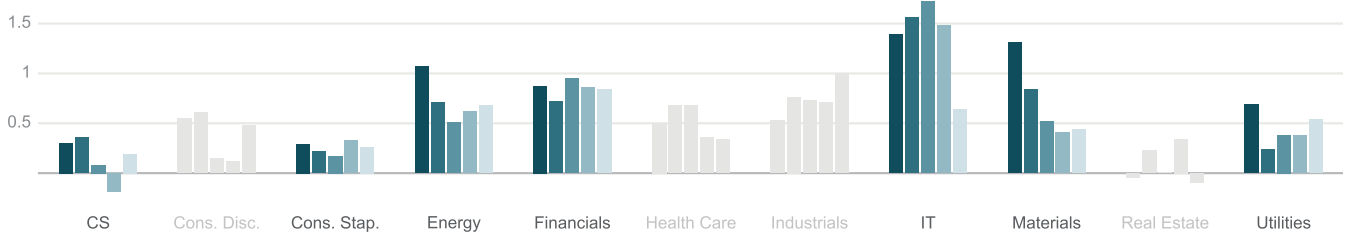
061 ● Working Capital to Sales

Earnings Quality L-S CAGR +7.2%



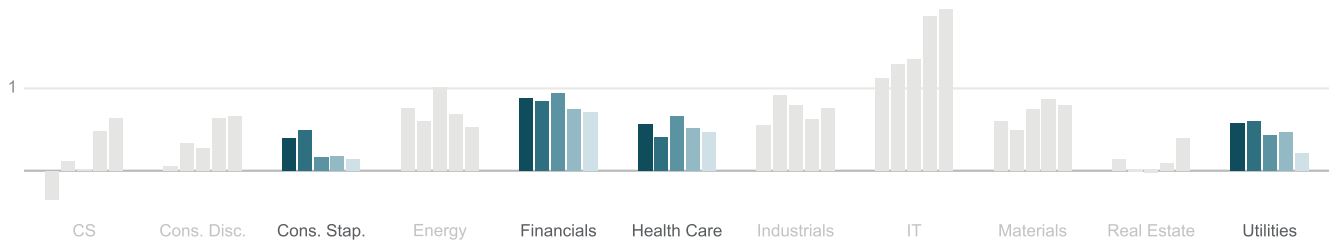
062 ● Receivables Turnover Ratio

Capital Efficiency L-S CAGR +7.2%



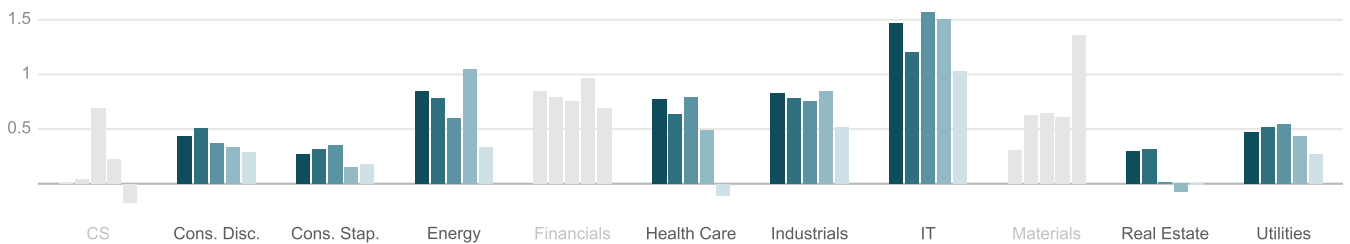
063 ● Dividends to Price Ratio

Valuation L-S CAGR +7.2%



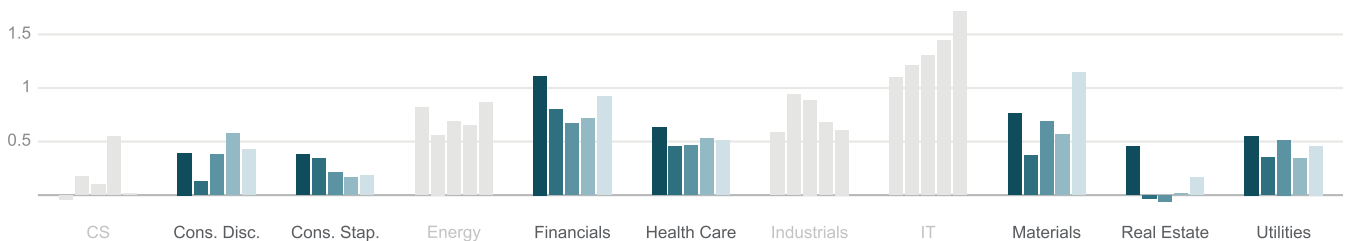
064 ● Sales to Invested Capital

Earnings Quality L-S CAGR +7.2%



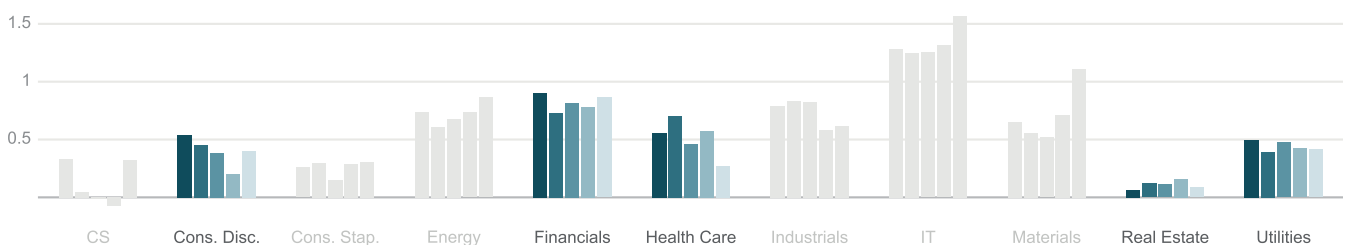
065 ● Adjusted EBIT to Price

Valuation L-S CAGR +7.1%



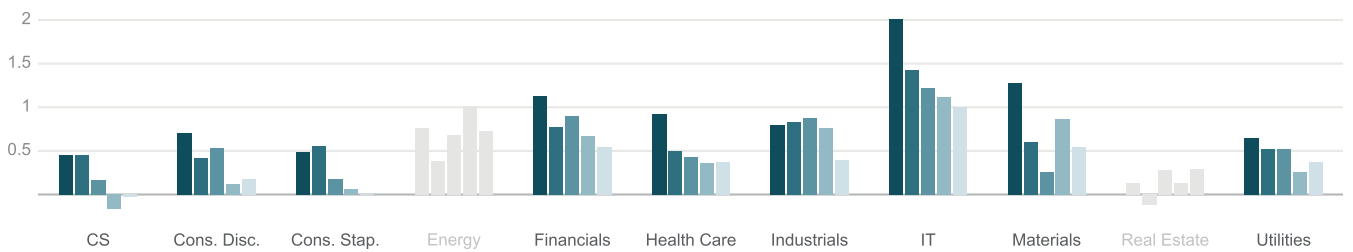
066 ● 3Y Chg in Cash Flow to Price

Historical Growth L-S CAGR +7.1%

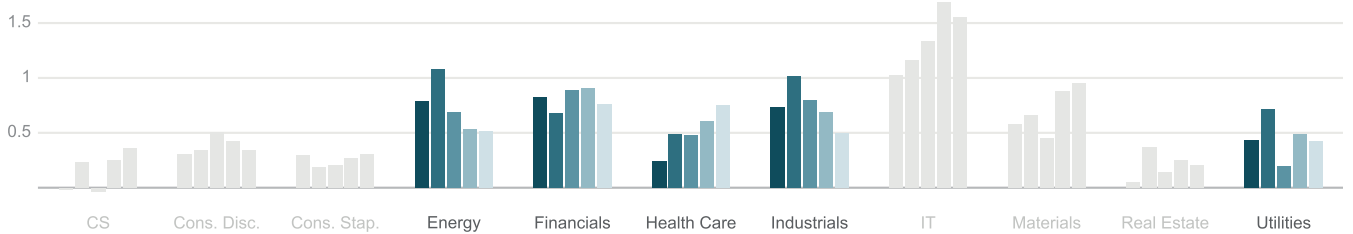


067 Street Revision Magnitude

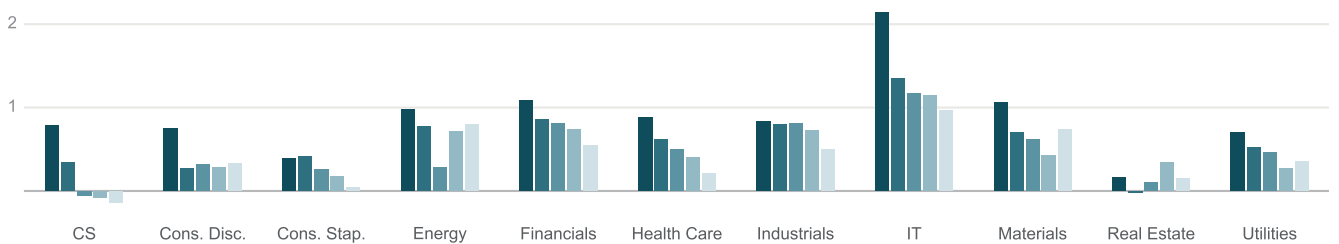
Analyst Expectations L-S CAGR +7.1%

**068 Capital Expenditure to Total Assets**

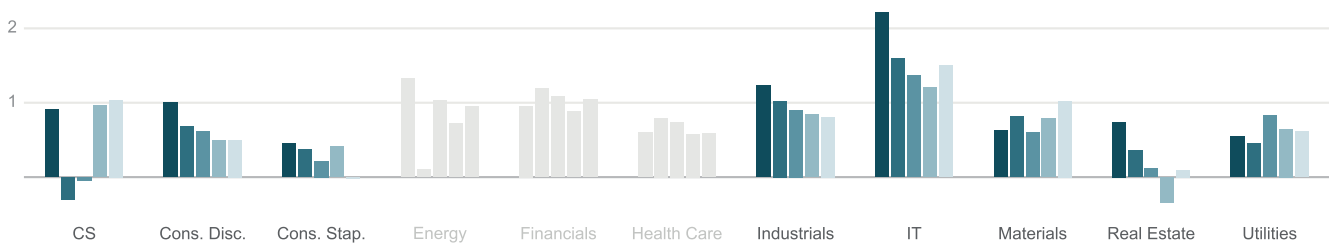
Capital Efficiency L-S CAGR +7.0%

**069 3M Revision in FY2 EPS Estimate**

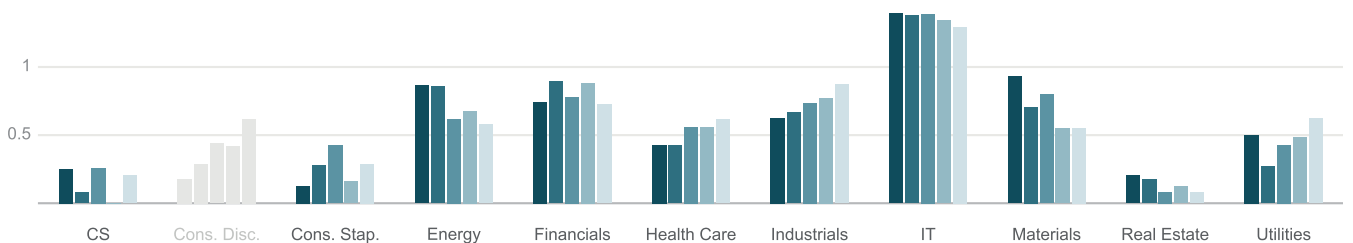
Analyst Expectations L-S CAGR +7.0%

**070 Adjusted 3Y Chg in Asset Adjusted Free Cash Flow**

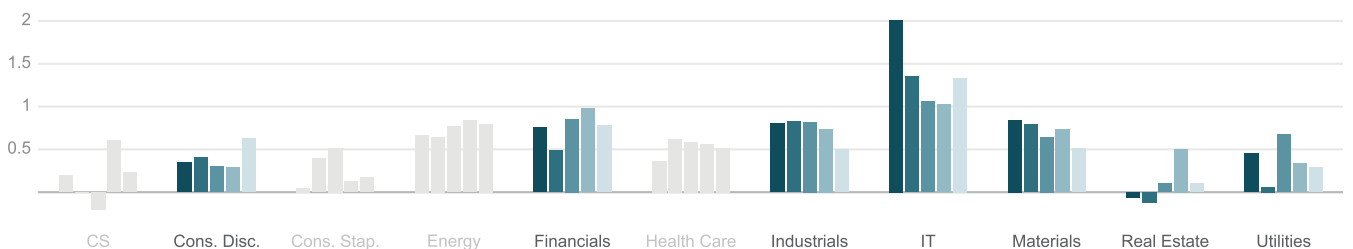
Historical Growth L-S CAGR +7.0%

**071 Analyst Dispersion for FY1 EPS**

Analyst Expectations L-S CAGR +7.0%

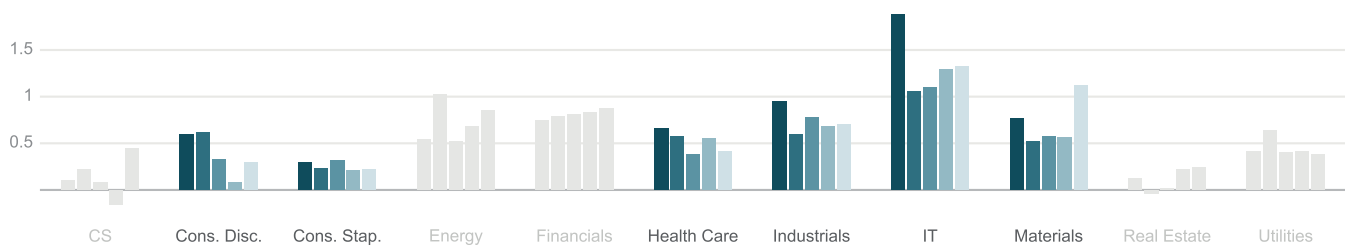
**072 Year over Year Change in SGA to Sales**

Earnings Quality L-S CAGR +6.9%



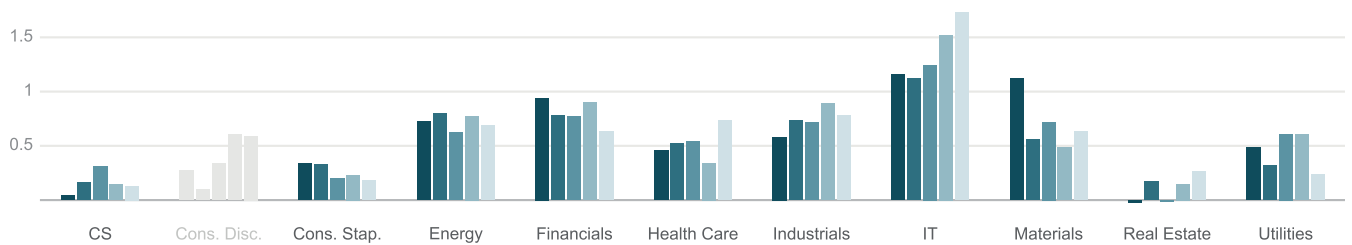
073 1Y Chg in Asset Adjusted Cash Flow

Historical Growth L-S CAGR +6.9%



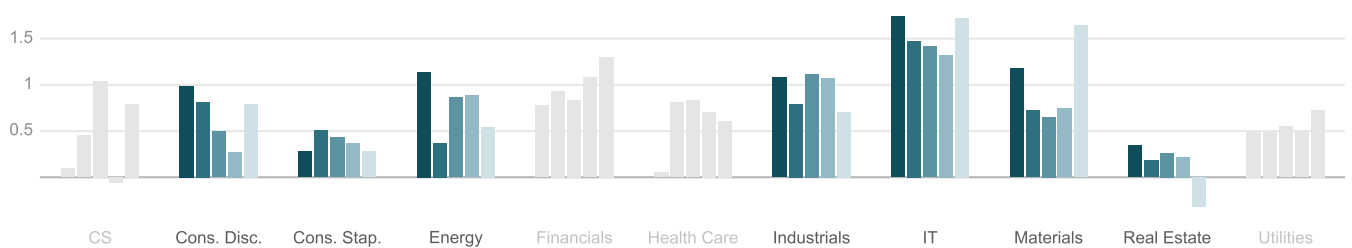
074 Tobin's Q Ratio

Capital Efficiency L-S CAGR +6.8%



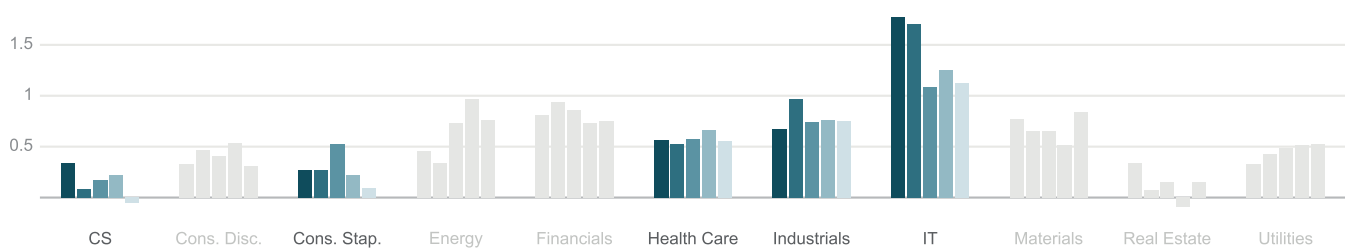
075 12-Quarter Trendline in Trailing 12-Month Earnings

Historical Growth L-S CAGR +6.8%



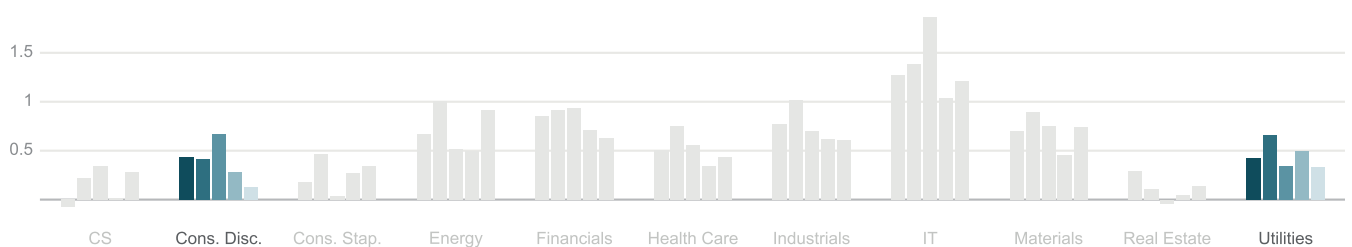
076 Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1

Historical Growth L-S CAGR +6.8%



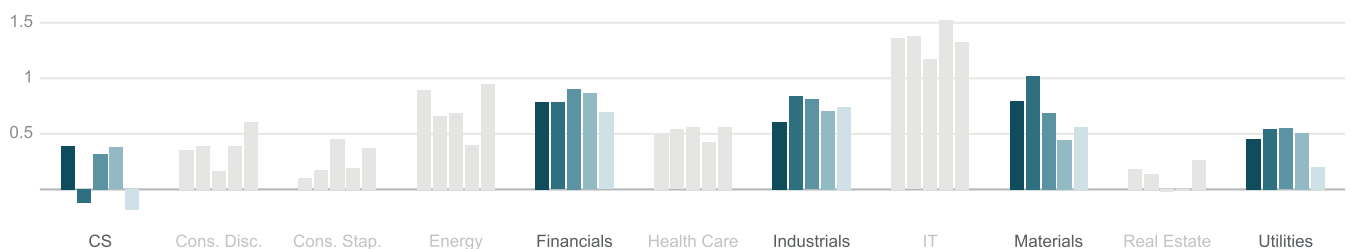
077 Institution Ownership Level - Active Manager

Price Momentum L-S CAGR +6.8%



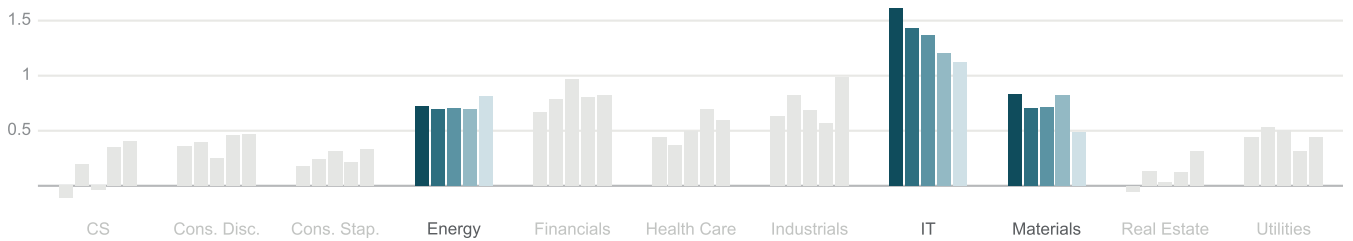
078 Debt to Assets Ratio

Capital Efficiency L-S CAGR +6.7%

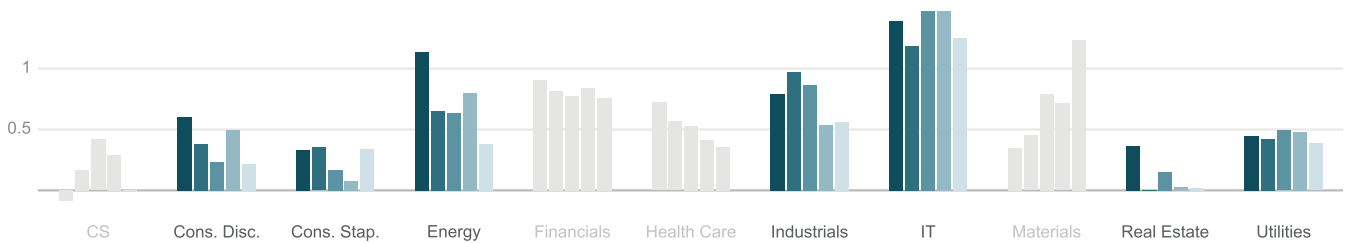


079 ● **Stability of Change in Number of Holders**

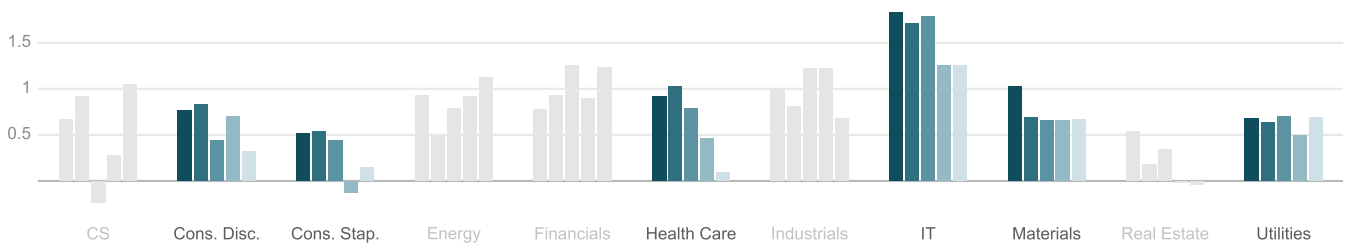
Price Momentum L-S CAGR +6.7%

080 ● **Equity Turnover**

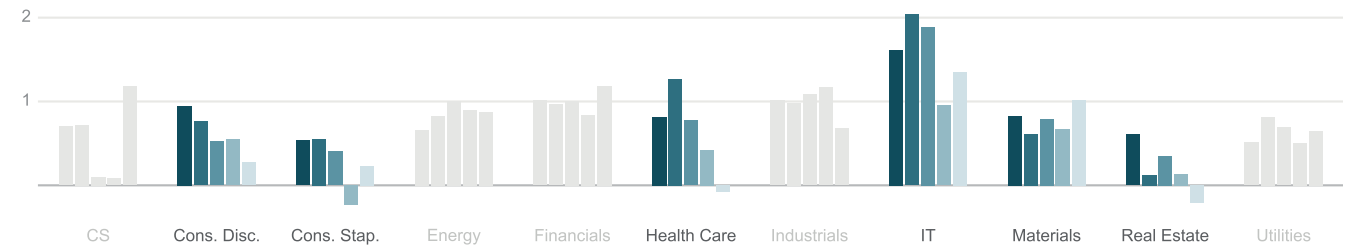
Earnings Quality L-S CAGR +6.7%

081 ● **Operating Cash Flow to Asset**

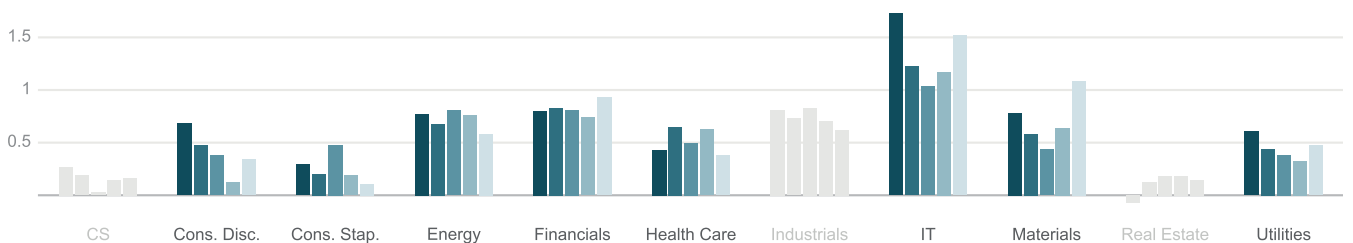
Earnings Quality L-S CAGR +6.6%

082 ● **Operating Cash Flow to Invested Capital**

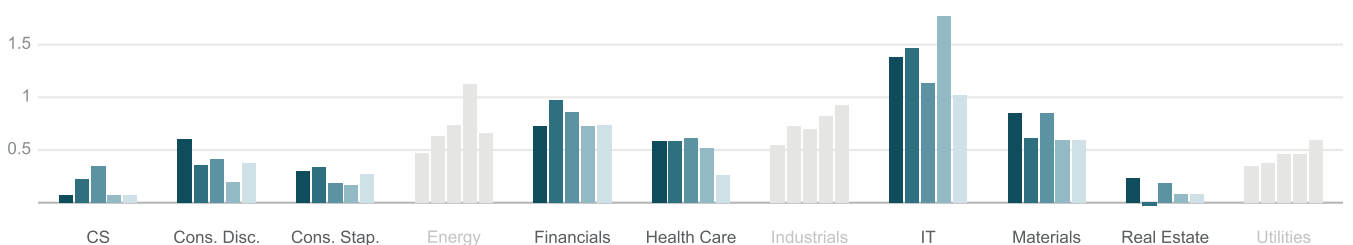
Earnings Quality L-S CAGR +6.6%

083 ● **Change EPS to Sales**

Earnings Quality L-S CAGR +6.6%

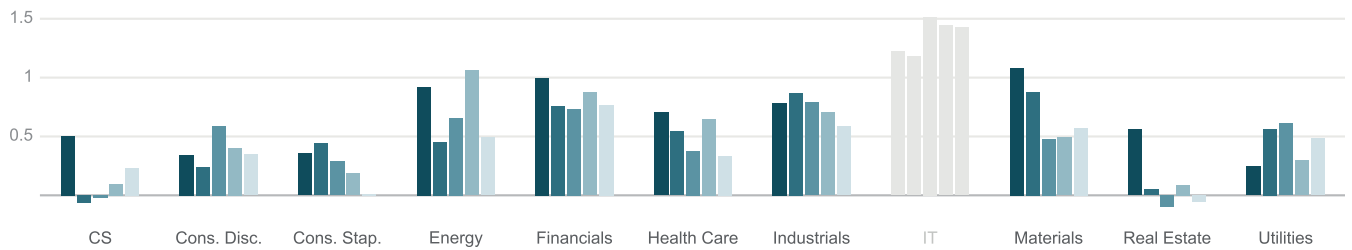
084 ● **Balance Sheet Accruals**

Earnings Quality L-S CAGR +6.6%

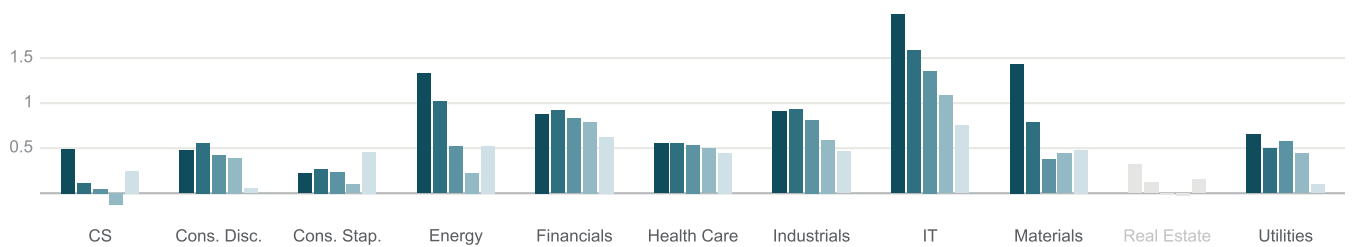


085 Operating Cash Flow to Enterprise Value

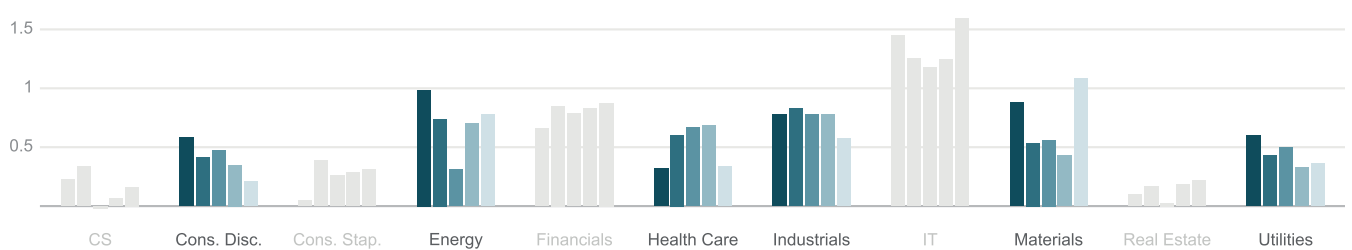
Valuation L-S CAGR +6.5%

**086 Percent Above 260-Day Low**

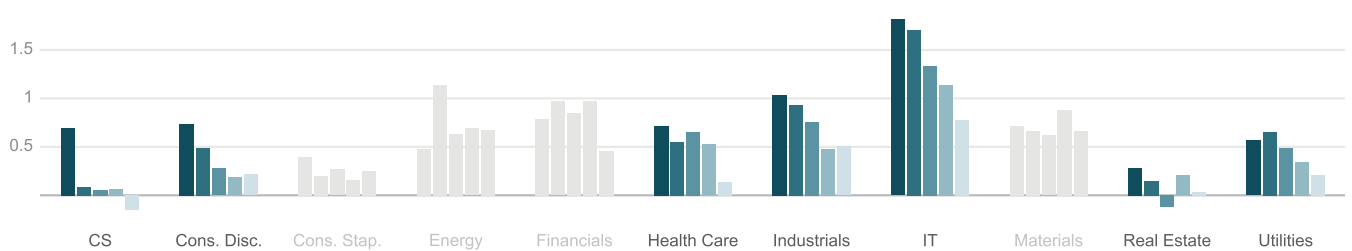
Price Momentum L-S CAGR +6.5%

**087 1Y Chg in Operating Margin**

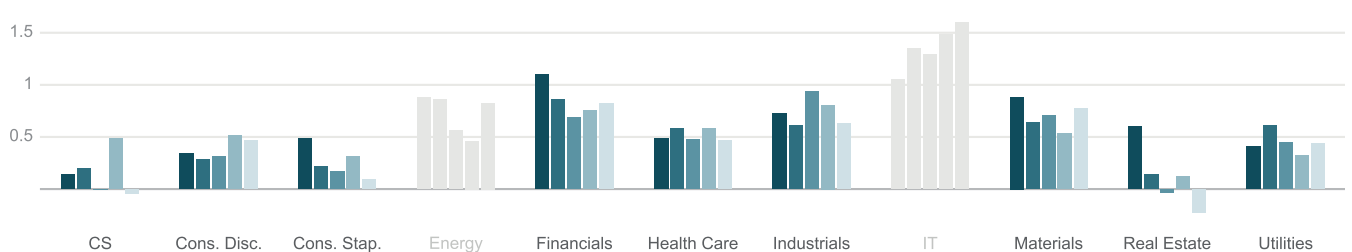
Historical Growth L-S CAGR +6.4%

**088 Amihud Illiquidity Measure**

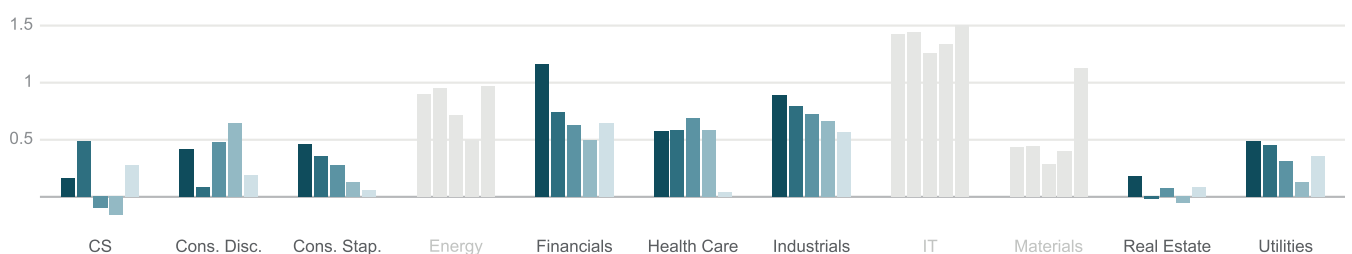
Volatility L-S CAGR +6.3%

**089 EBITDA to Price**

Valuation L-S CAGR +6.3%

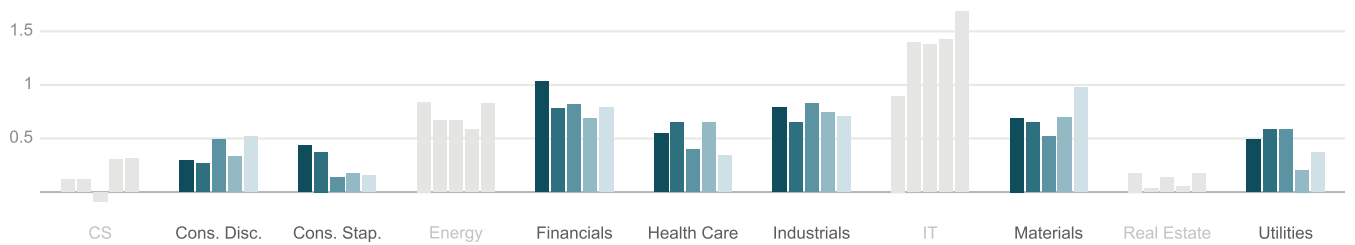
**090 Growth Flow to Price**

Valuation L-S CAGR +6.3%



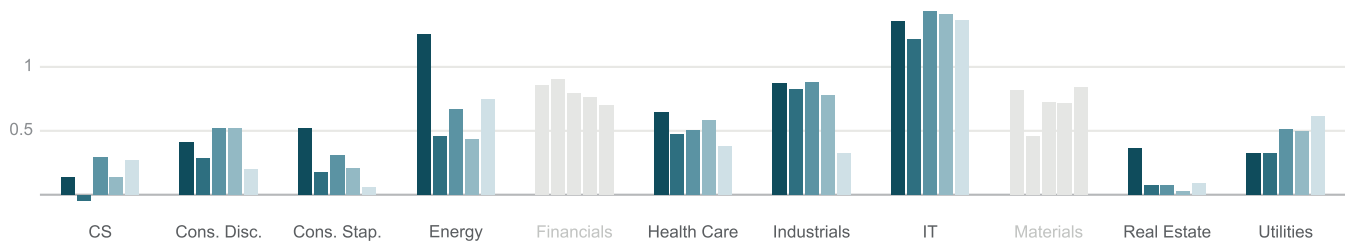
091 ● Net Cash Flow to Price

Valuation L-S CAGR +6.2%



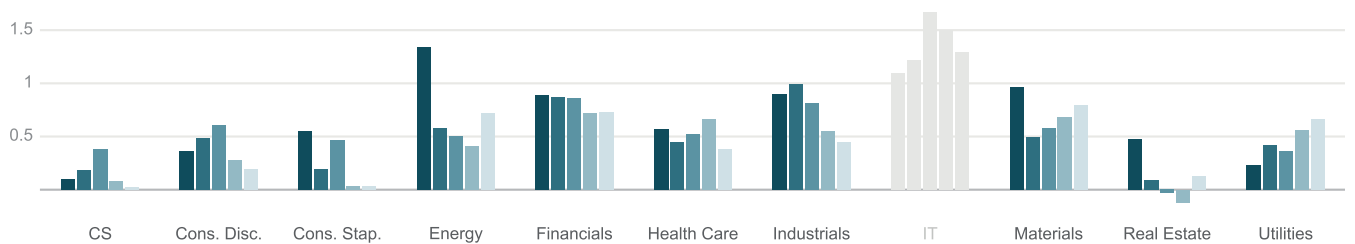
092 ● Forward Free Cash Flow to Price Ratio

Valuation L-S CAGR +6.2%



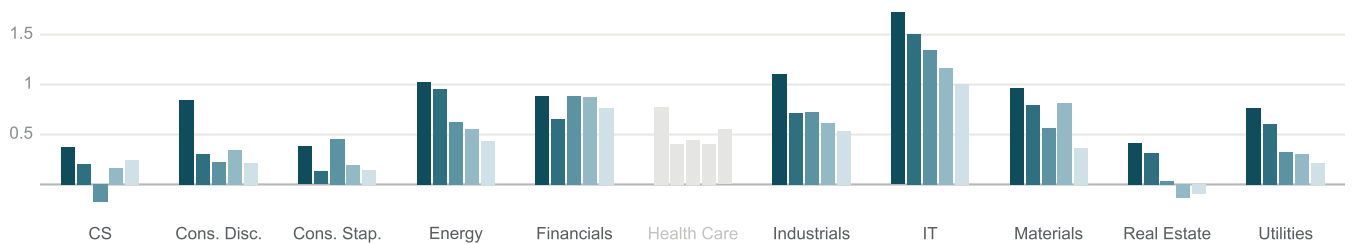
093 ● Free Cash Flow to Price

Valuation L-S CAGR +6.2%



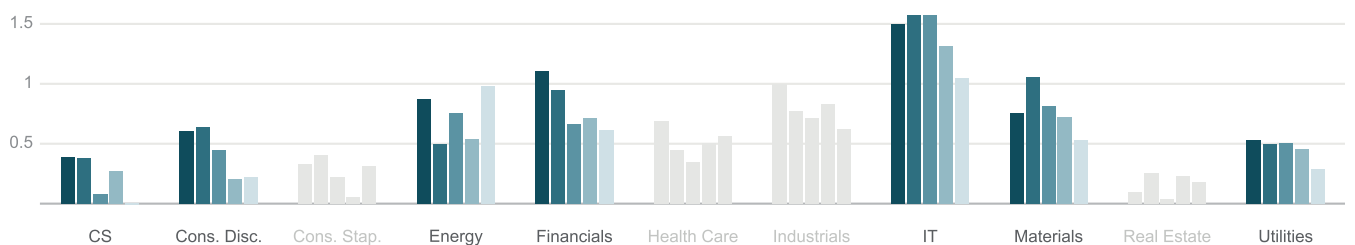
094 ● Slope of 52W Price Trend

Price Momentum L-S CAGR +6.2%



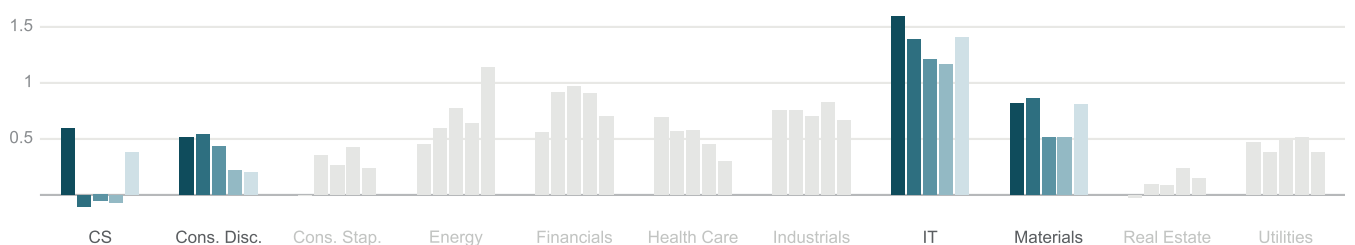
095 ● Standardized Unexpected Earnings

Analyst Expectations L-S CAGR +6.1%



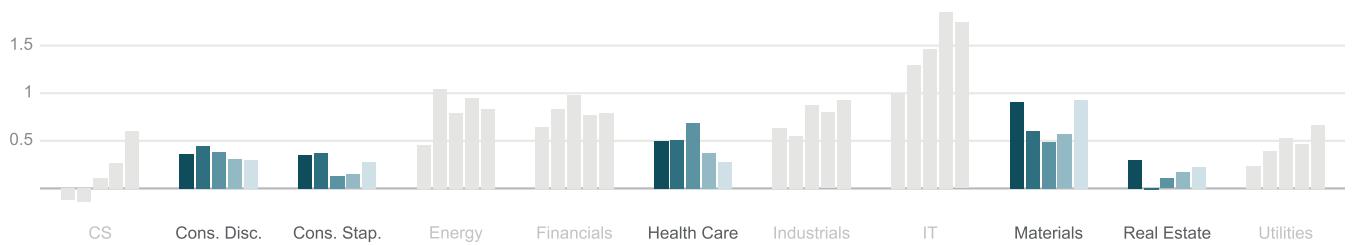
096 ● Solvency Ratio

Valuation L-S CAGR +6.1%



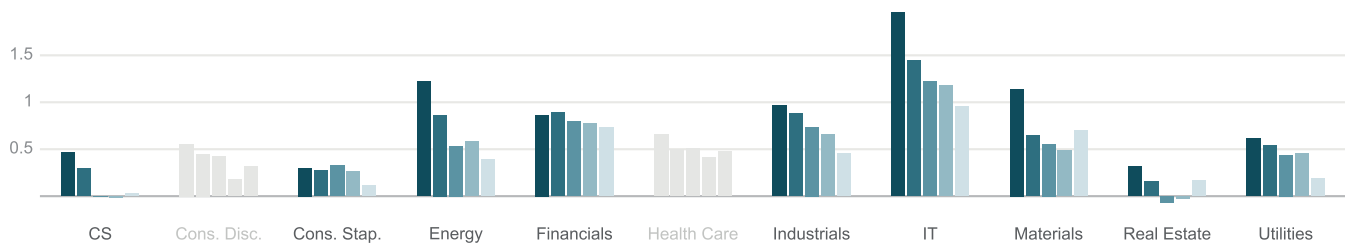
097 ● Inventory Turnover

Earnings Quality L-S CAGR +6.1%



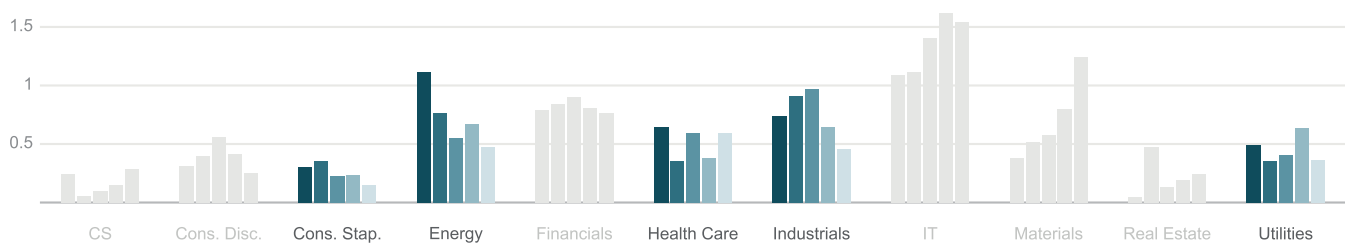
098 ● 39W Lagged Return

Price Momentum L-S CAGR +6.0%



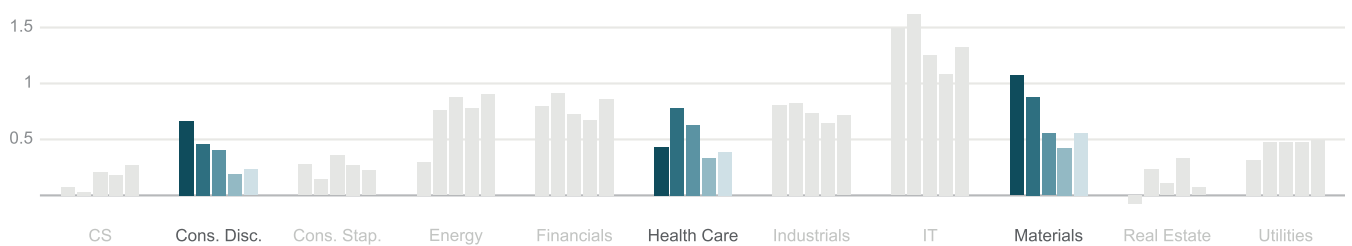
099 ● Capital Expenditure to Sales

Earnings Quality L-S CAGR +6.0%



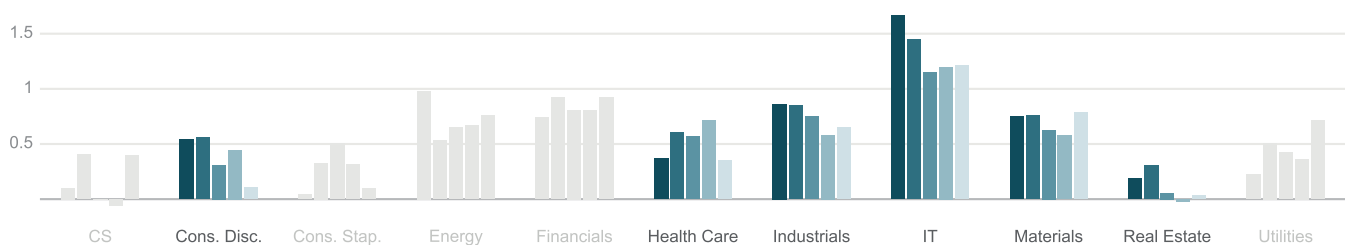
100 ● Operating Margin

Earnings Quality L-S CAGR +6.0%



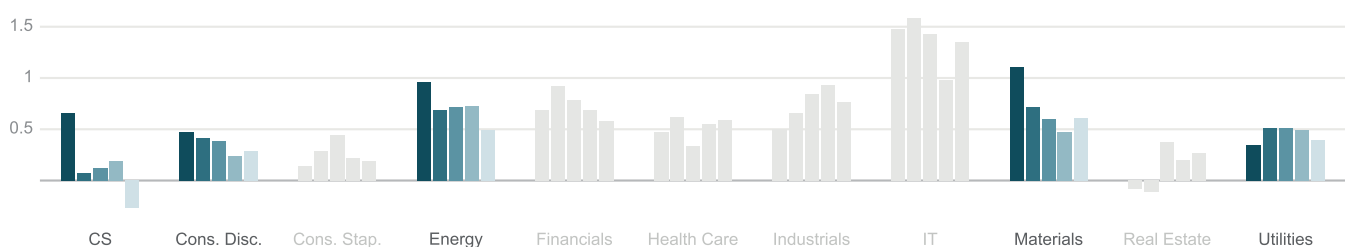
101 ● 1Y Chg in Operating Cash Flow per Share

Historical Growth L-S CAGR +5.9%



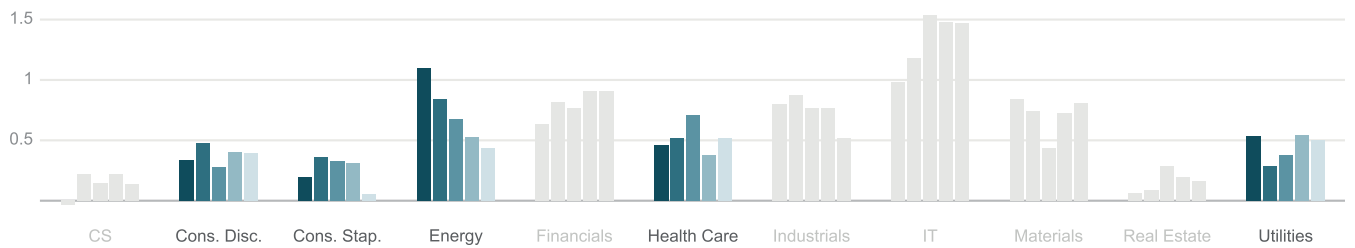
102 ● Cash Ratio

Earnings Quality L-S CAGR +5.9%



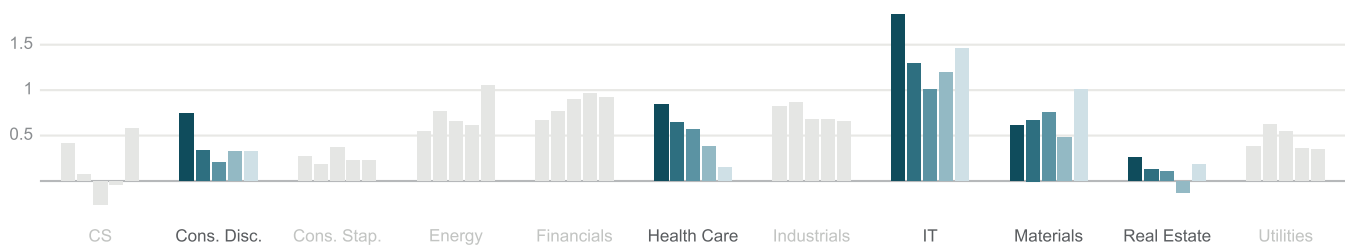
103 ● Depreciation to Capex Ratio

Earnings Quality L-S CAGR +5.9%



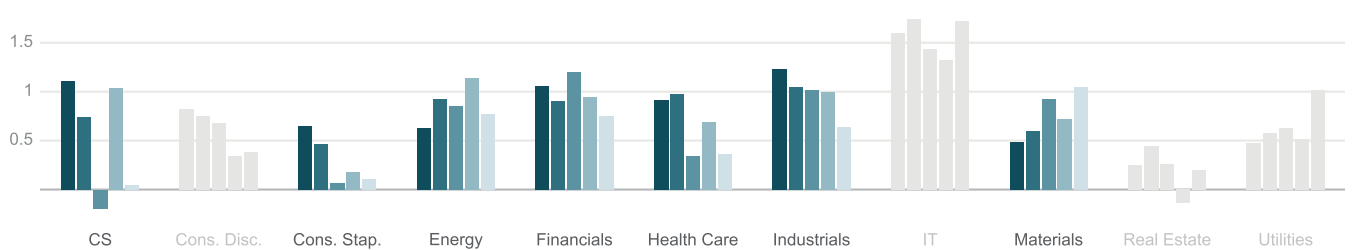
104 ● EBIT to Assets

Capital Efficiency L-S CAGR +5.9%



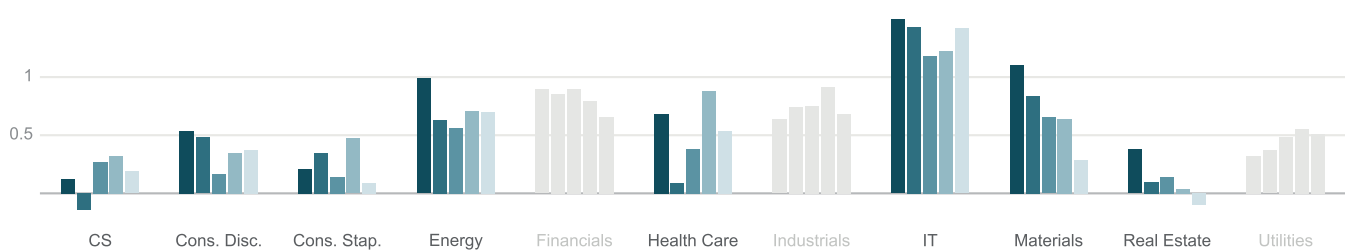
105 ● Short Interest Ratio

Price Momentum L-S CAGR +5.9%



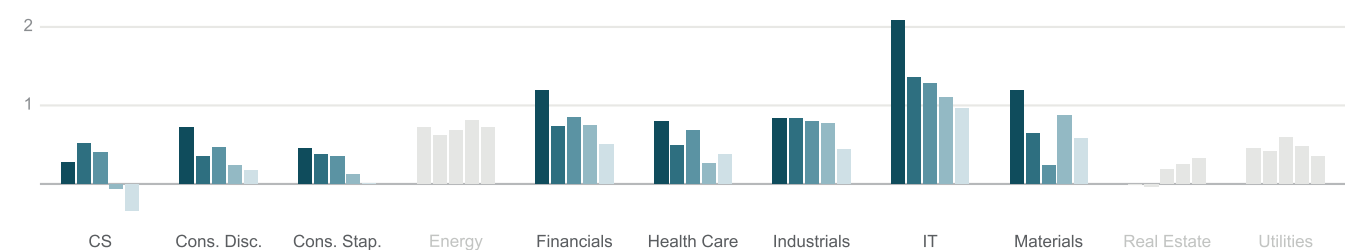
106 ● Adjusted Accruals

Earnings Quality L-S CAGR +5.9%



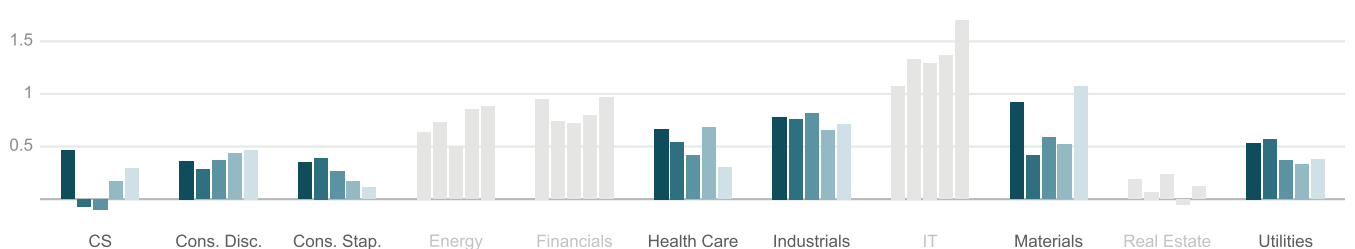
107 ● Adjusted Revision Magnitude

Analyst Expectations L-S CAGR +5.8%



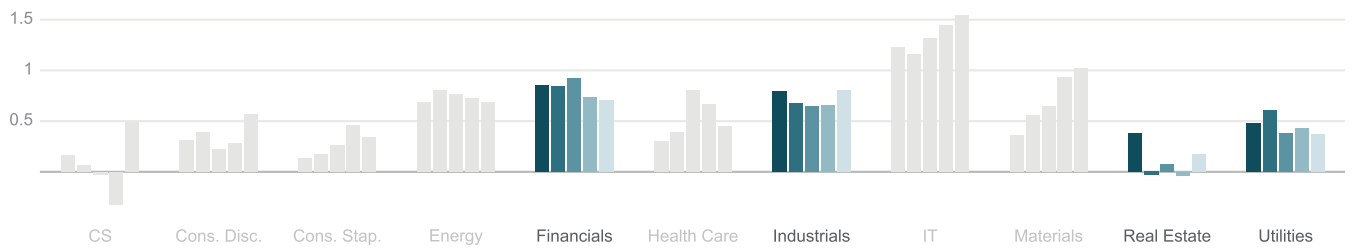
108 ● Net Cash Flow to Enterprise Value

Valuation L-S CAGR +5.8%

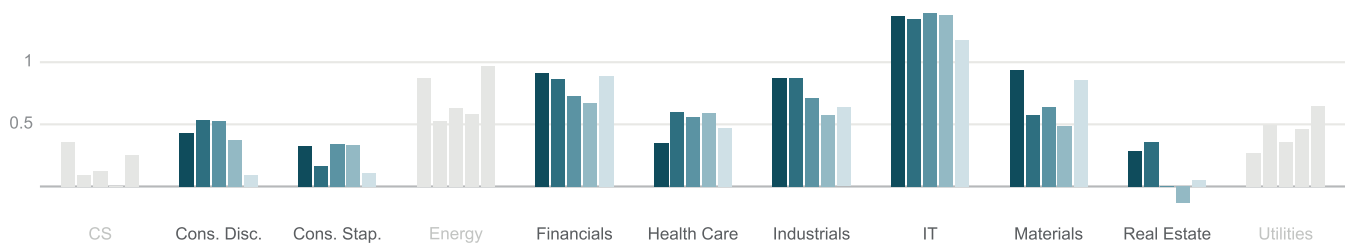


109 Dividend Growth

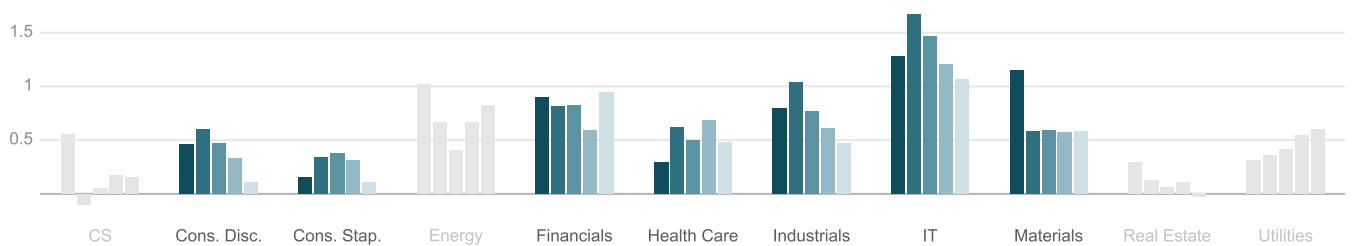
Historical Growth L-S CAGR +5.8%

**110 1Y Chg in Operating Cash Flow to Price**

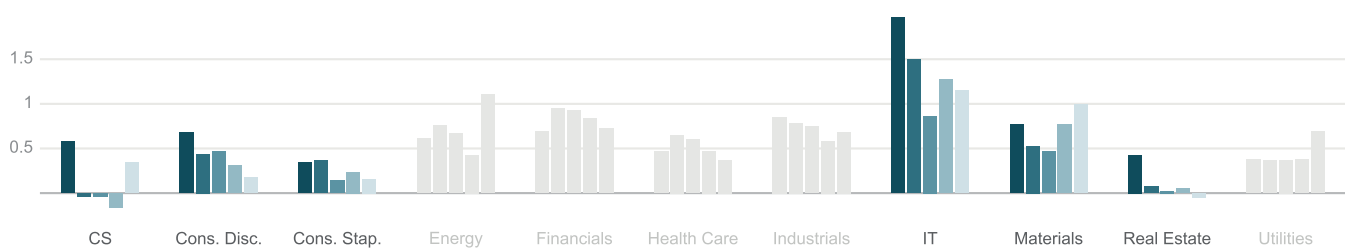
Historical Growth L-S CAGR +5.7%

**111 1Y Chg in Free Cash Flow to Price**

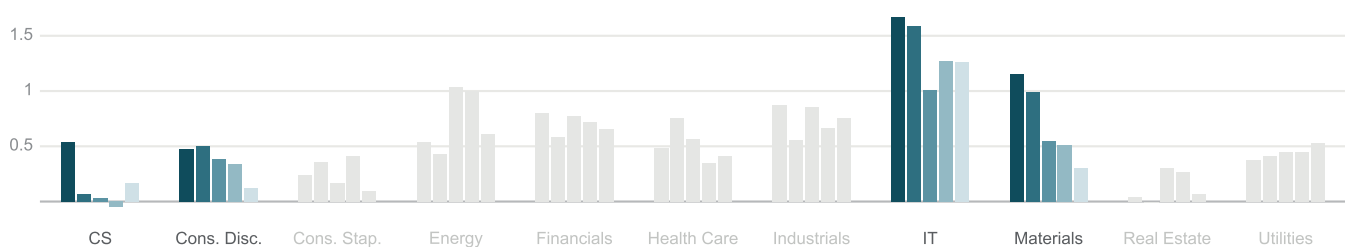
Historical Growth L-S CAGR +5.7%

**112 3Y Chg in Asset Adjusted Operating Cash Flow**

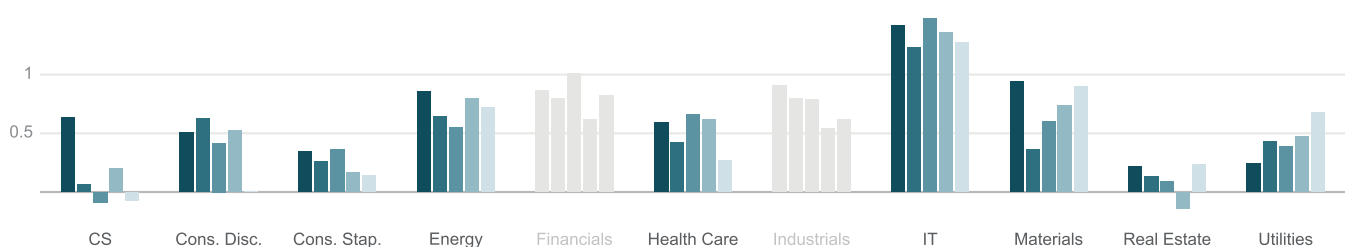
Historical Growth L-S CAGR +5.7%

**113 Operating Cash Flow Ratio**

Capital Efficiency L-S CAGR +5.6%

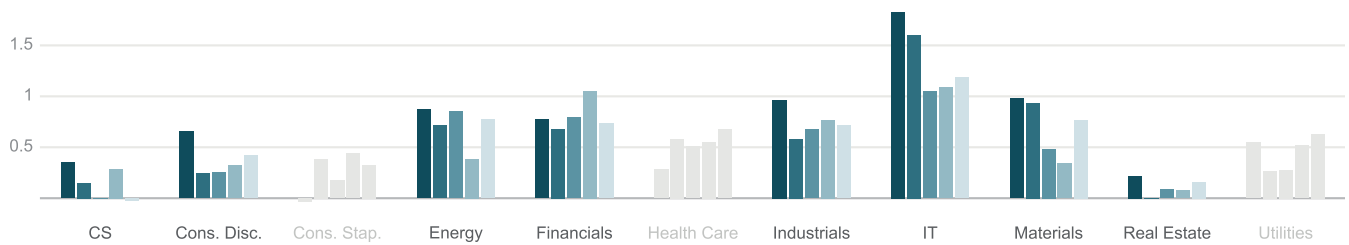
**114 3Y Chg in Free Cash Flow to Price**

Historical Growth L-S CAGR +5.6%



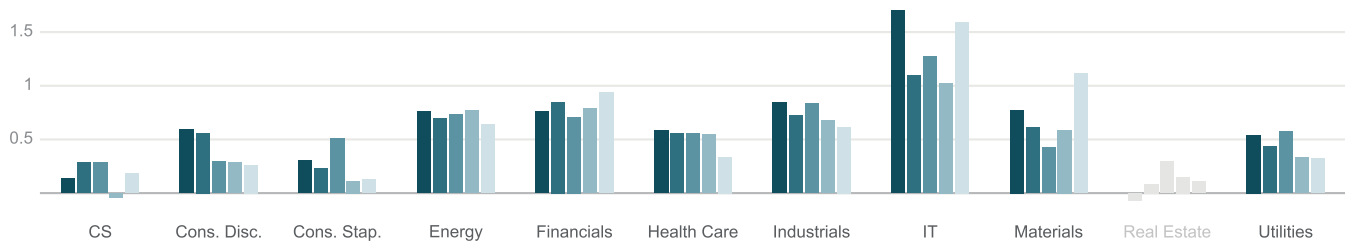
115 ● 52-Week Volume Price Trend with 20-Day Lag

Price Momentum L-S CAGR +5.6%



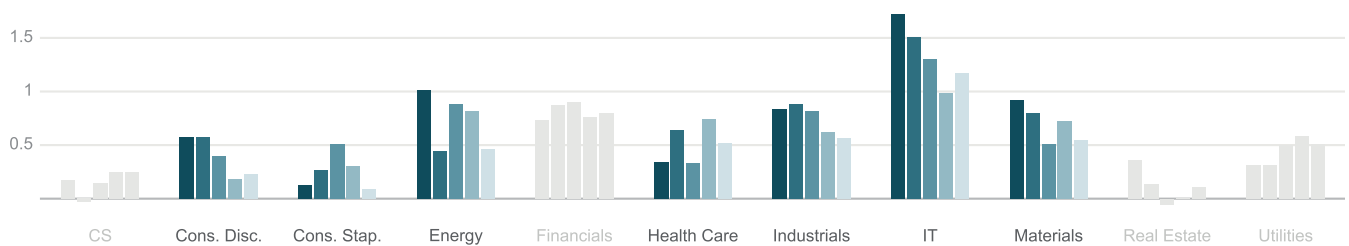
116 ● Year over Year Change of EPS to Sales

Earnings Quality L-S CAGR +5.6%



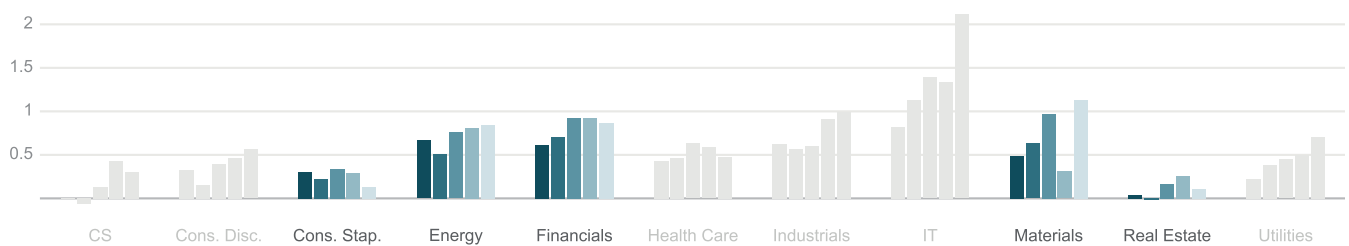
117 ● 1Y Chg in Asset Adjusted Free Cash Flow

Historical Growth L-S CAGR +5.6%



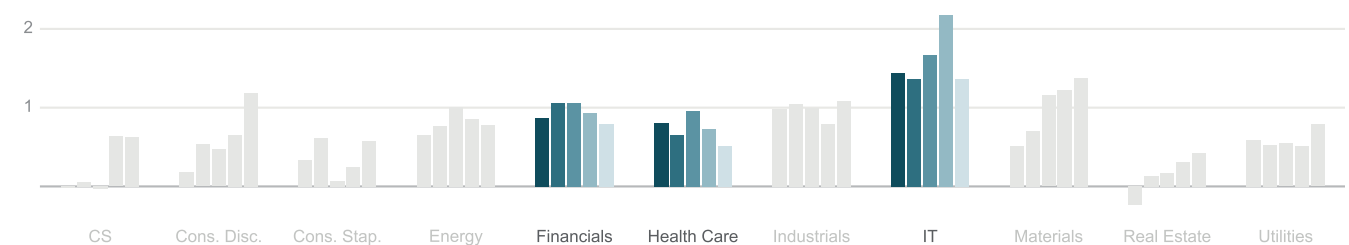
118 ● Adjusted 6 Month Average Share Turn Over

Price Momentum L-S CAGR +5.5%



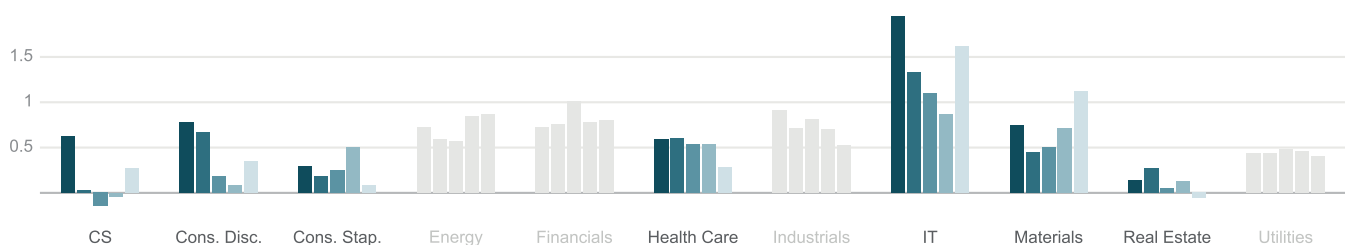
119 ● Indicated Annual Dividends to Price Ratio

Valuation L-S CAGR +5.5%



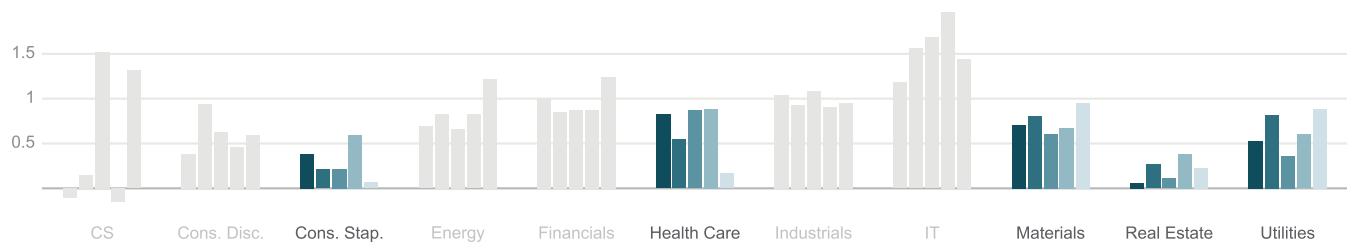
120 ● 3Y Chg in Asset Adjusted EPS

Historical Growth L-S CAGR +5.5%



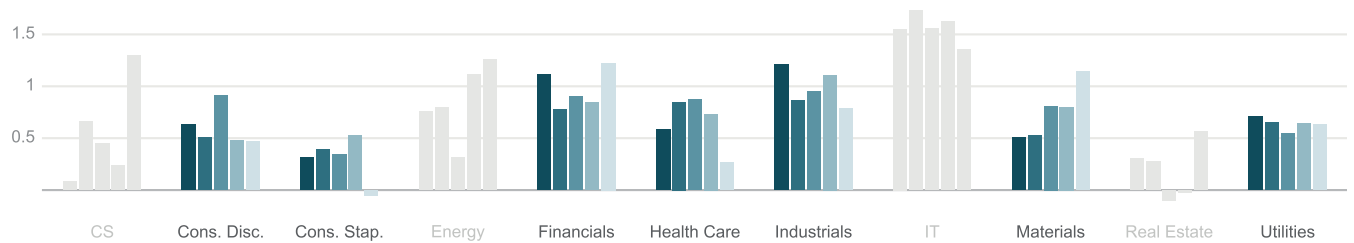
121 ● EBIT to Price

Valuation L-S CAGR +5.5%



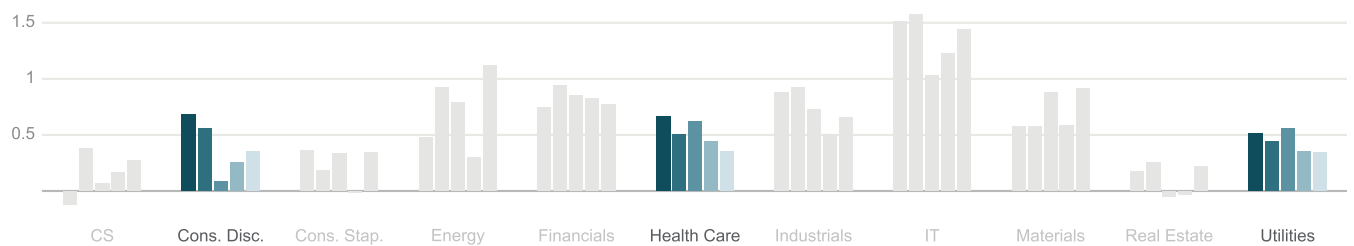
122 ● Adj Growth Flow to Price

Valuation L-S CAGR +5.5%



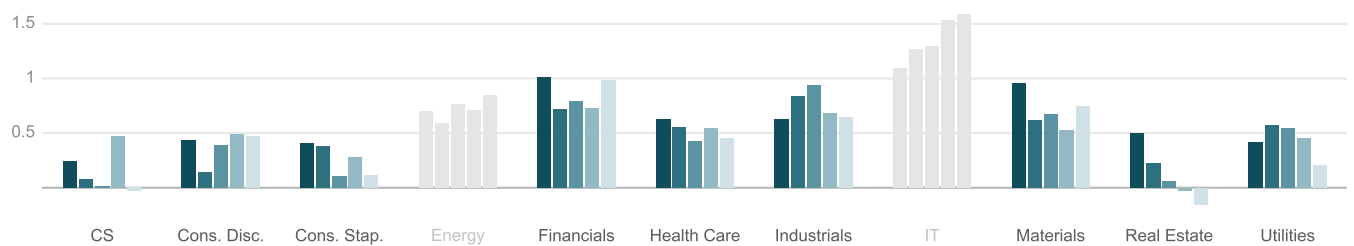
123 ● Cash Flow to Equity

Valuation L-S CAGR +5.5%



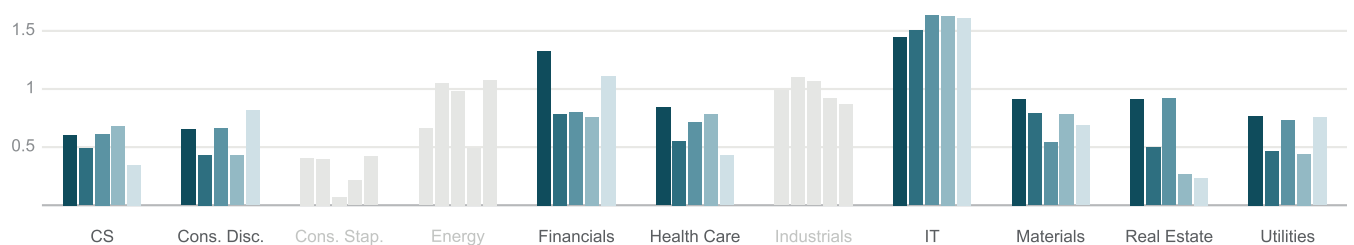
124 ● EBITDA to Enterprise Value

Valuation L-S CAGR +5.4%



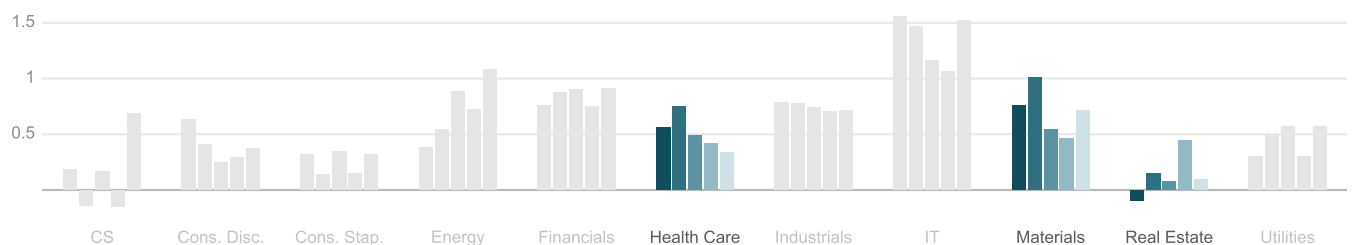
125 ● Adj. EBITDA to Enterprise Value

Valuation L-S CAGR +5.4%



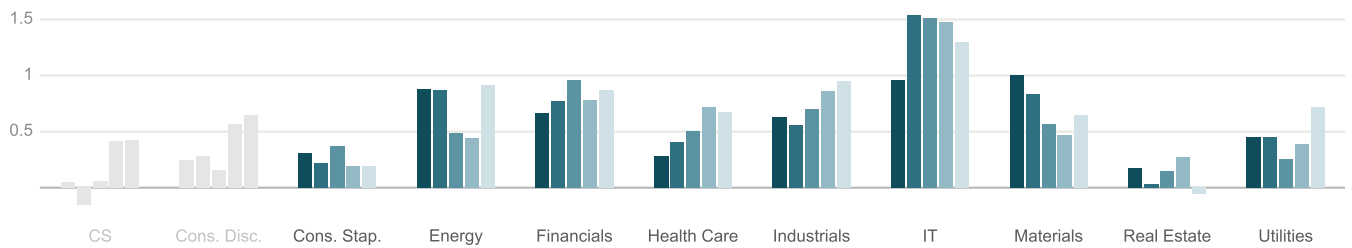
126 ● EBIT Margin

Earnings Quality L-S CAGR +5.4%



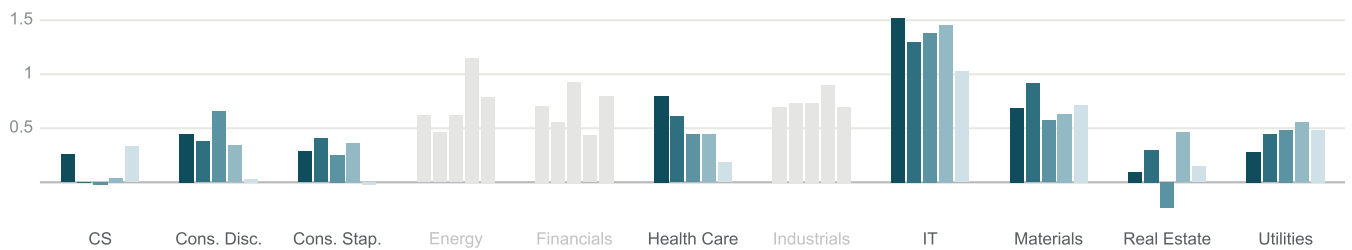
127 ● Log of Unadjusted Stock Price

Price Momentum L-S CAGR +5.4%



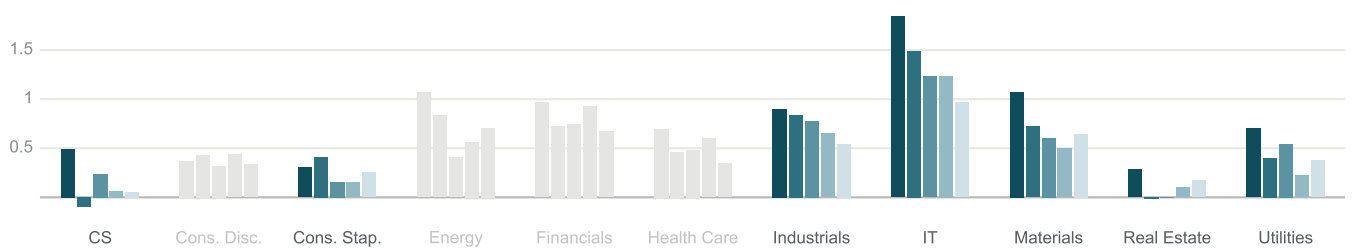
128 ● Adjusted Interest Coverage Ratio

Capital Efficiency L-S CAGR +5.3%



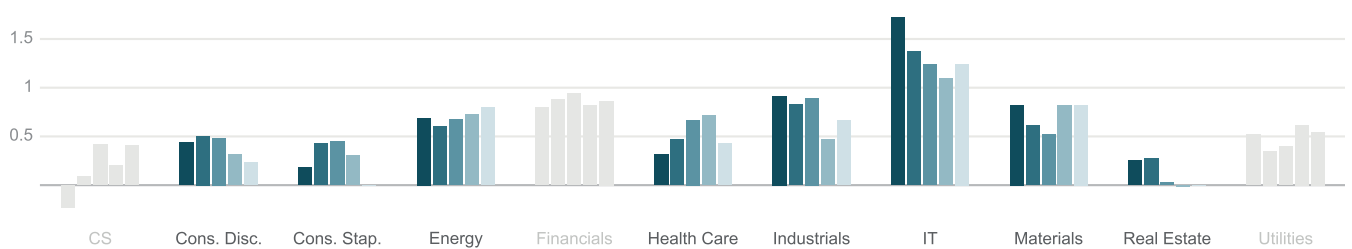
129 ● 15/36 Week Price Ratio

Price Momentum L-S CAGR +5.3%



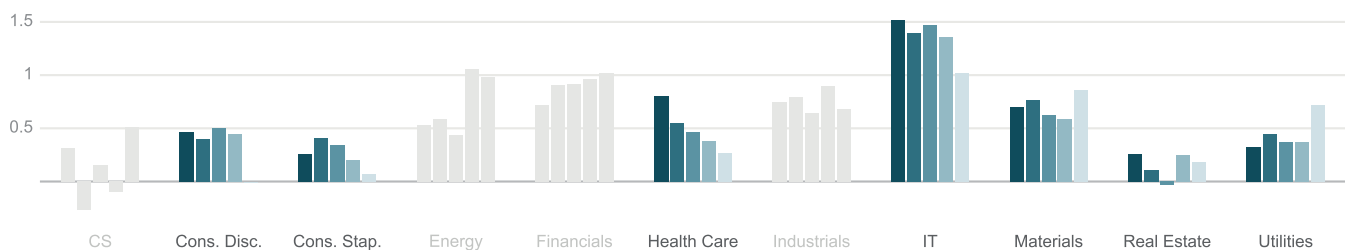
130 ● 1Y Chg in Free Cash Flow per Share

Historical Growth L-S CAGR +5.3%



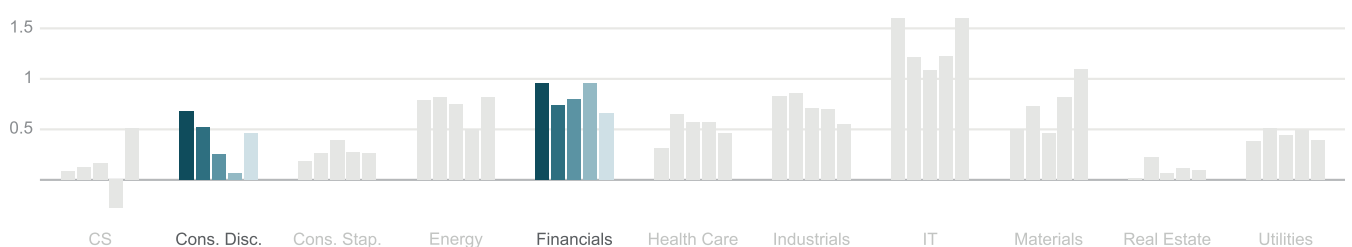
131 ● Interest Coverage Ratio

Capital Efficiency L-S CAGR +5.3%



132 ● 3Y Chg in Cash Flow per Share

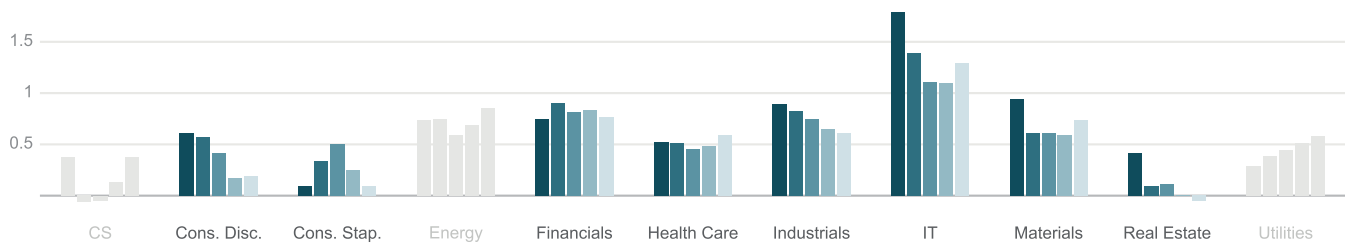
Historical Growth L-S CAGR +5.3%



133 ● 1Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

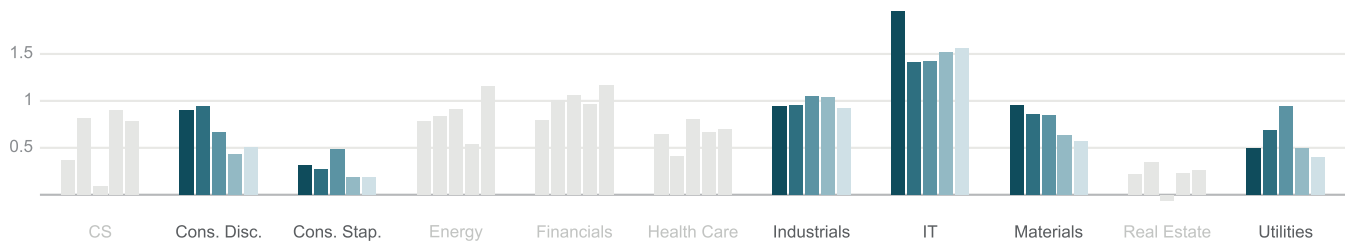
L-S CAGR +5.3%



134 ● Free Cash Flow Stability

Earnings Quality

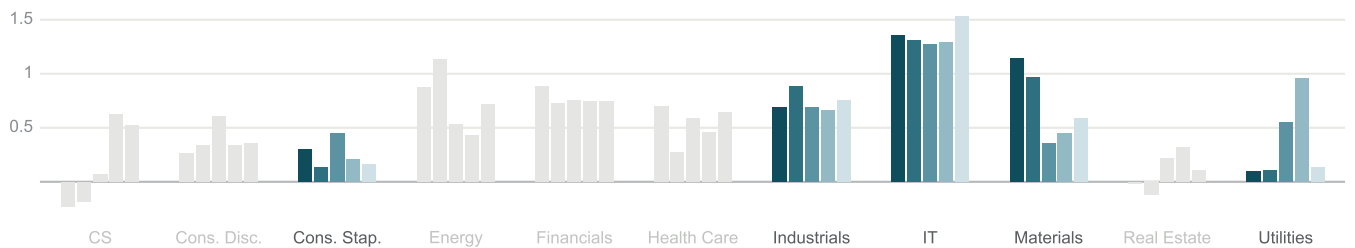
L-S CAGR +5.3%



135 ● SG&A to Sales

Earnings Quality

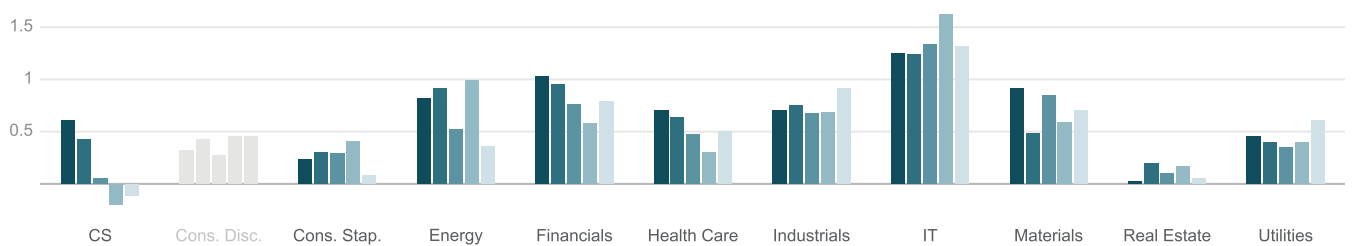
L-S CAGR +5.3%



136 ● Cash to EV Ratio

Valuation

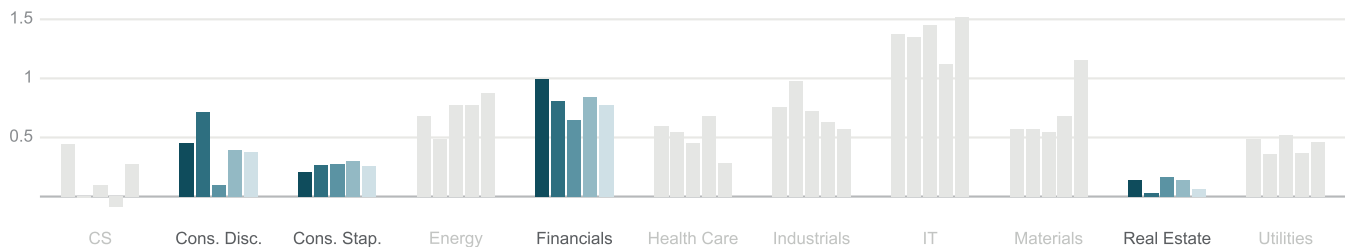
L-S CAGR +5.3%



137 ● 3Y Chg in Earnings to Price

Historical Growth

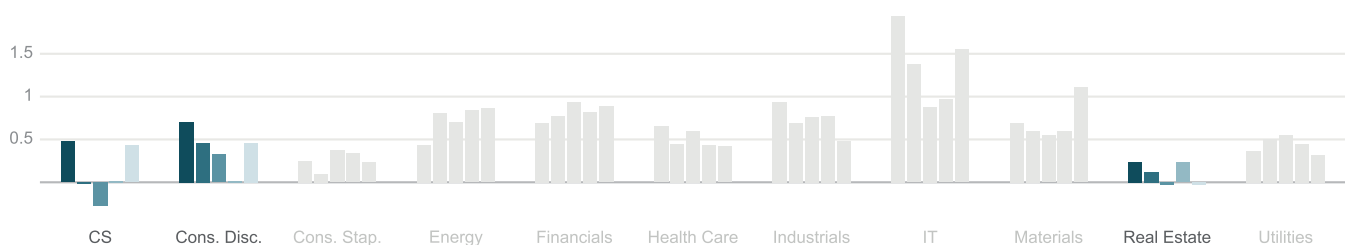
L-S CAGR +5.2%



138 ● 3Y Chg in Asset Adjusted Cash Flow

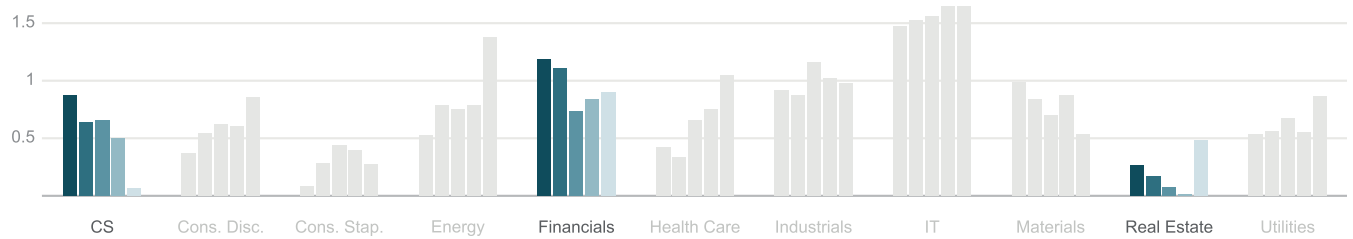
Historical Growth

L-S CAGR +5.2%

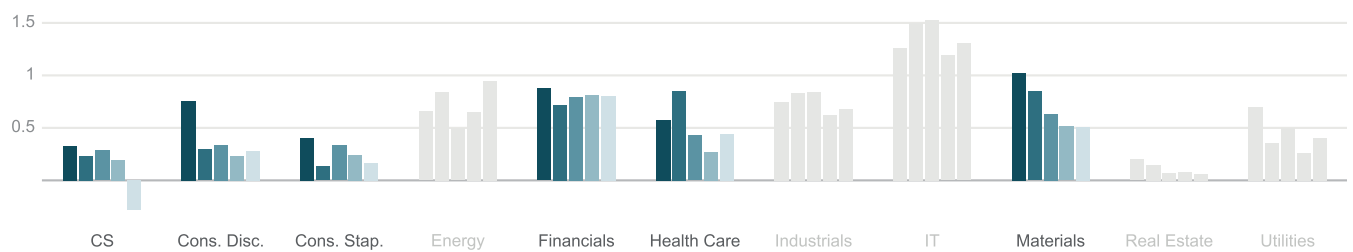


139 ● **Adjusted Book to Enterprise Value**

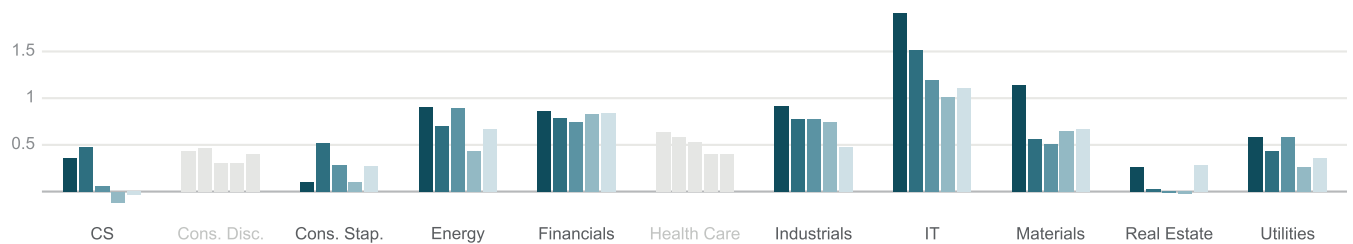
Valuation L-S CAGR +5.2%

140 ● **Working Capital Accruals**

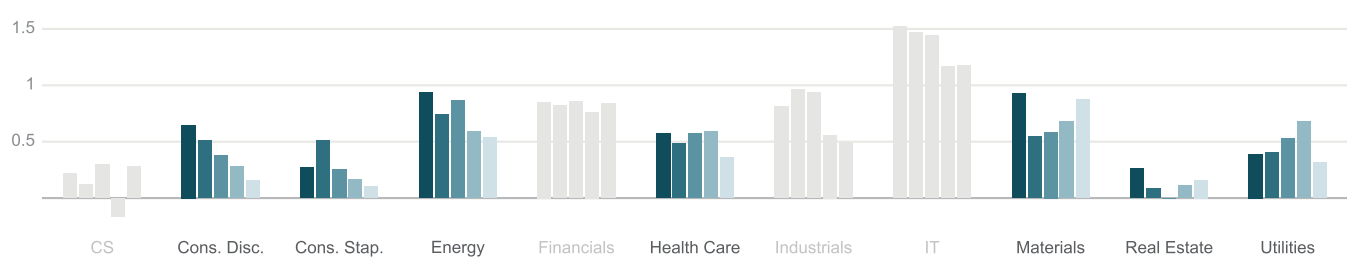
Earnings Quality L-S CAGR +5.1%

141 ● **9 Month Price Momentum**

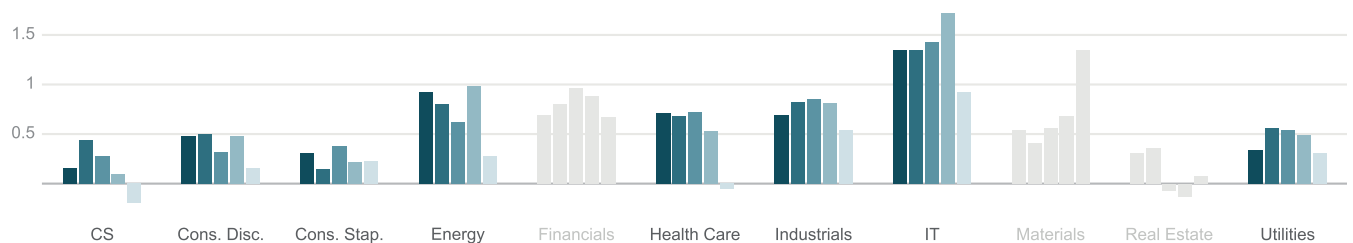
Price Momentum L-S CAGR +5.1%

142 ● **Free Cash Flow to Sales**

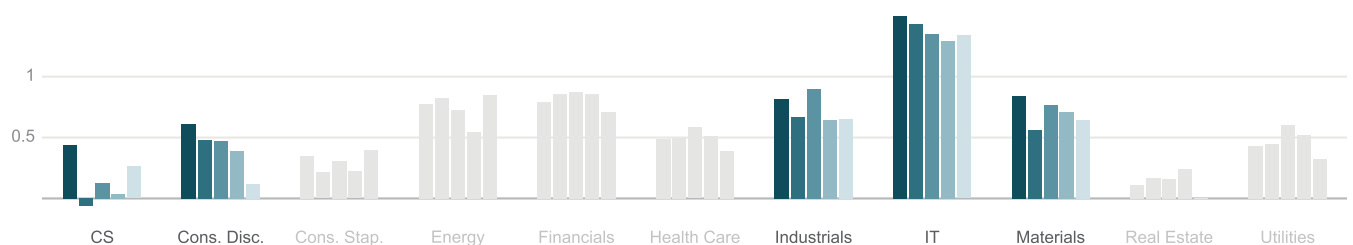
Earnings Quality L-S CAGR +5.1%

143 ● **Asset Turnover**

Earnings Quality L-S CAGR +5.1%

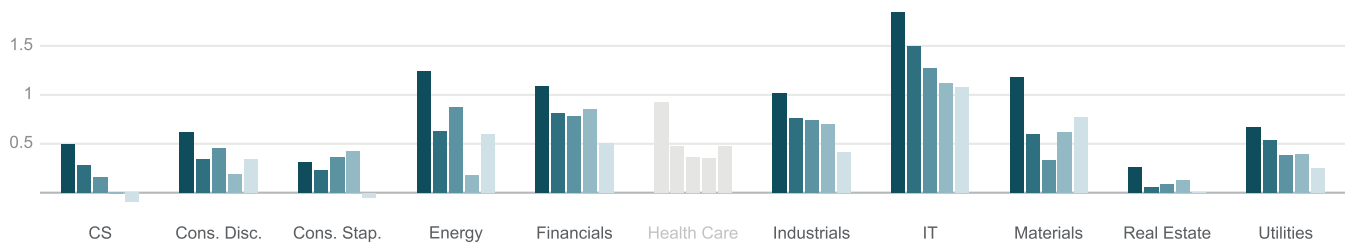
144 ● **EPS Stability**

Historical Growth L-S CAGR +5.0%



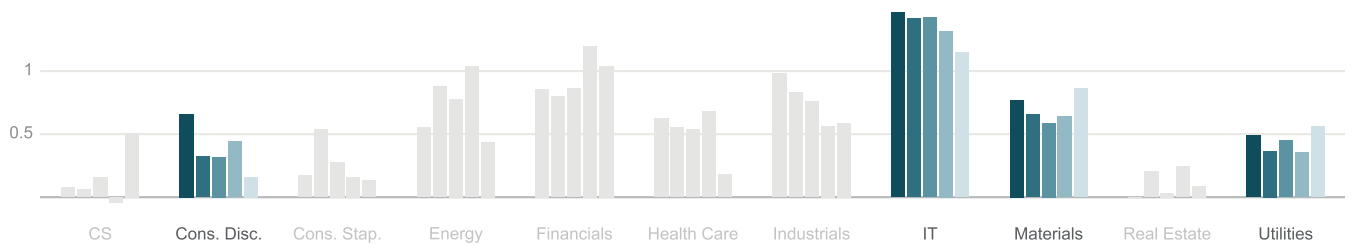
145 ● 6M Chg in Target Price

Analyst Expectations L-S CAGR +4.9%



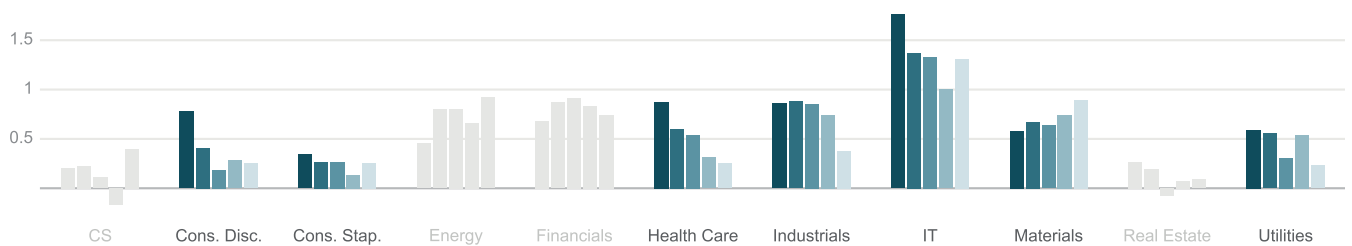
146 ● Cash Burn Rate

Earnings Quality L-S CAGR +4.9%



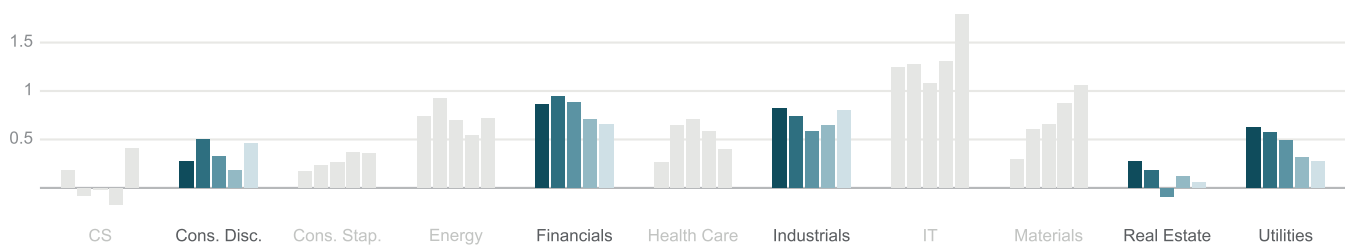
147 ● Cash Flow to Invested Capital

Earnings Quality L-S CAGR +4.9%



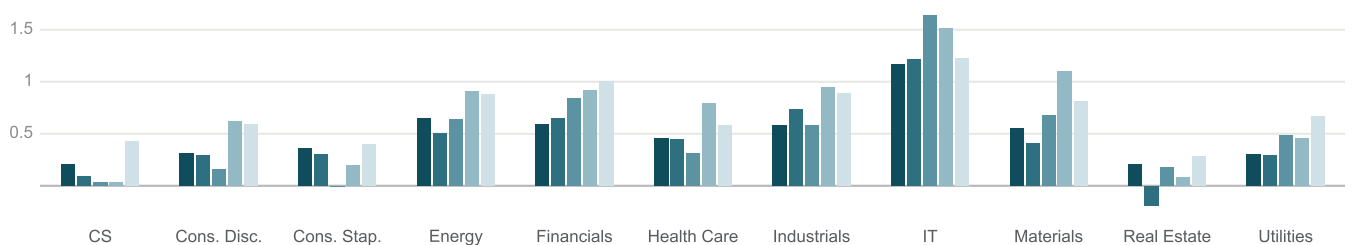
148 ● Dividend per Share Growth

Historical Growth L-S CAGR +4.8%



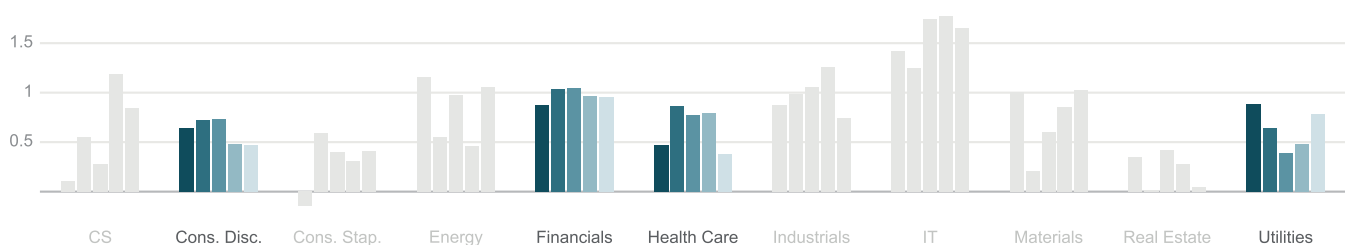
149 ● 1Y Chg in Sales Turnover

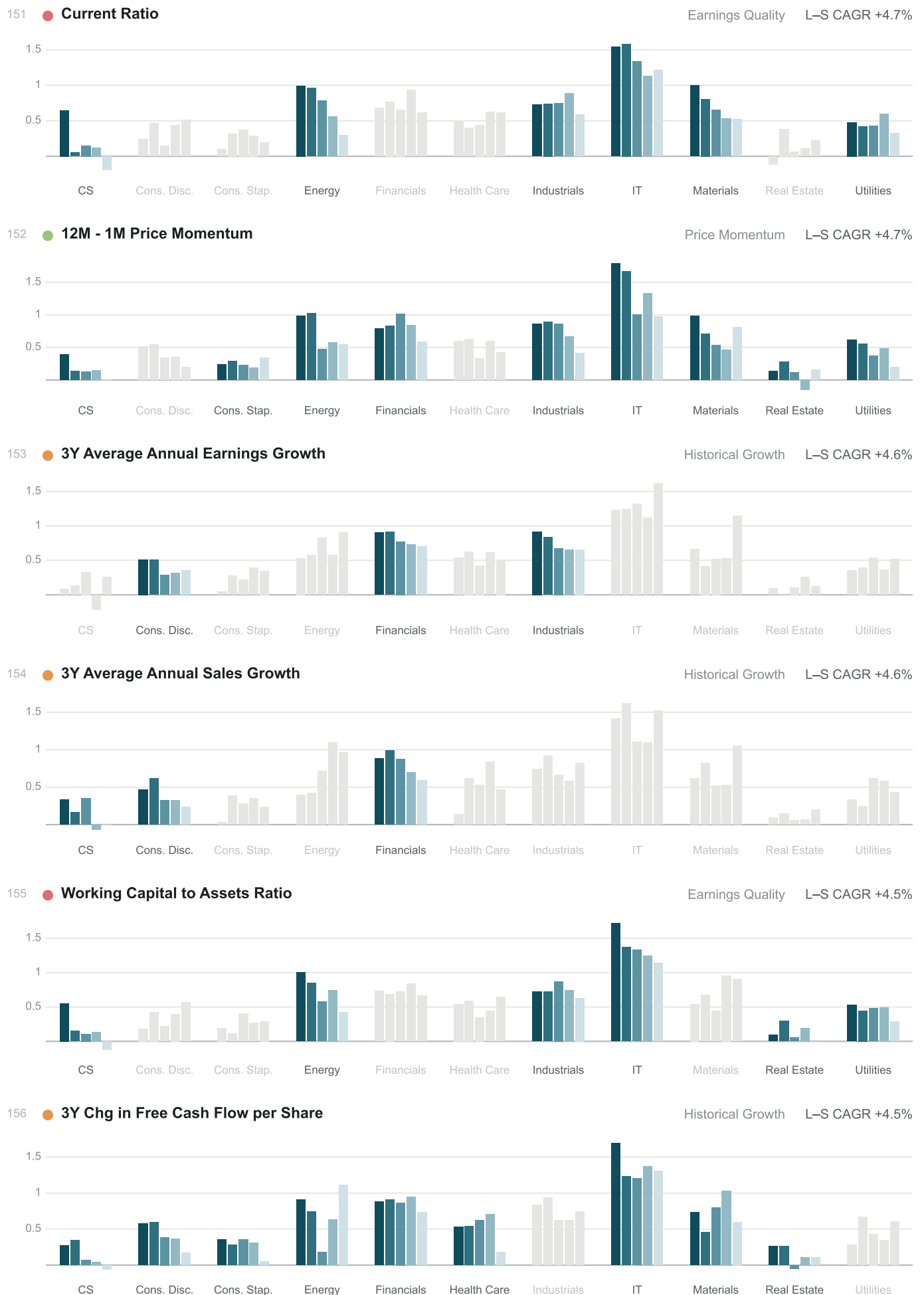
Historical Growth L-S CAGR +4.8%



150 ● Adj Forward Free Cash Flow to Price Ratio

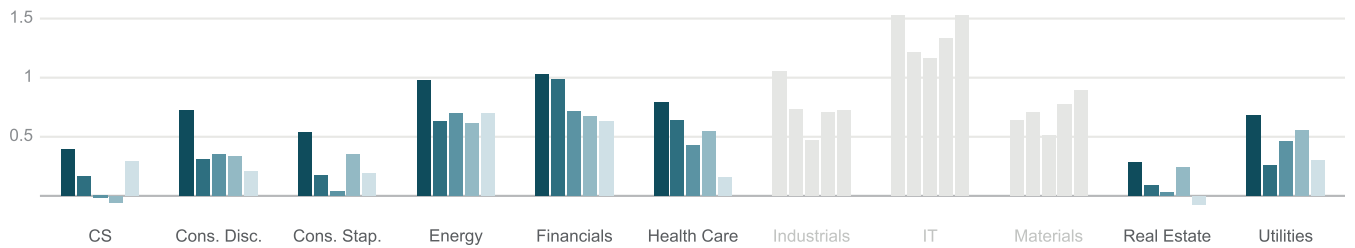
Valuation L-S CAGR +4.8%





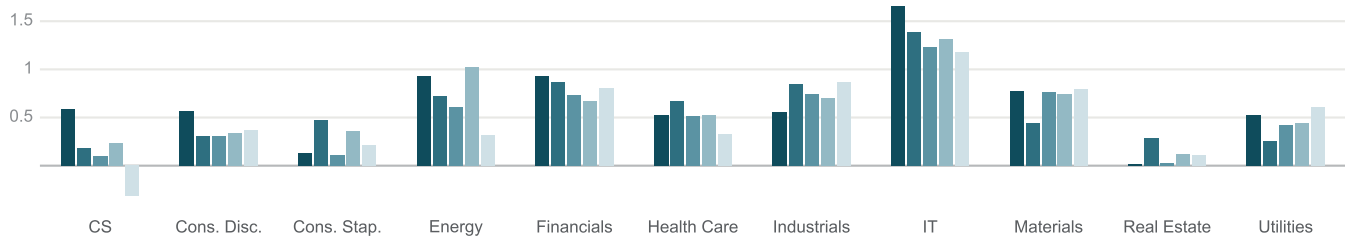
157 ● 1Y Chg in Shares Outstanding

Capital Efficiency L-S CAGR +4.5%



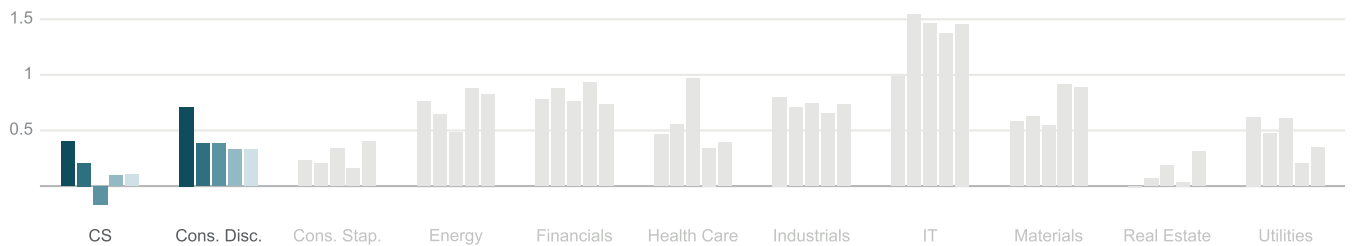
158 ● Cash to Total Assets

Valuation L-S CAGR +4.5%



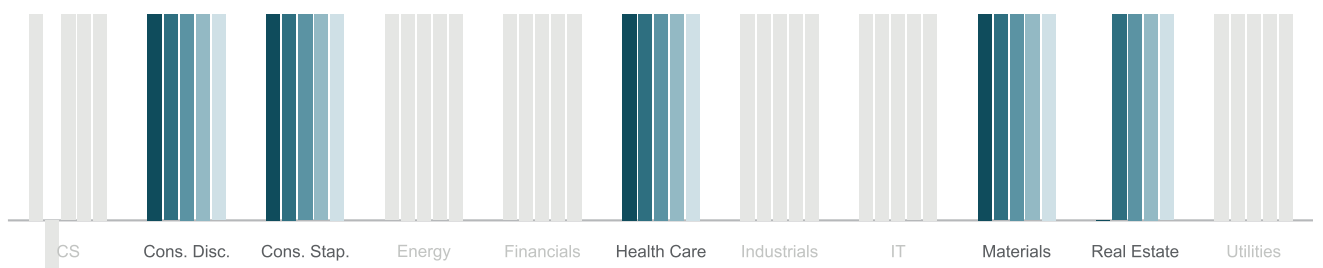
159 ● Net Income Stability

Earnings Quality L-S CAGR +4.4%



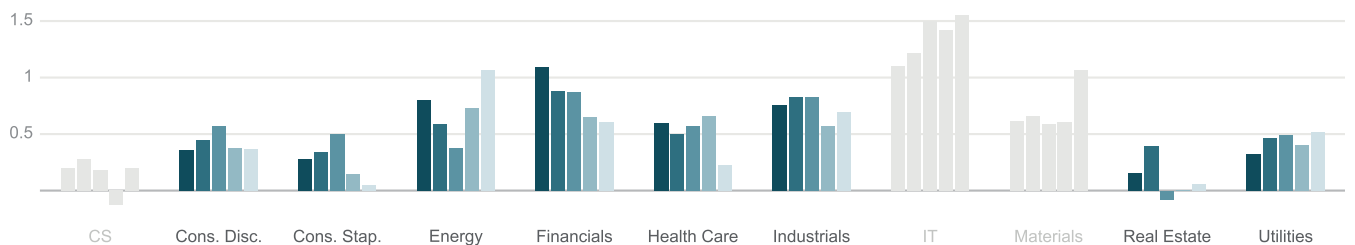
160 ● Inventory to Assets Ratio

Earnings Quality L-S CAGR +4.4%



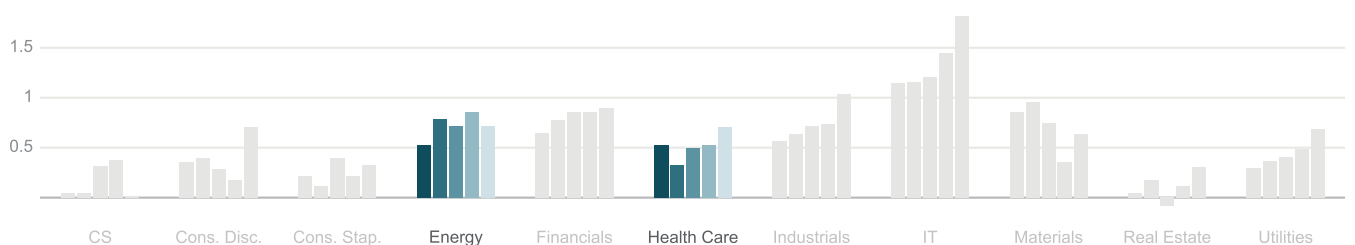
161 ● 3Y Chg in Sales to Price

Historical Growth L-S CAGR +4.4%



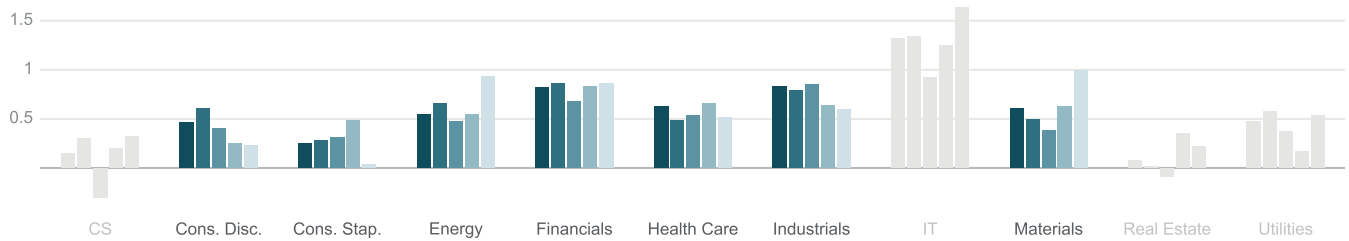
162 ● Sharpe Ratio

Price Momentum L-S CAGR +4.3%



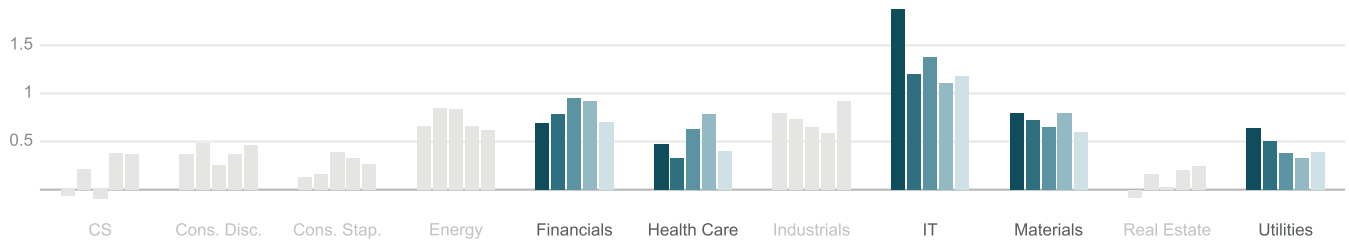
163 ● 1Y Chg in Earnings per Share

Historical Growth L-S CAGR +4.3%



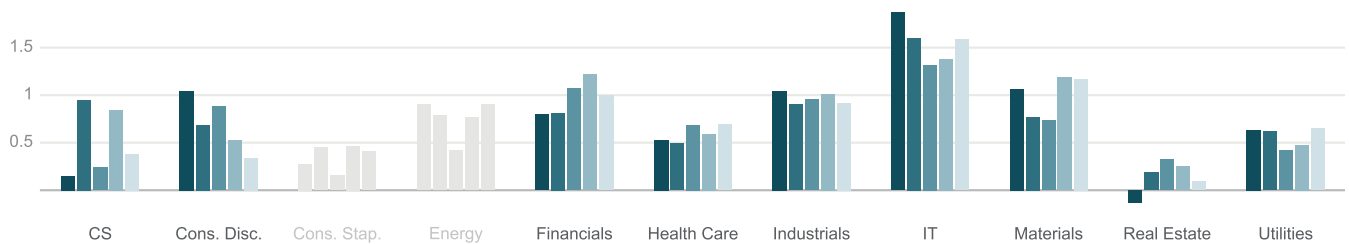
164 ● Change in Number of Holders

Price Momentum L-S CAGR +4.2%



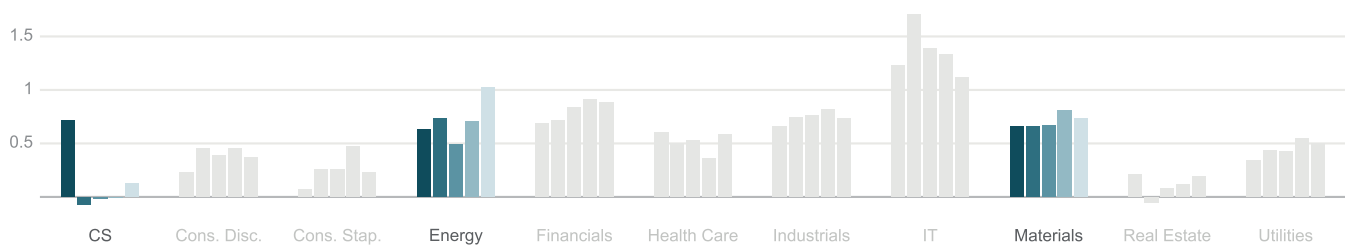
165 ● Return on Equity Stability

Historical Growth L-S CAGR +4.2%



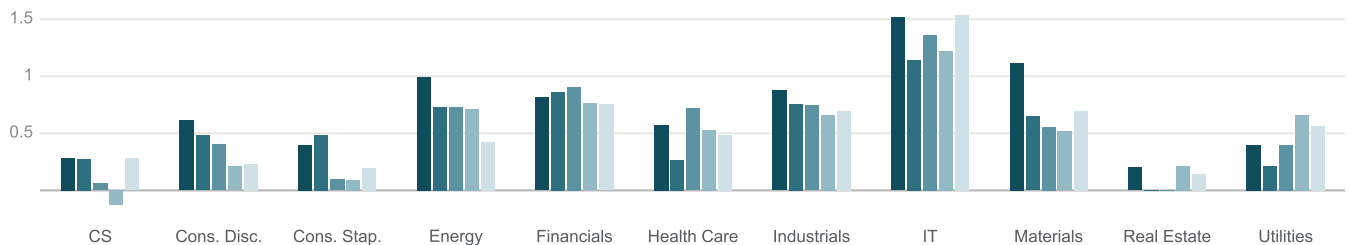
166 ● Net Current Assets to Price Ratio

Valuation L-S CAGR +4.2%



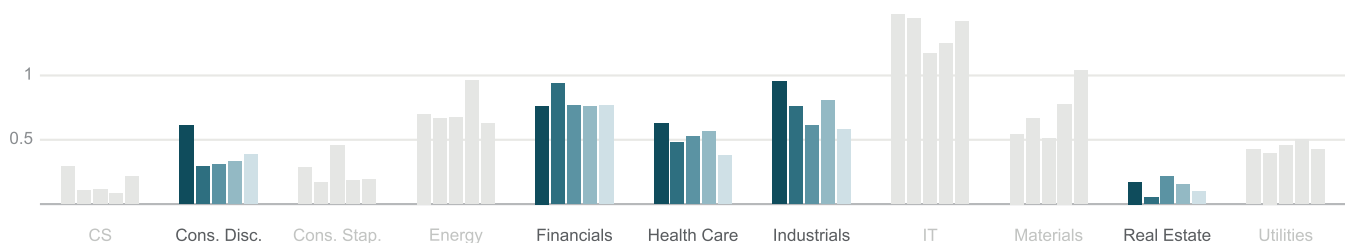
167 ● Accrual Ratio - Cash Flows

Earnings Quality L-S CAGR +4.2%



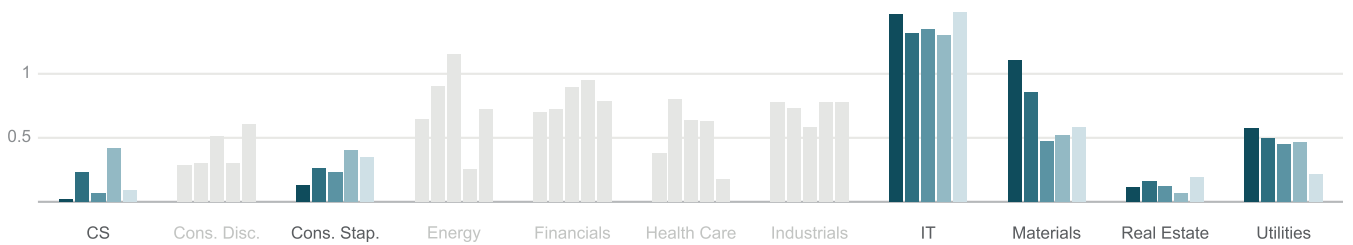
168 ● ROA 60 Month Slope

Earnings Quality L-S CAGR +4.2%

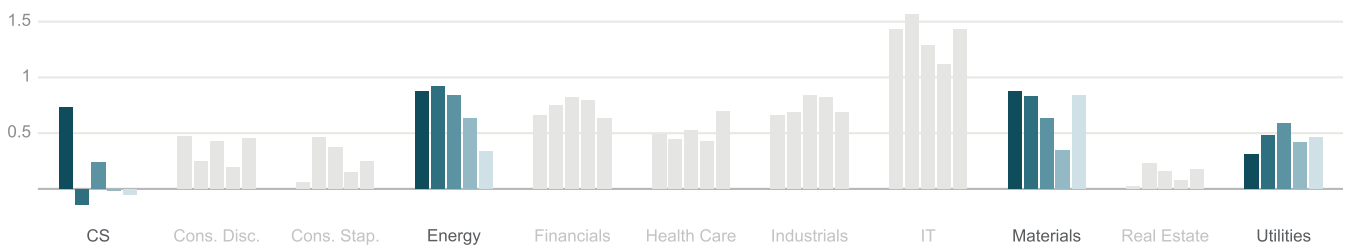


169 ● **3M Momentum in Trailing 12M Sales**

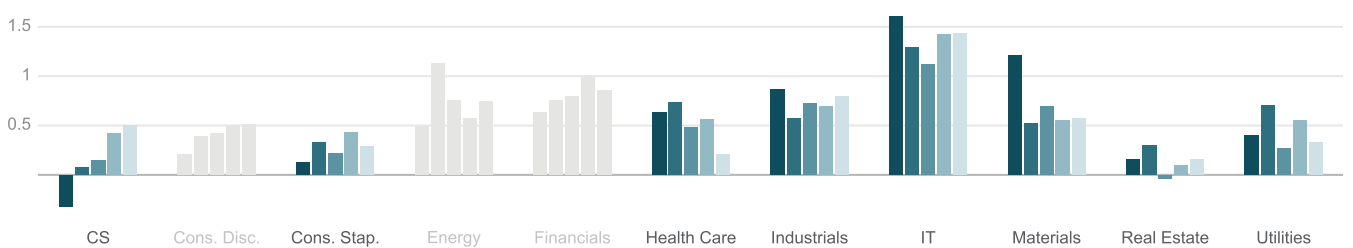
Historical Growth L-S CAGR +4.1%

170 ● **Quick Ratio**

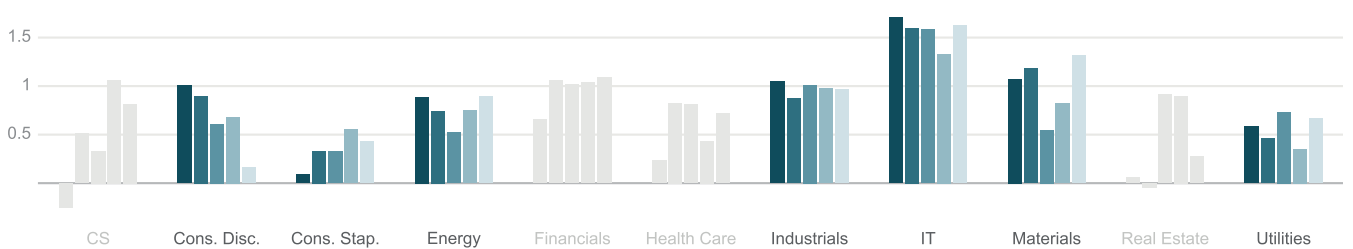
Earnings Quality L-S CAGR +4.1%

171 ● **6M Momentum in Trailing 12M Sales**

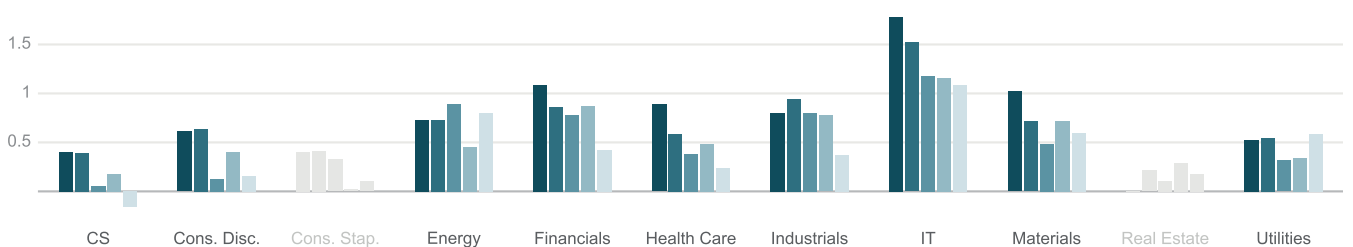
Historical Growth L-S CAGR +4.1%

172 ● **Return on Assets Stability**

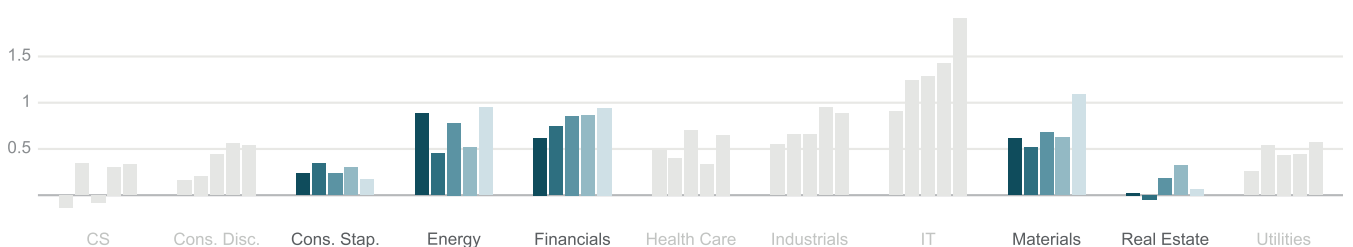
Historical Growth L-S CAGR +4.0%

173 ● **Number of EPS FY1 Revisions**

Analyst Expectations L-S CAGR +4.0%

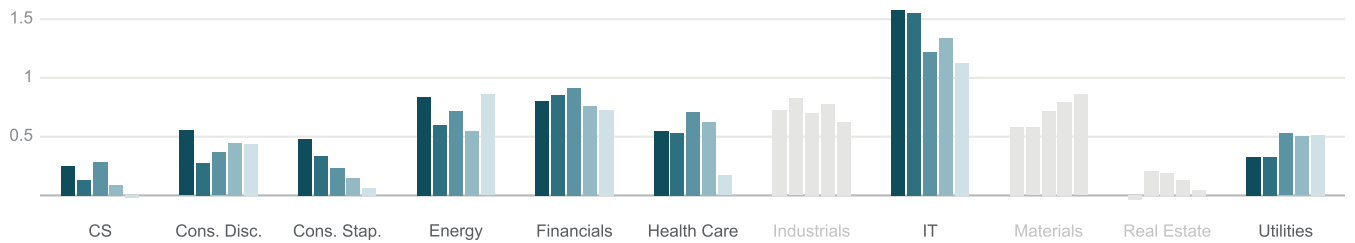
174 ● **Share Turnover**

Price Momentum L-S CAGR +4.0%



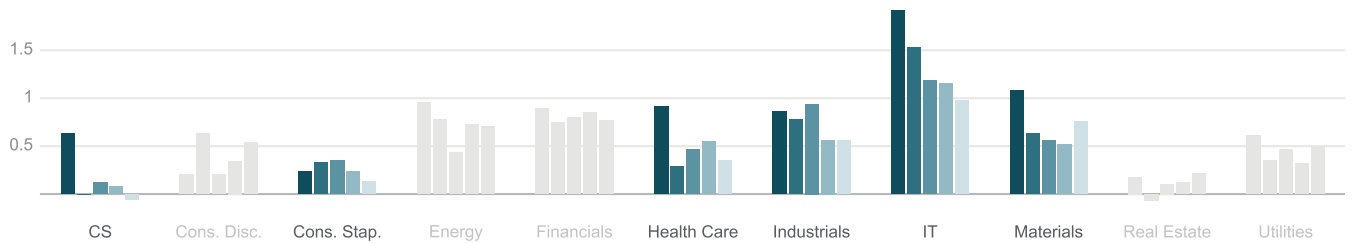
175 ● 3Y Chg in Sales Turnover

Historical Growth L-S CAGR +4.0%



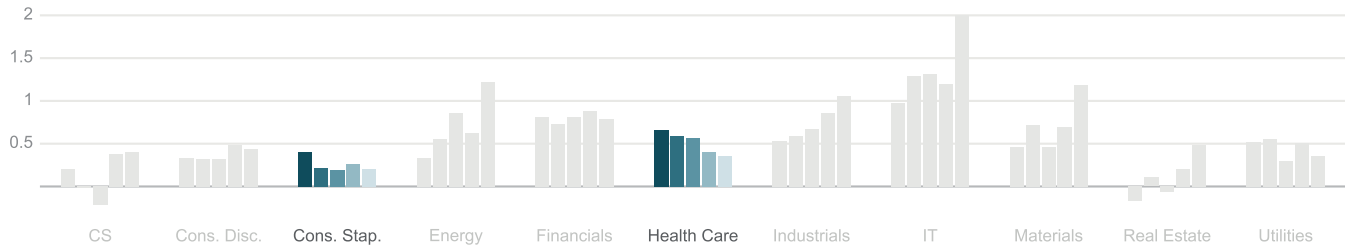
176 ● 26W Relative Price Strength

Price Momentum L-S CAGR +4.0%



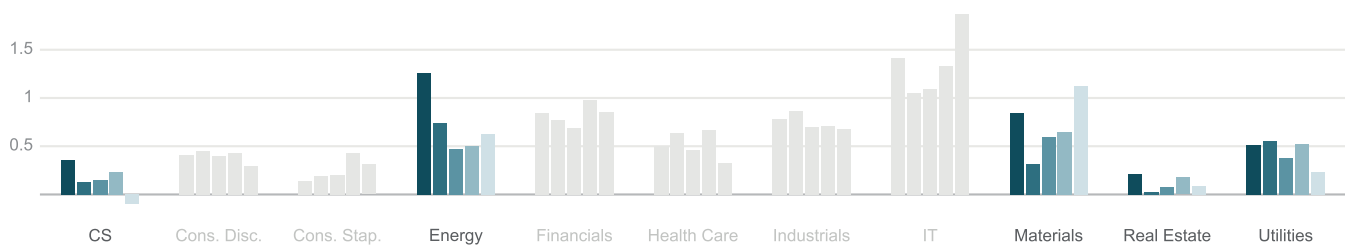
177 ● 60 Day Coefficient of Variation of Volume to Price

Price Momentum L-S CAGR +4.0%



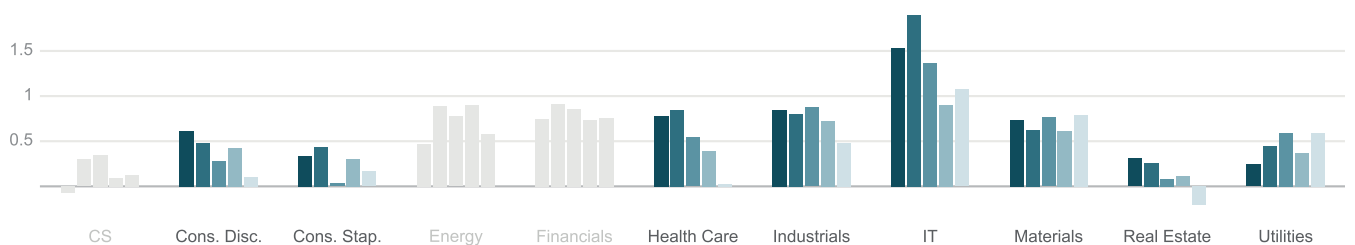
178 ● Sales to Gross Profit Margin Growth

Earnings Quality L-S CAGR +3.9%



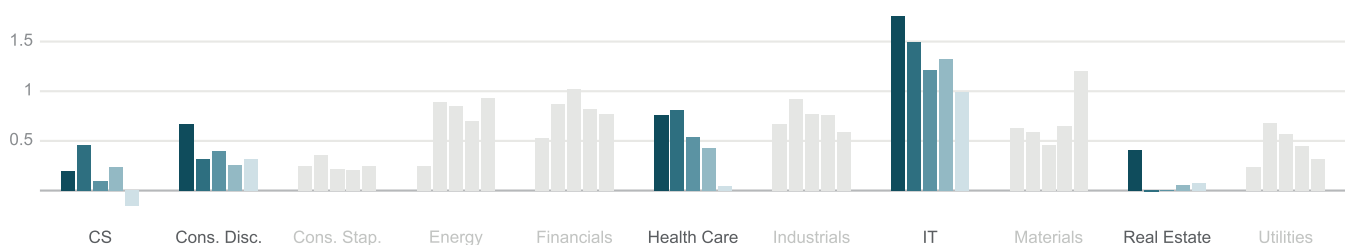
179 ● Cash Flow Return on Invested Capital

Capital Efficiency L-S CAGR +3.9%



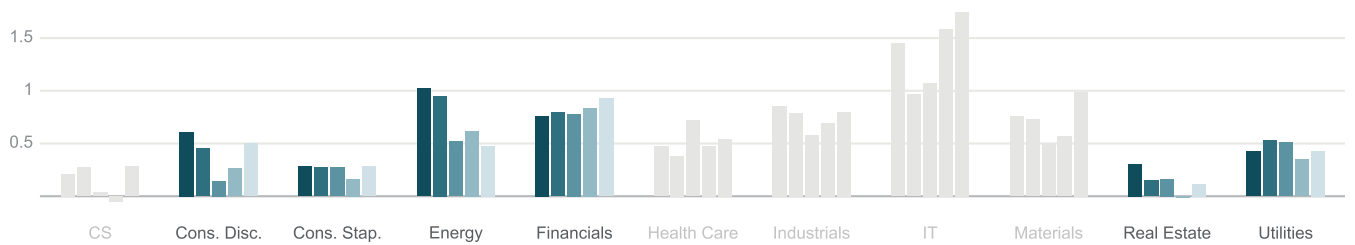
180 ● Gross Profit to Assets

Capital Efficiency L-S CAGR +3.9%

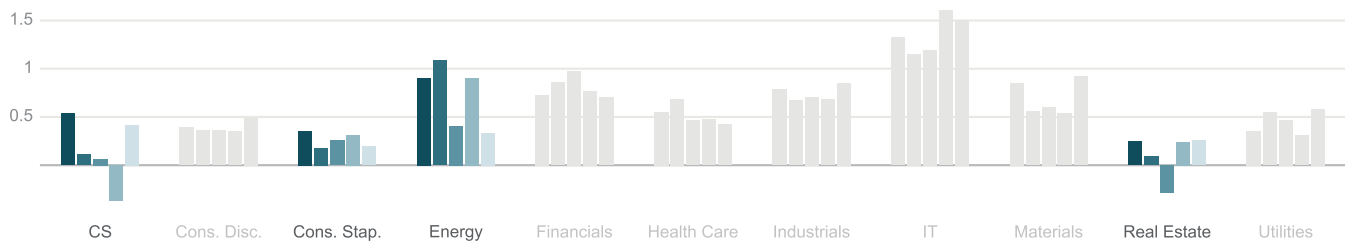


181 ● **Asset Adjusted Capital Investments**

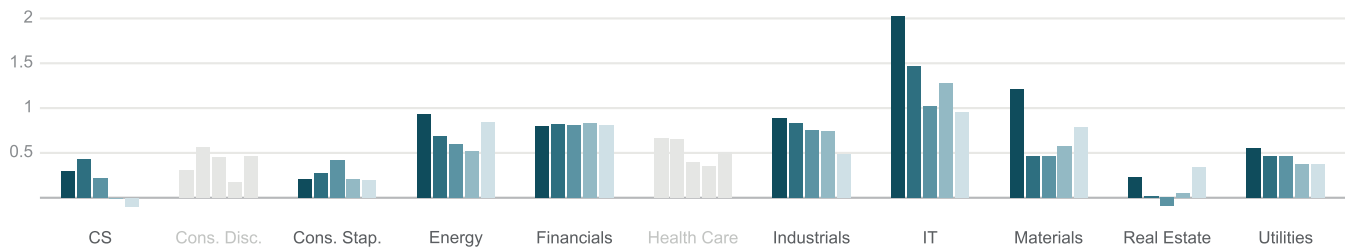
Earnings Quality L-S CAGR +3.9%

182 ● **Retained Earnings to Total Assets**

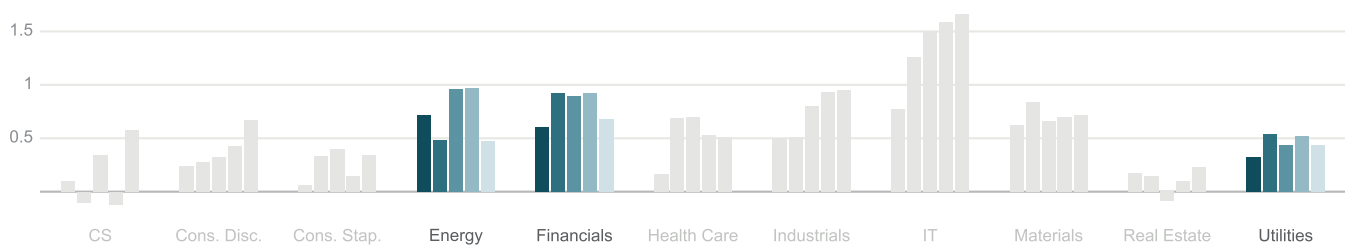
Earnings Quality L-S CAGR +3.9%

183 ● **4W to 52W Price Oscillator**

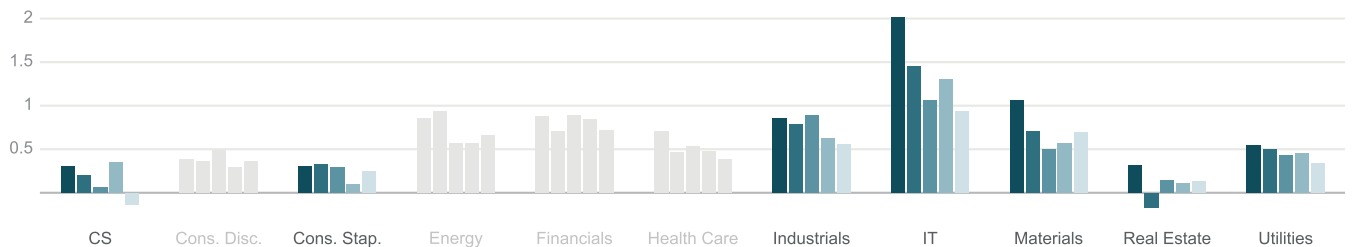
Price Momentum L-S CAGR +3.9%

184 ● **Log of Market Capitalization Cubed**

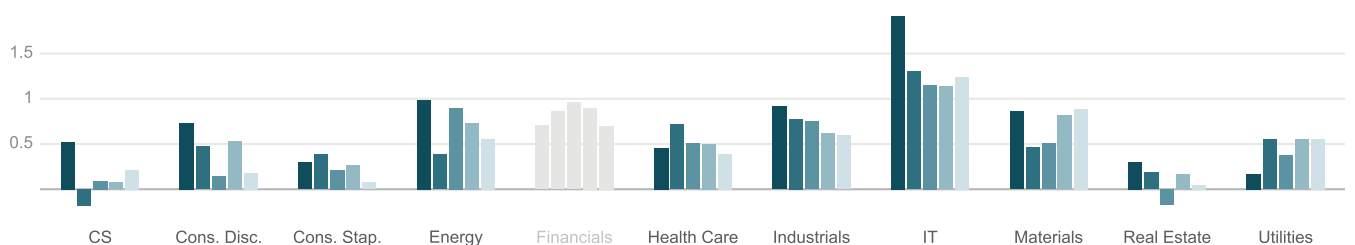
Size L-S CAGR +3.9%

185 ● **50 Day to 200 Day Stock Price**

Price Momentum L-S CAGR +3.9%

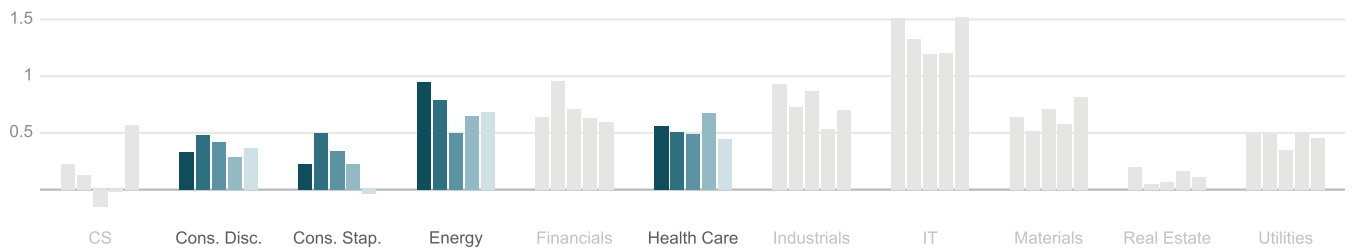
186 ● **3Y Chg in Asset Adjusted Free Cash Flow**

Historical Growth L-S CAGR +3.8%



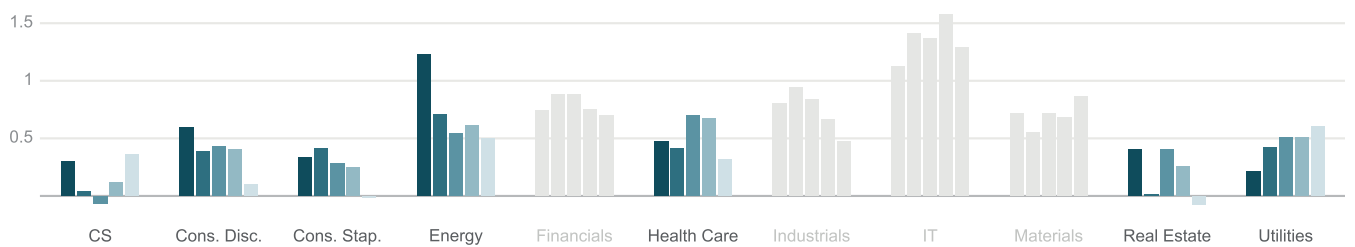
187 ● Degree of Financial Leverage

Capital Efficiency L-S CAGR +3.7%



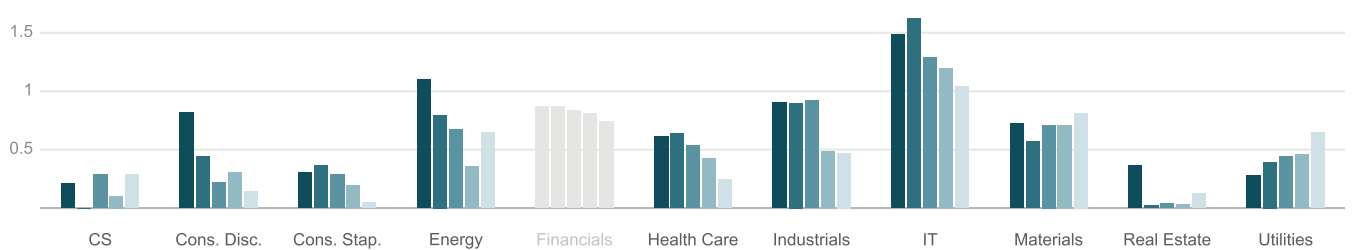
188 ● Capital Acquisition Ratio

Capital Efficiency L-S CAGR +3.7%



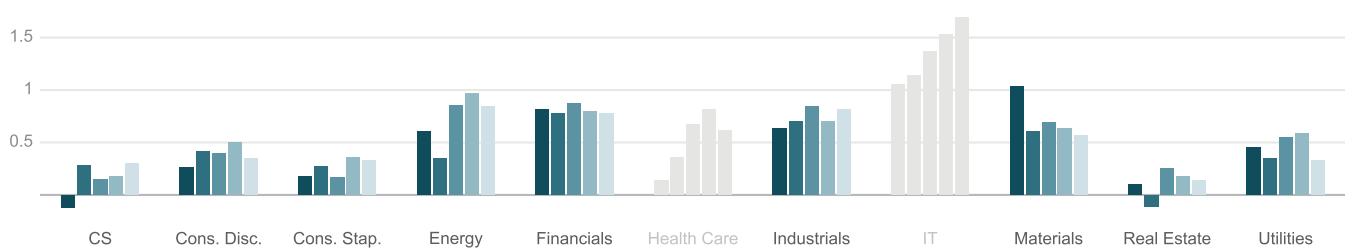
189 ● Free Cash Flow to Common Equity

Valuation L-S CAGR +3.7%



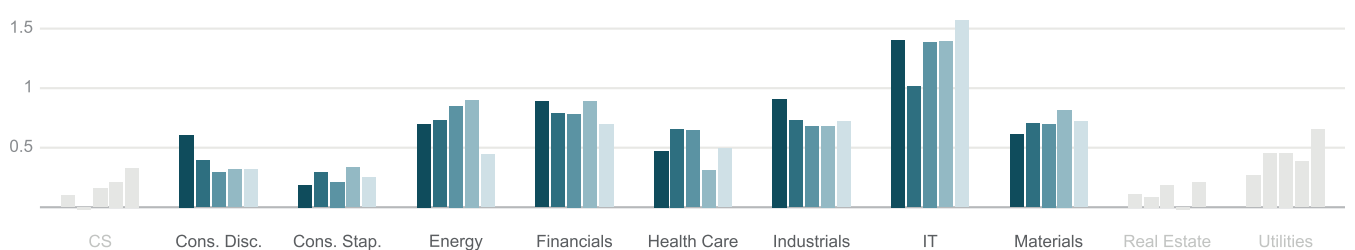
190 ● Foreign Institution Ownership

Price Momentum L-S CAGR +3.6%



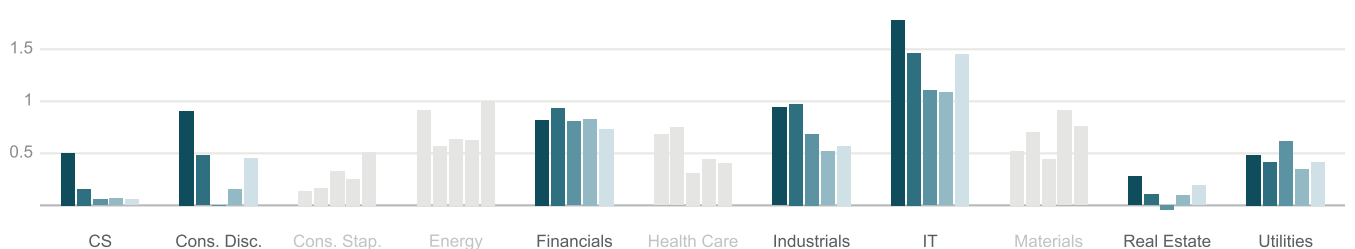
191 ● 1Y Chg in Debt to Assets

Volatility L-S CAGR +3.6%



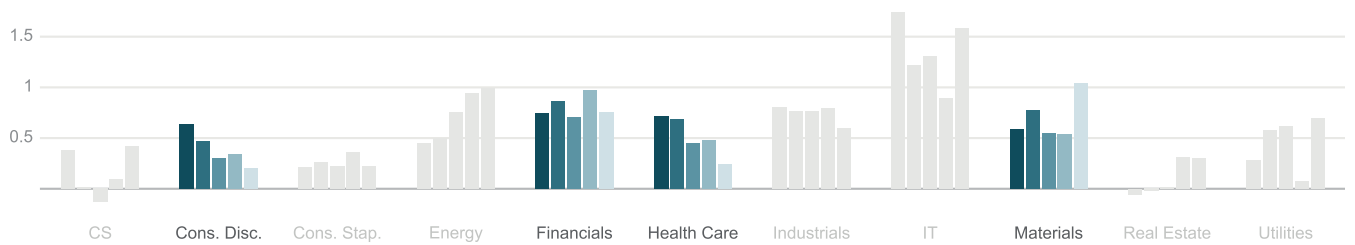
192 ● Sustainable Growth Rate

Historical Growth L-S CAGR +3.6%

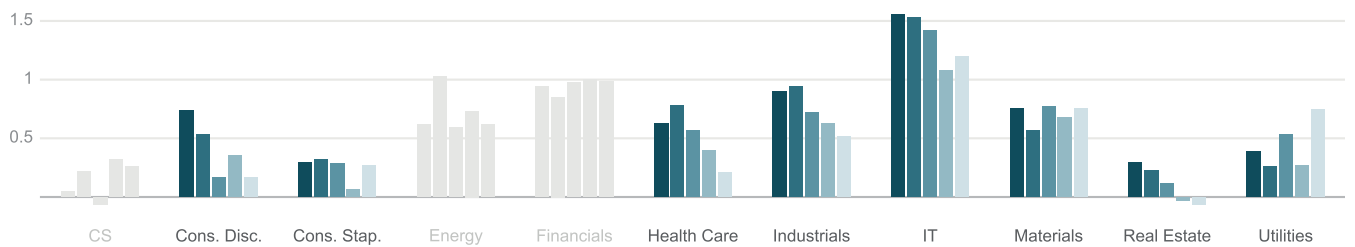


193 ● **Net Profit Margin**

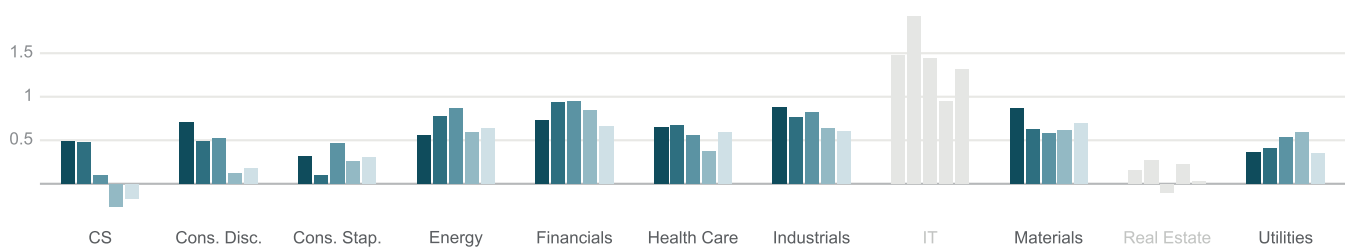
Earnings Quality L-S CAGR +3.5%

194 ● **Operating Cash Flow to Equity**

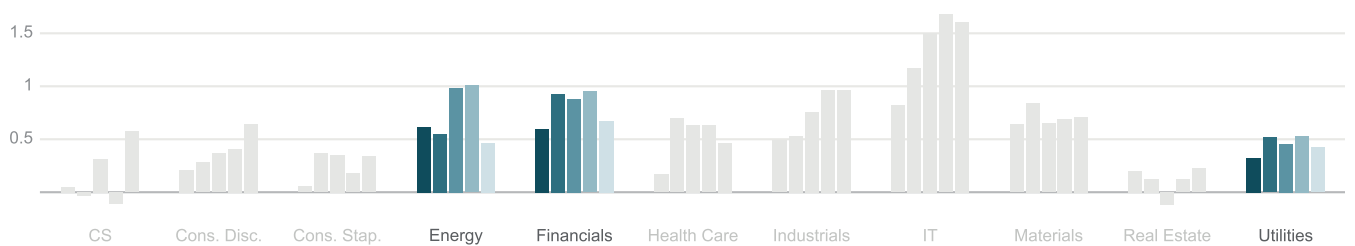
Capital Efficiency L-S CAGR +3.5%

195 ● **Payout Ratio**

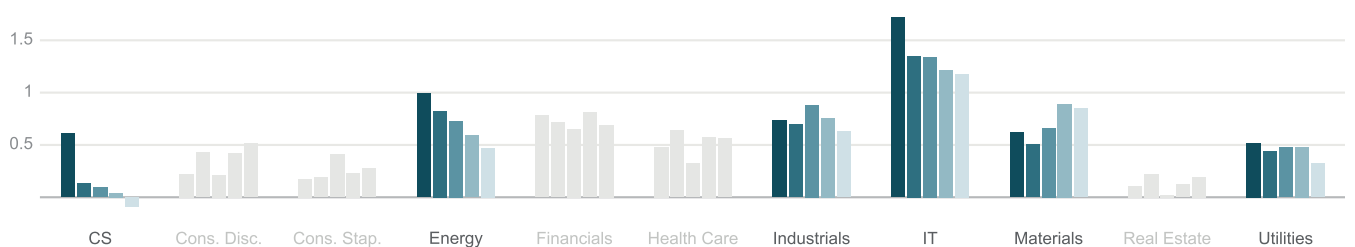
Capital Efficiency L-S CAGR +3.5%

196 ● **Log Market Cap**

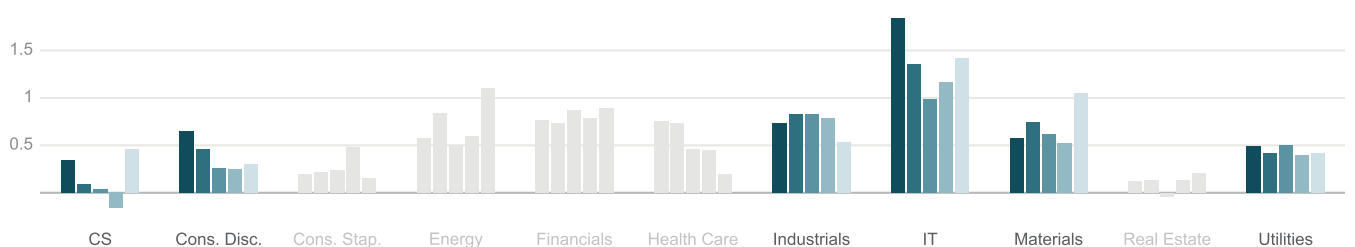
Size L-S CAGR +3.5%

197 ● **Working Capital to Total Assets**

Earnings Quality L-S CAGR +3.5%

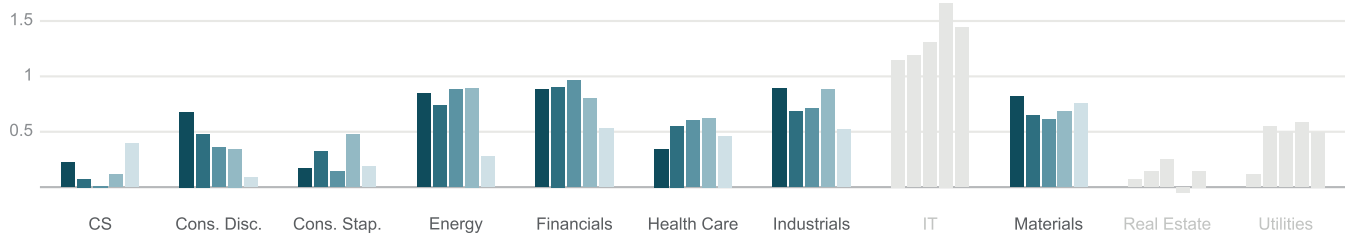
198 ● **Pretax Income to Net Operating Assets**

Earnings Quality L-S CAGR +3.4%



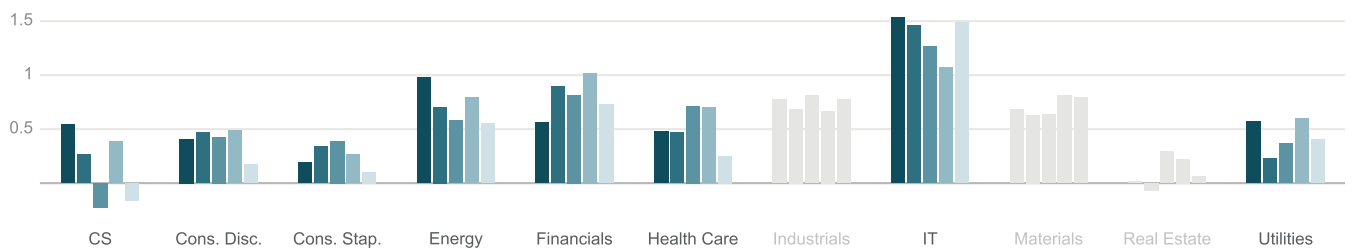
199 ● Risk Adjusted Relative Strength

Price Momentum L-S CAGR +3.4%



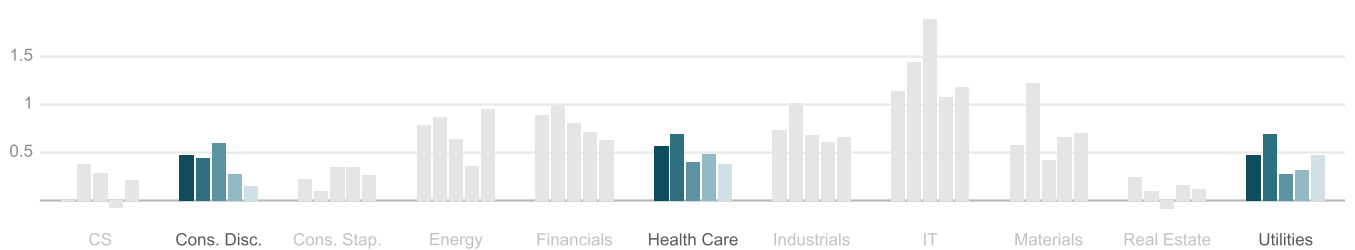
200 ● Asset Growth - 1 Year

Capital Efficiency L-S CAGR +3.4%



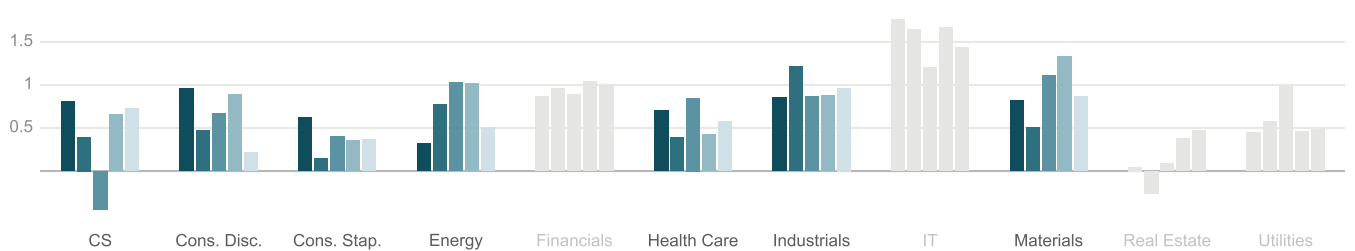
201 ● Institution Ownership Level

Price Momentum L-S CAGR +3.3%



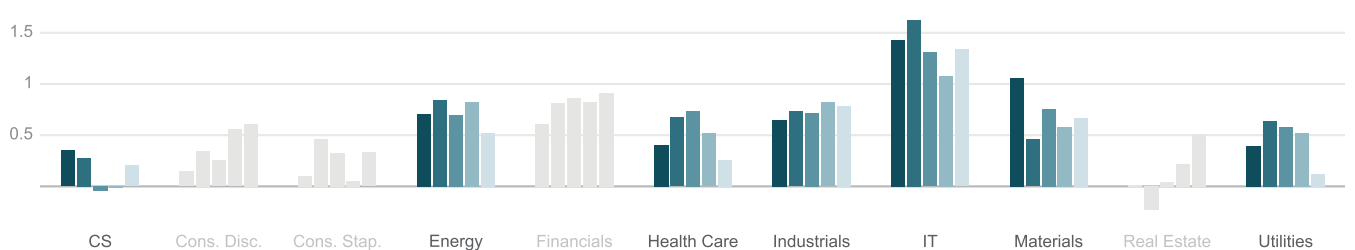
202 ● Num of Quarters of Declines in Receivables and Inventories/TTM Sales

Historical Growth L-S CAGR +3.3%



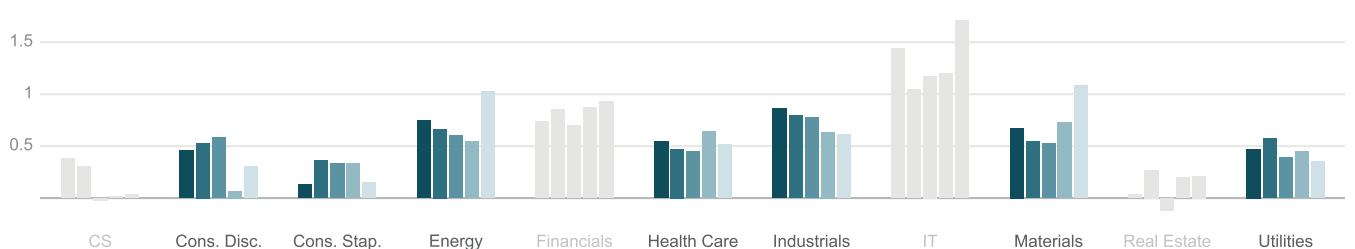
203 ● 10 Day MACD Trend

Price Momentum L-S CAGR +3.3%



204 ● 1Y Chg in Cash Flow per Share

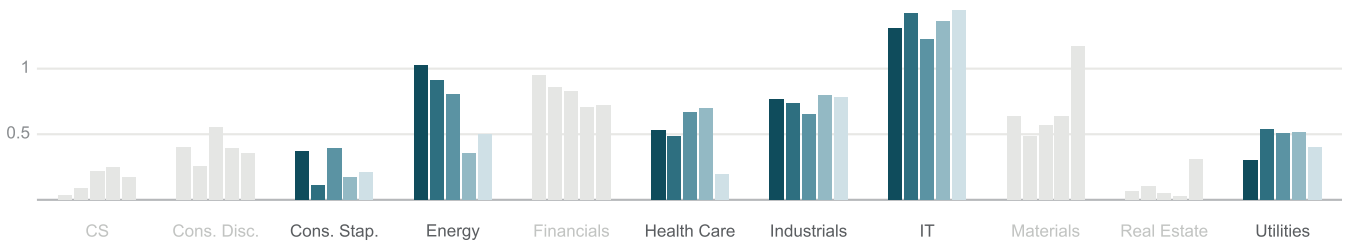
Historical Growth L-S CAGR +3.3%



205 ● **Trailing 12-Month Overhead/Trailing 12-Month Sales**

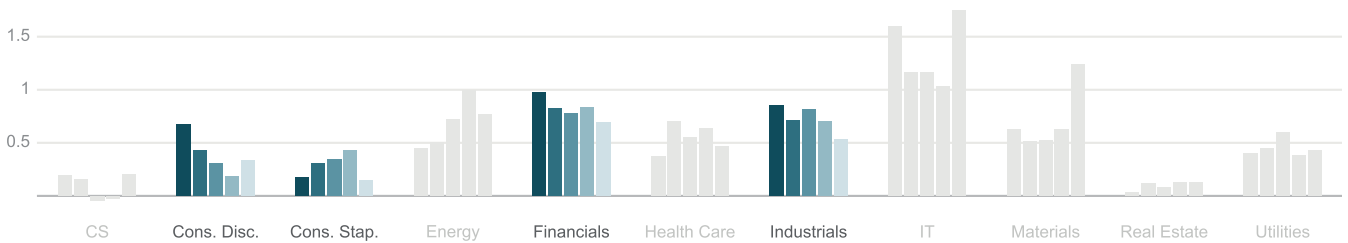
Historical Growth

L-S CAGR +3.3%

206 ● **3Y Chg in Earnings per Share**

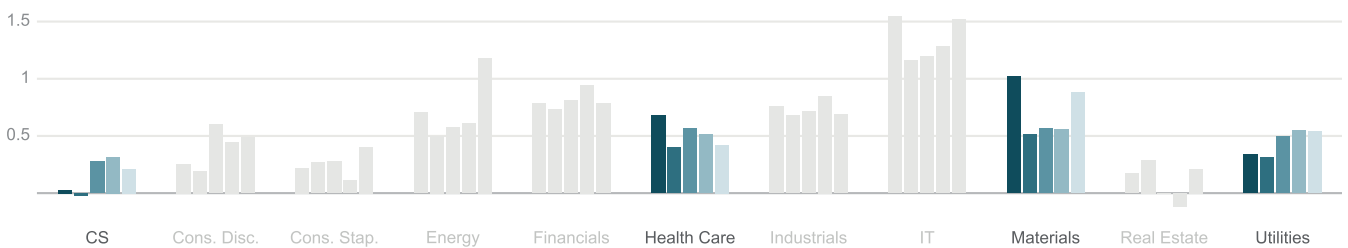
Historical Growth

L-S CAGR +3.2%

207 ● **1Y Chg in Sales to Earnings**

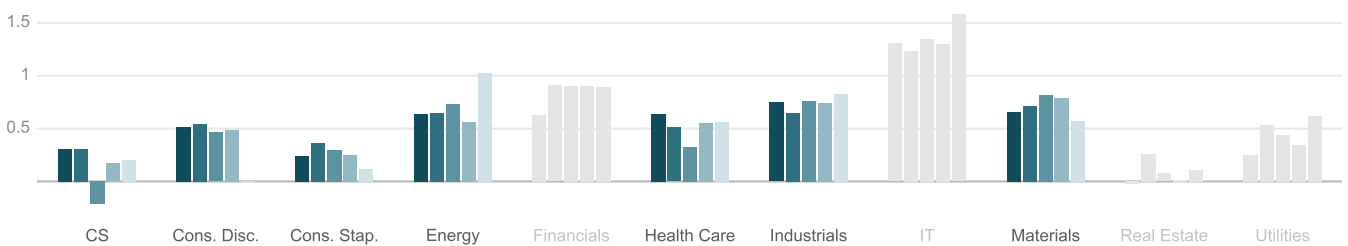
Earnings Quality

L-S CAGR +3.2%

208 ● **Unexpected Receivables Change**

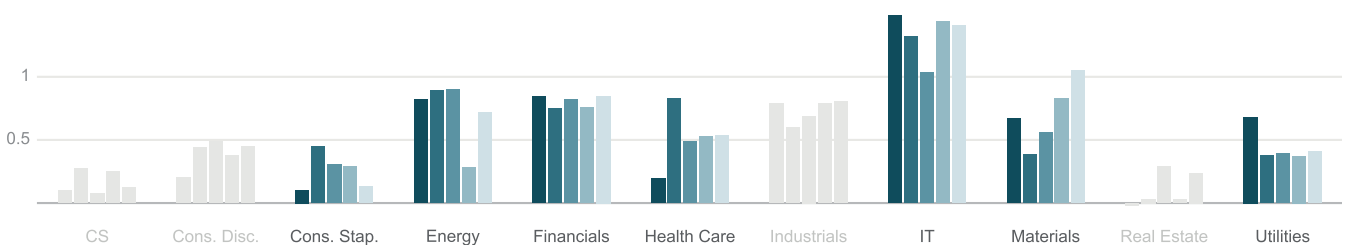
Earnings Quality

L-S CAGR +3.2%

209 ● **1Y Chg in EPS to Operating Cash Flow**

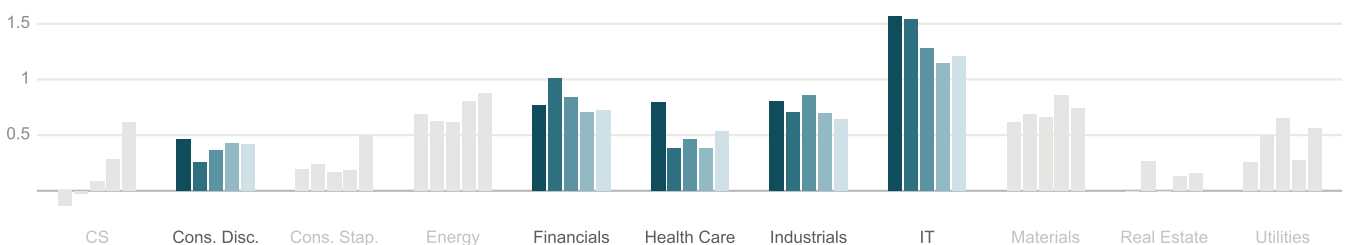
Earnings Quality

L-S CAGR +3.2%

210 ● **6M Chg in 18M CAPM Alpha**

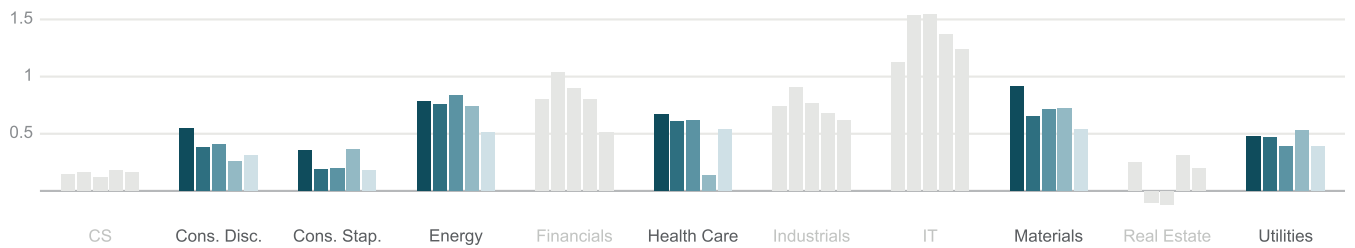
Price Momentum

L-S CAGR +3.2%

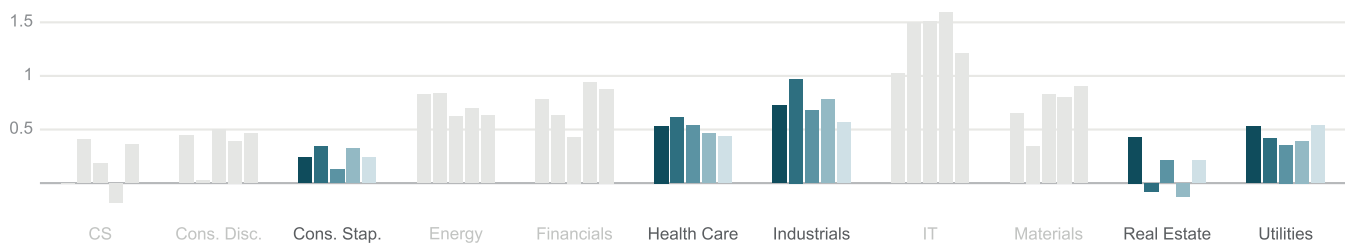


211 ● **Institution Ownership Concentration**

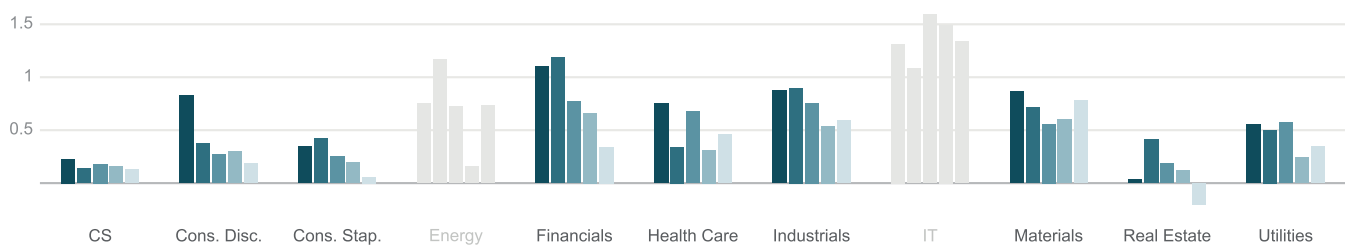
Price Momentum L-S CAGR +3.1%

212 ● **Working Capital Turnover Ratio**

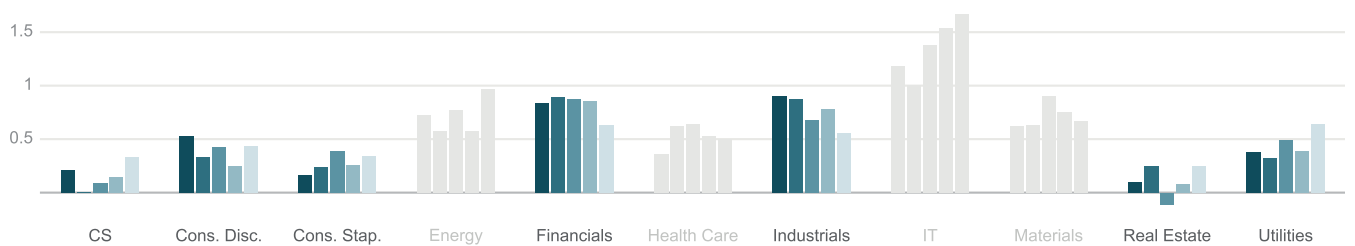
Capital Efficiency L-S CAGR +3.1%

213 ● **Target Price Gap to 6M EMA**

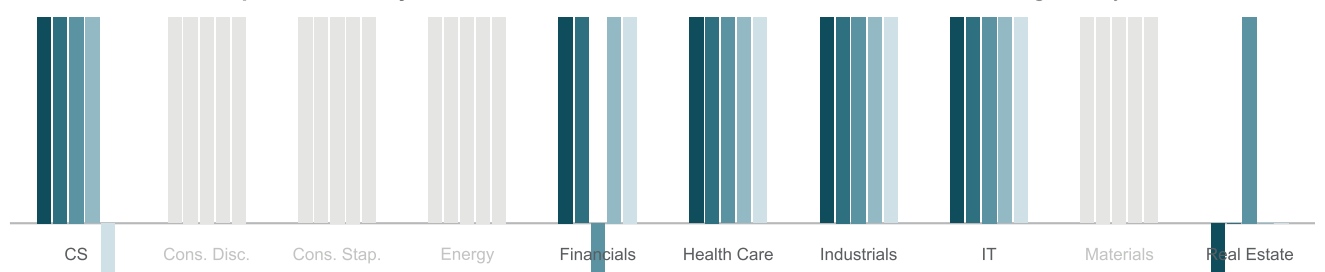
Analyst Expectations L-S CAGR +3.1%

214 ● **Year over Year Change of Total Debt**

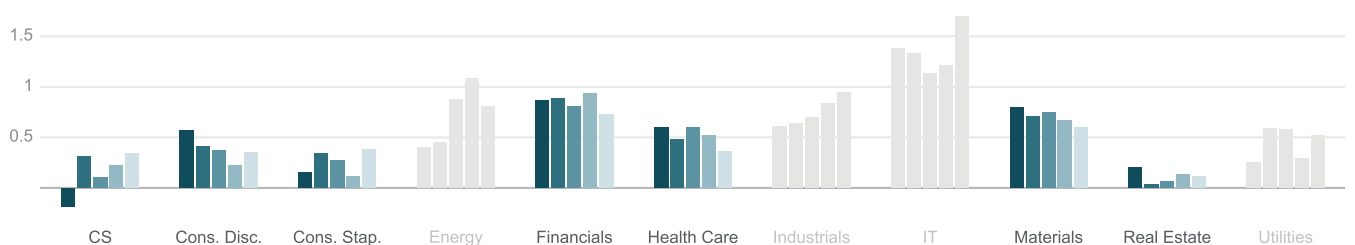
Capital Efficiency L-S CAGR +3.0%

215 ● **Research & Development Intensity**

Earnings Quality L-S CAGR +3.0%

216 ● **Cash Conversion Cycle**

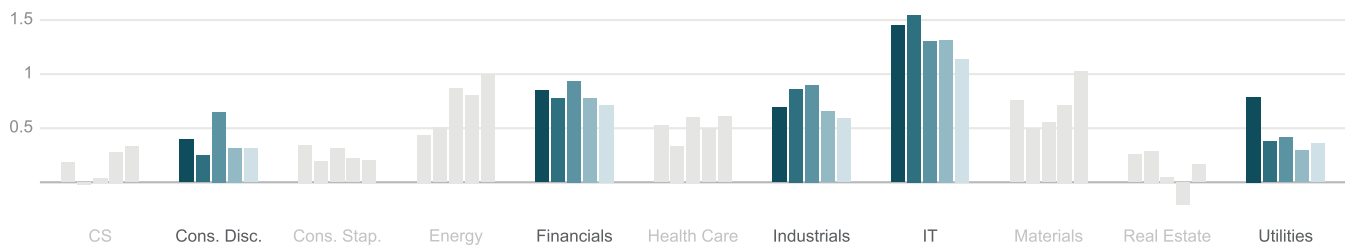
Earnings Quality L-S CAGR +2.9%



217 ● **6M Chg in 36M CAPM Alpha**

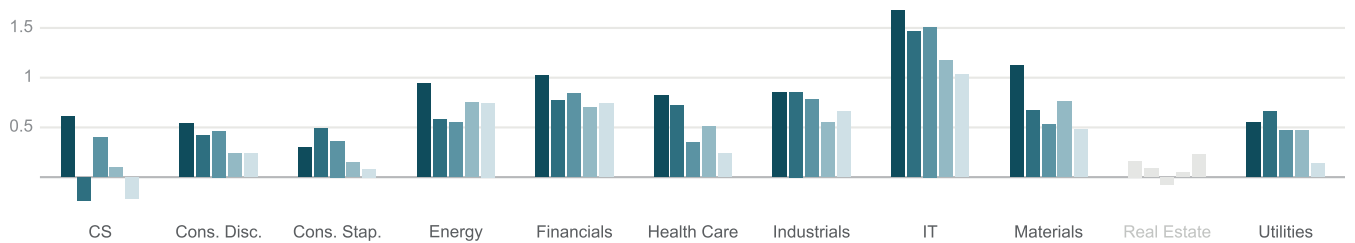
Price Momentum

L-S CAGR +2.9%

218 ● **Adj Number of EPS FY2 Revisions**

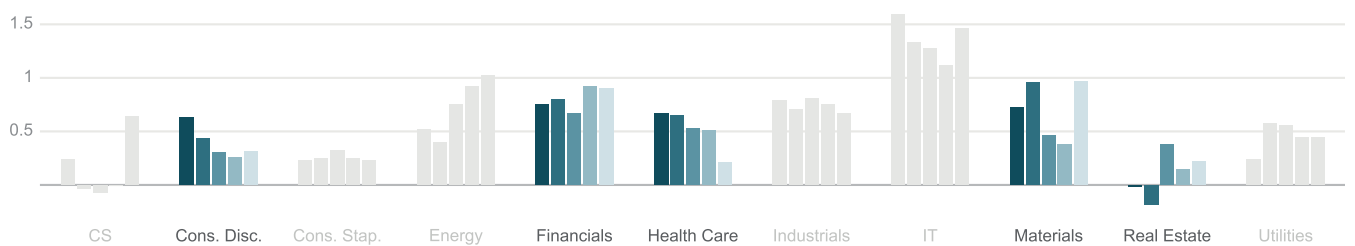
Analyst Expectations

L-S CAGR +2.8%

219 ● **Pretax Profit Margin**

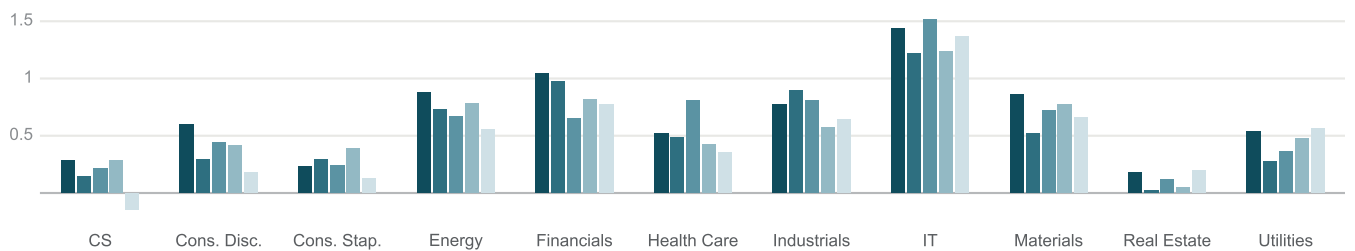
Capital Efficiency

L-S CAGR +2.8%

220 ● **Accrual Ratio - Balance Sheet**

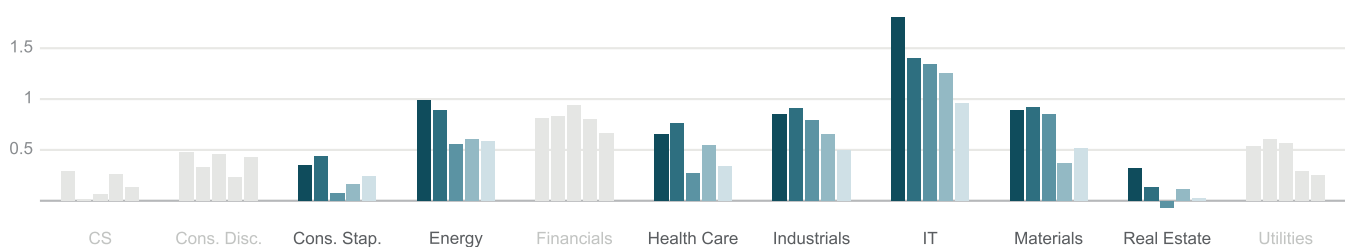
Earnings Quality

L-S CAGR +2.7%

221 ● **180D Price TStat**

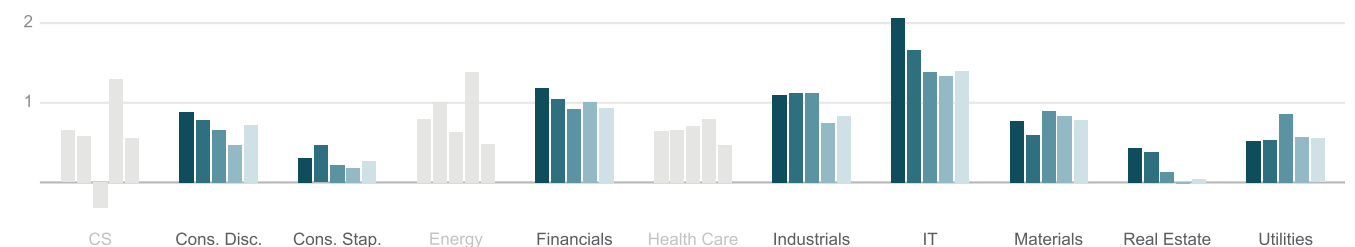
Price Momentum

L-S CAGR +2.6%

222 ● **Operating Cash Flow Stability**

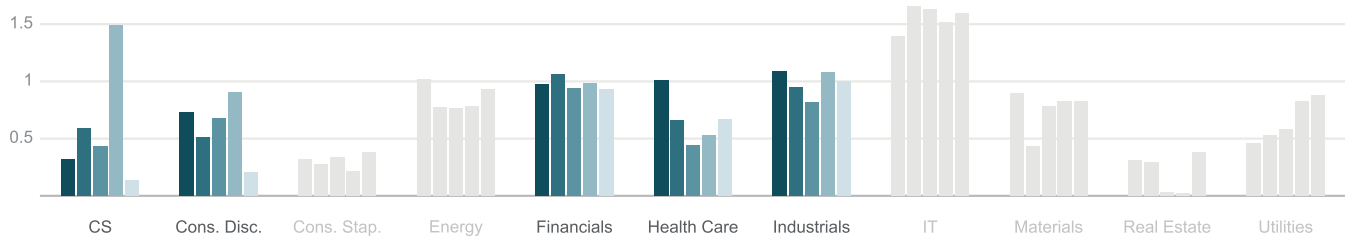
Earnings Quality

L-S CAGR +2.6%

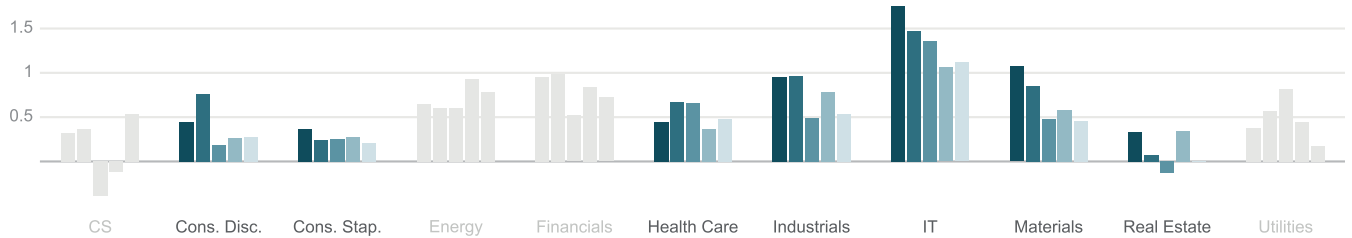


223 ● **Change of Ownership Level - Hedge Fund**

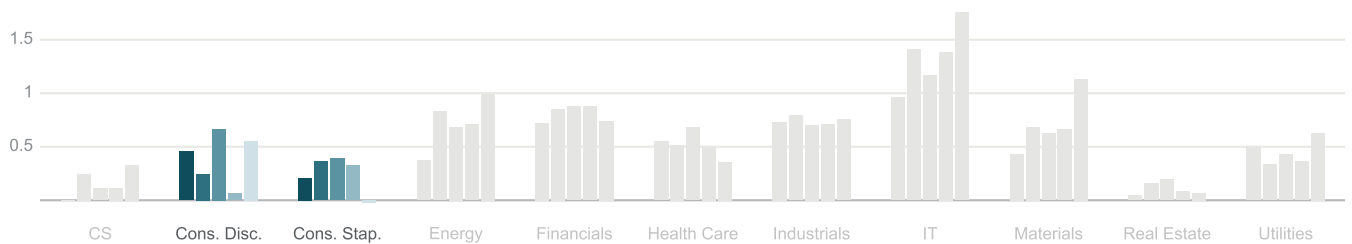
Price Momentum L-S CAGR +2.6%

224 ● **6 Month Average Share Turn Over**

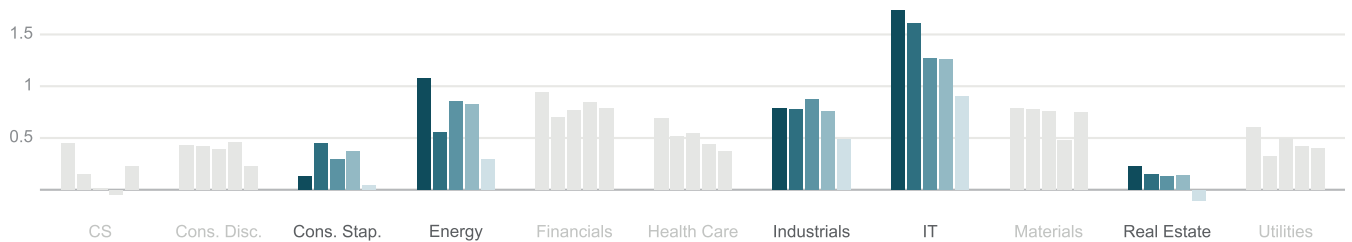
Price Momentum L-S CAGR +2.5%

225 ● **Change of Sales to Change of EPS**

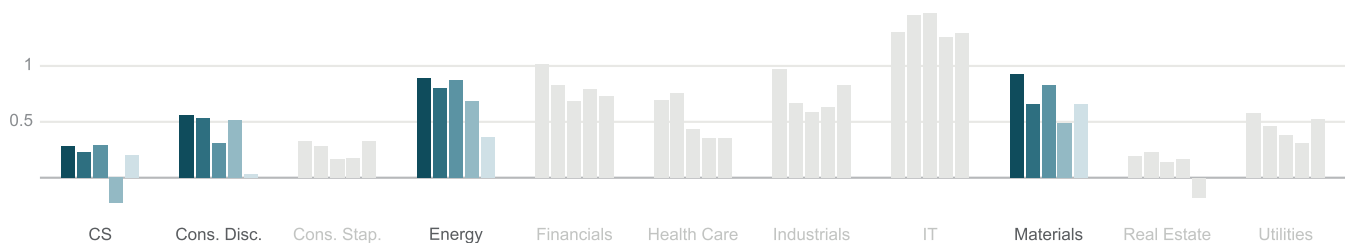
Earnings Quality L-S CAGR +2.5%

226 ● **26W Price Stat**

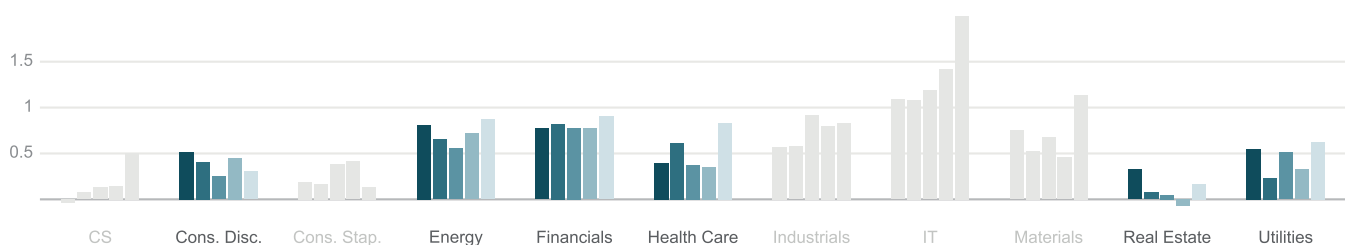
Price Momentum L-S CAGR +2.4%

227 ● **Ownership Breadth Stability of Institution Ownership by Buyers**

Price Momentum L-S CAGR +2.4%

228 ● **6 Months Price Reversal**

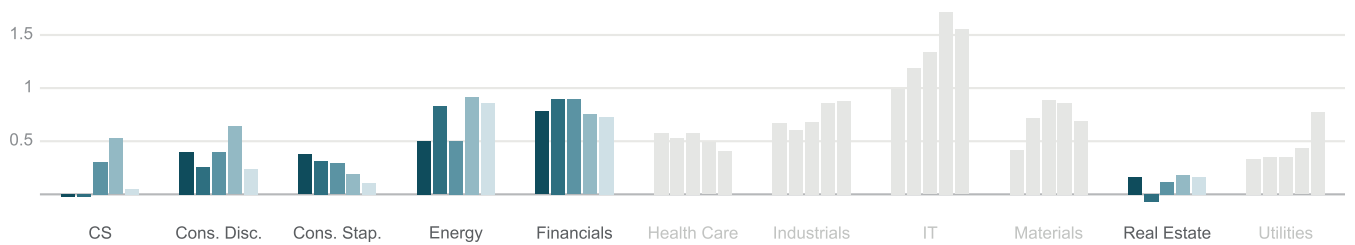
Price Momentum L-S CAGR +2.2%



229 ● Ownership Breadth Stability of Institution Ownership by Holders

Price Momentum

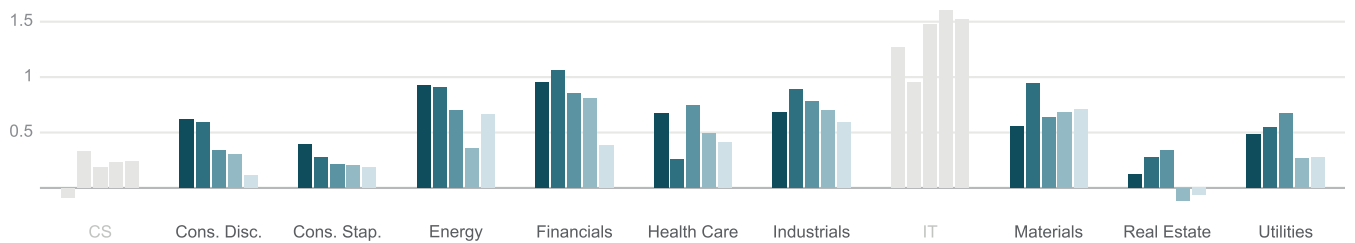
L-S CAGR +2.2%



230 ● 6M Chg in Target Price Gap

Analyst Expectations

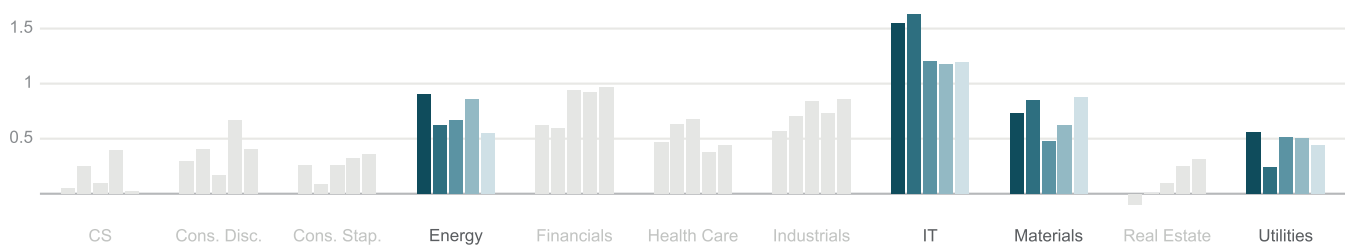
L-S CAGR +2.2%



231 ● Change in Number of Buyers

Price Momentum

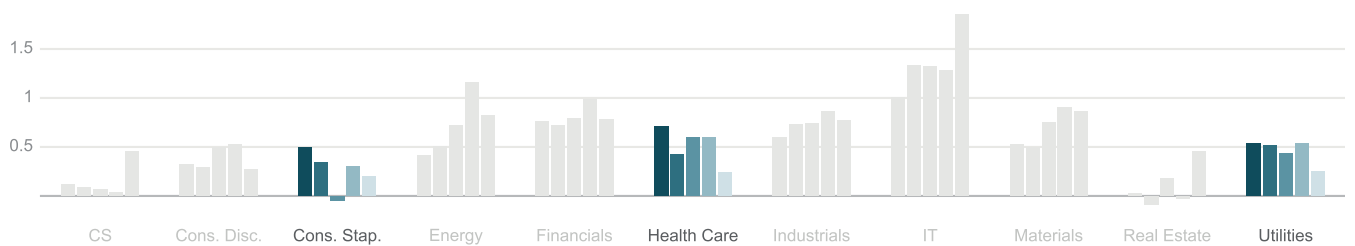
L-S CAGR +2.2%



232 ● 30 Day Coefficient of Variation of Volume to Price

Price Momentum

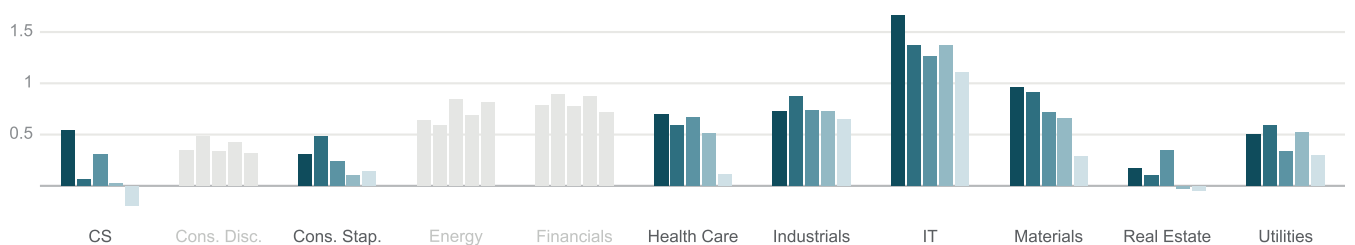
L-S CAGR +2.2%



233 ● 1Y Chg in Amihud

Volatility

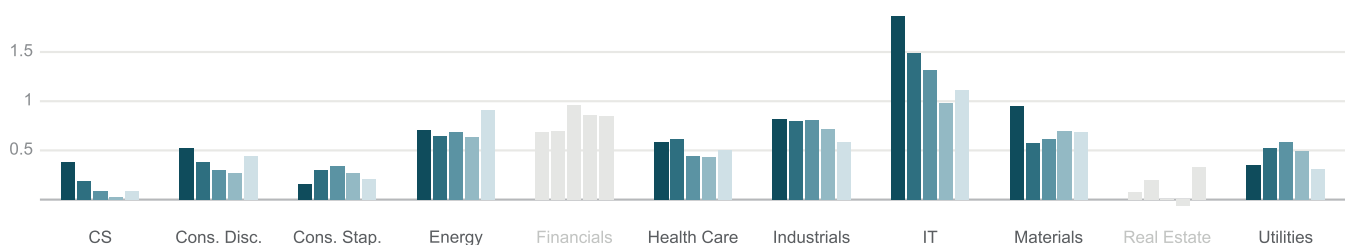
L-S CAGR +2.2%



234 ● 52W High Low

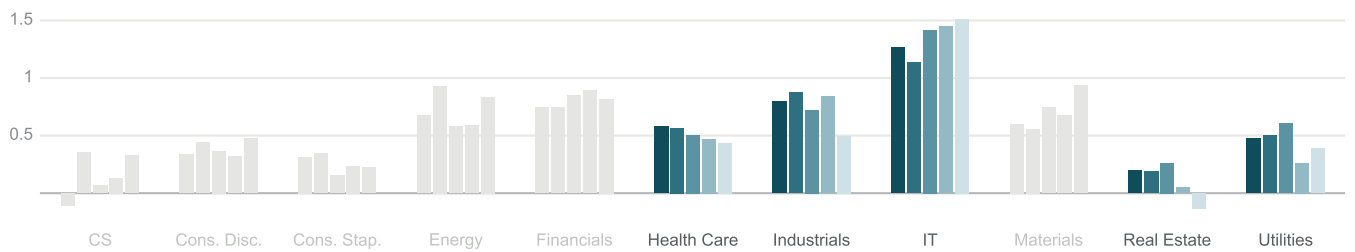
Price Momentum

L-S CAGR +2.2%

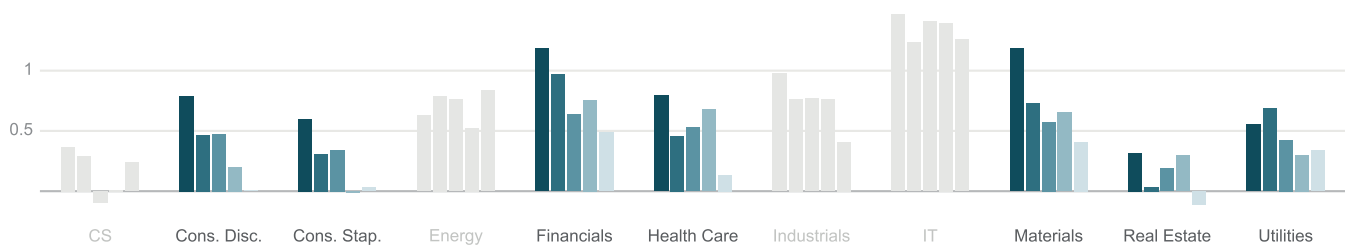


235 ● **Cash to Sales**

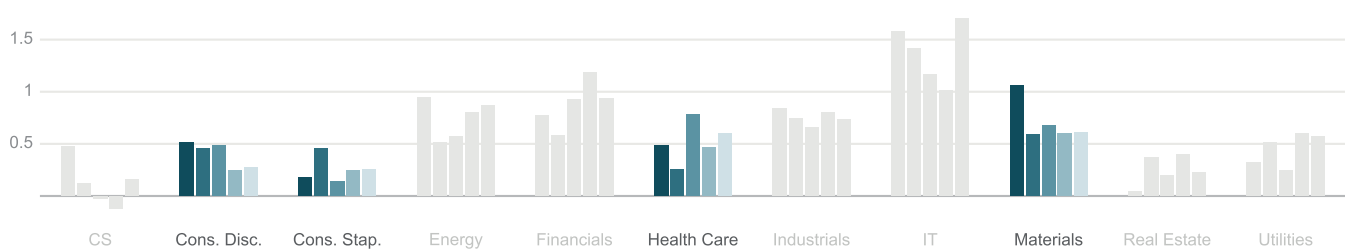
Earnings Quality L-S CAGR +2.2%

236 ● **8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

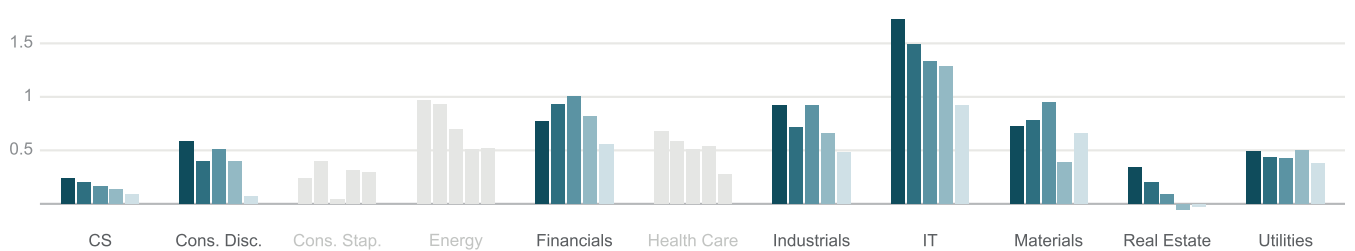
Valuation L-S CAGR +2.1%

237 ● **Year over Year Change of Inventory to Assets**

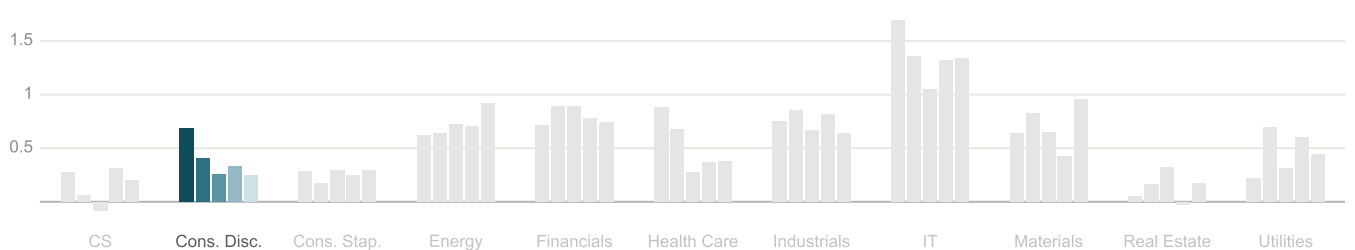
Earnings Quality L-S CAGR +2.1%

238 ● **Volatility adjusted 12M return**

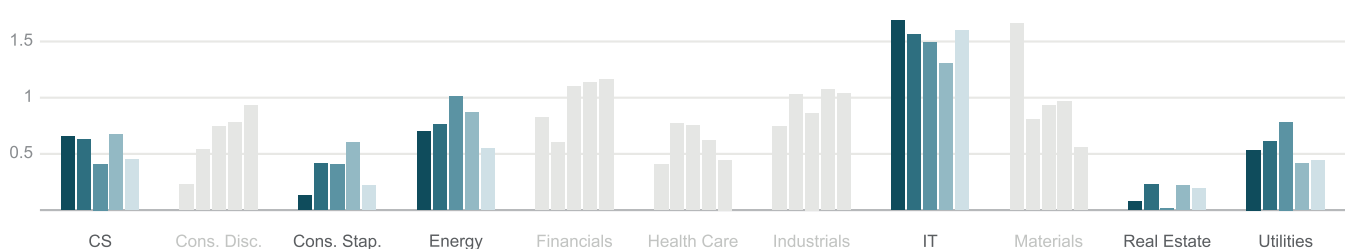
Price Momentum L-S CAGR +2.0%

239 ● **Normalized ROA**

Capital Efficiency L-S CAGR +2.0%

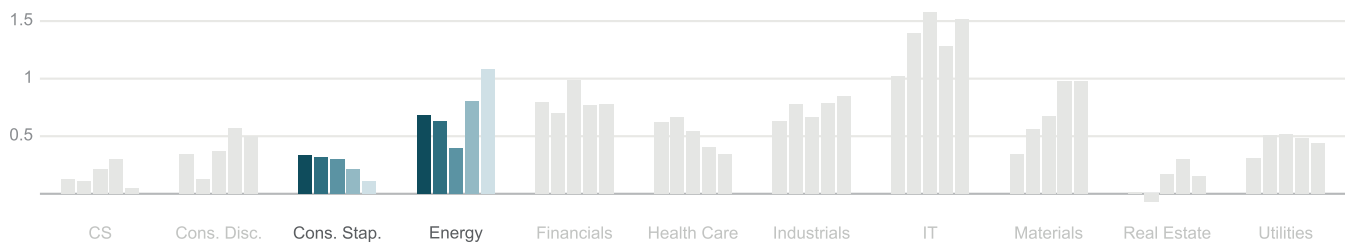
240 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**

Historical Growth L-S CAGR +2.0%



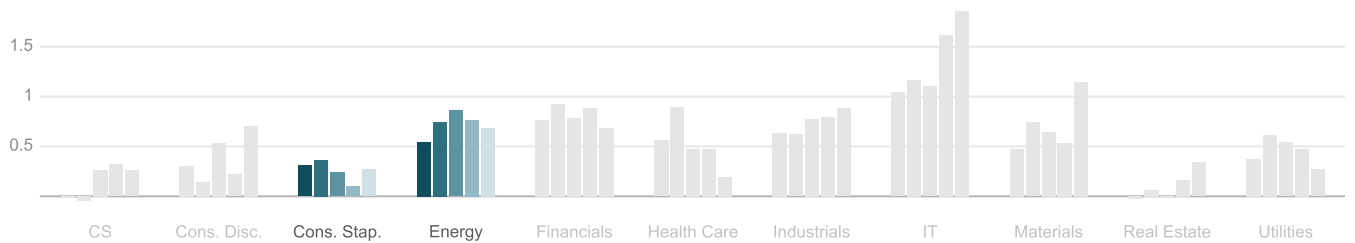
241 ● 130 Day Minimum Return

Volatility L-S CAGR +1.9%



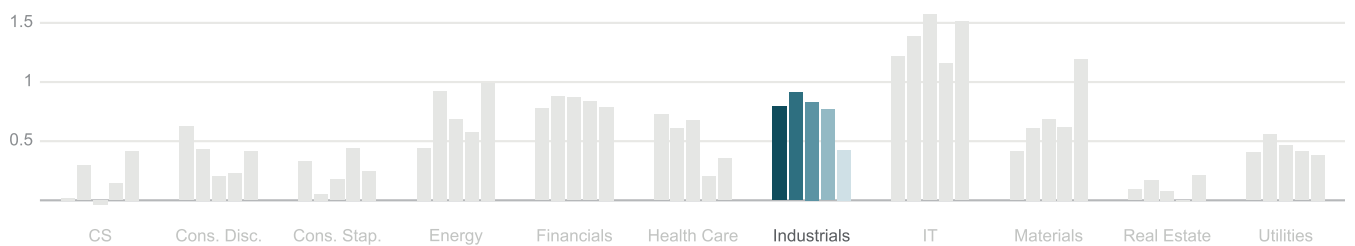
242 ● Max Daily Return in the Past 6 Months

Price Momentum L-S CAGR +1.9%



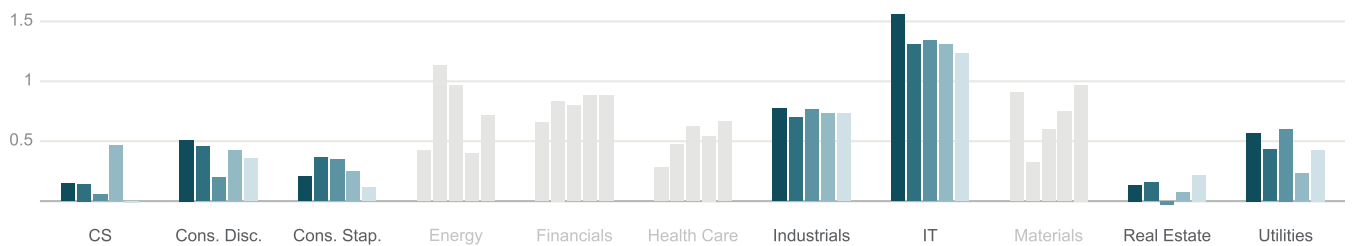
243 ● Return on Net Tangible Assets

Valuation L-S CAGR +1.9%



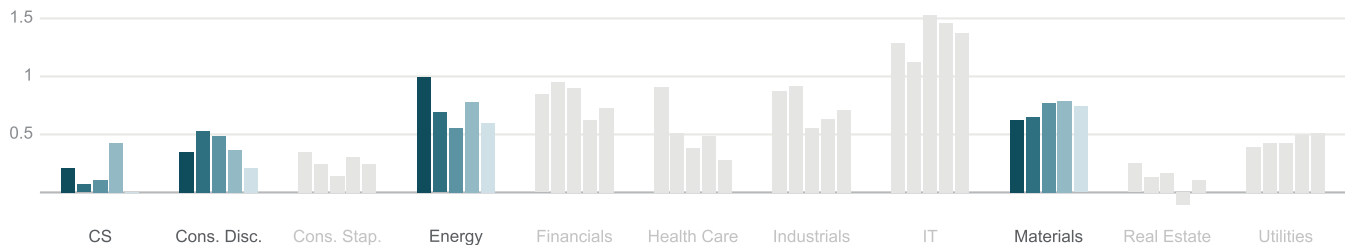
244 ● Change of Ownership Level - All Managers

Price Momentum L-S CAGR +1.8%



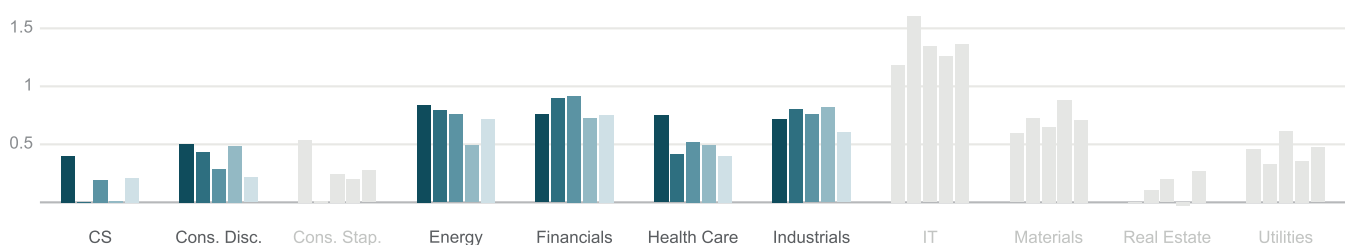
245 ● Institution Ownership Turnover

Price Momentum L-S CAGR +1.7%



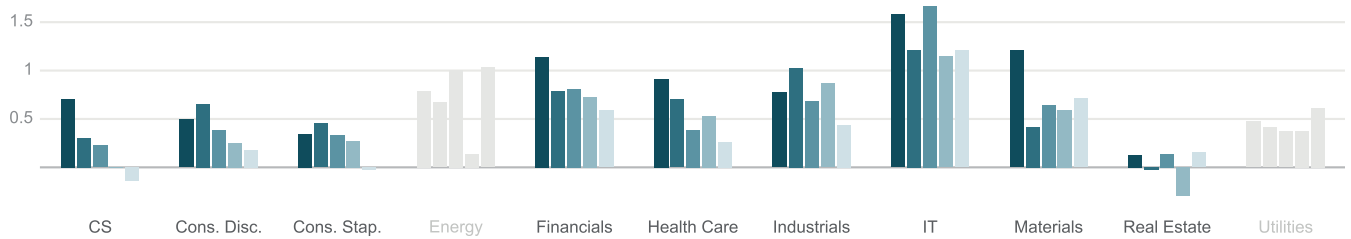
246 ● 6M Chg in 12M CAPM Alpha

Price Momentum L-S CAGR +1.7%

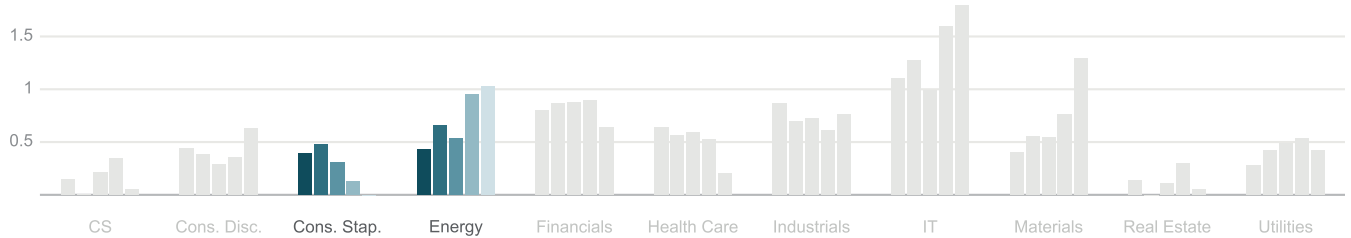


247 ● **Adjusted Number of EPS FY1 Revisions**

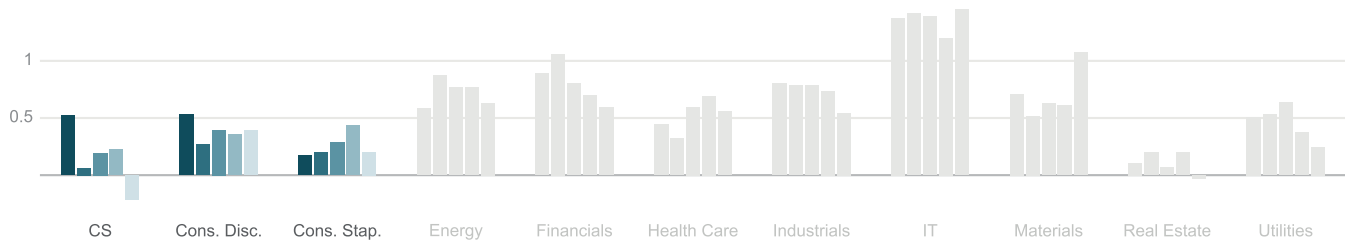
Analyst Expectations L-S CAGR +1.7%

248 ● **24M Residual Return Variance**

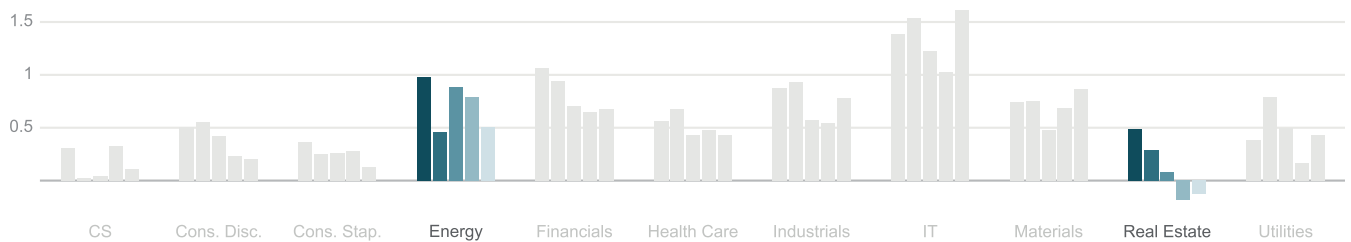
Price Momentum L-S CAGR +1.6%

249 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

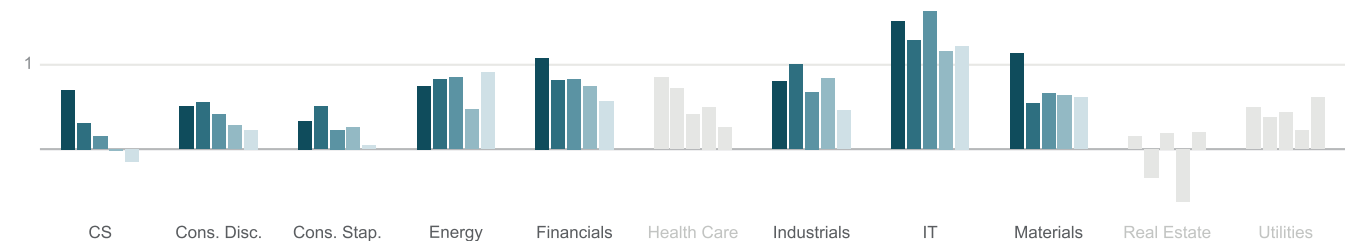
Analyst Expectations L-S CAGR +1.6%

250 ● **14-Day Relative Strength Index**

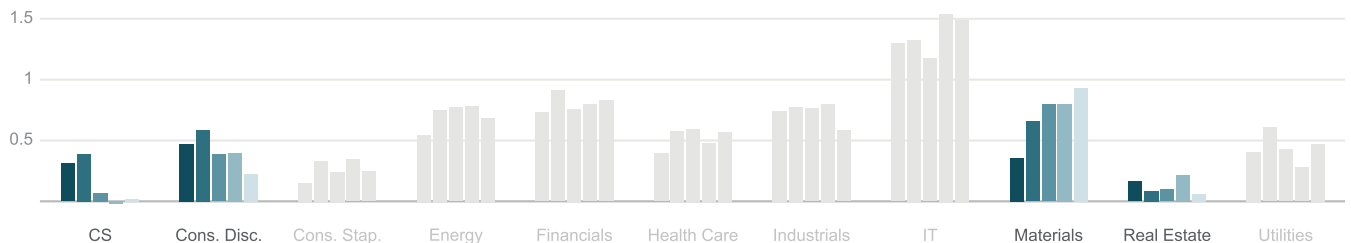
Price Momentum L-S CAGR +1.6%

251 ● **Analyst Earnings Estimate Diffusion**

Analyst Expectations L-S CAGR +1.6%

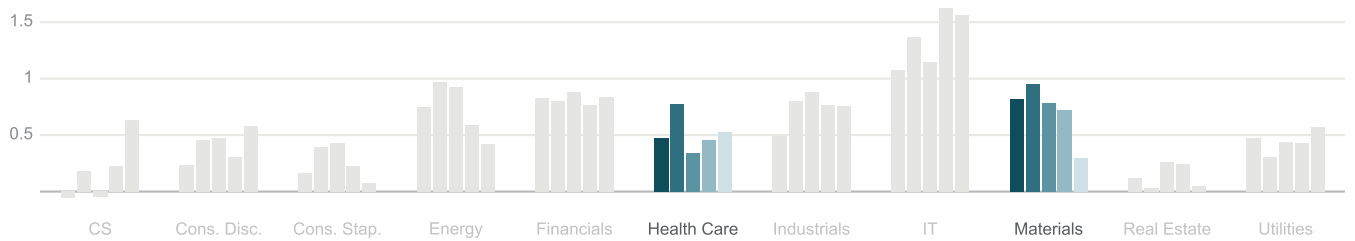
252 ● **3Y Chg in Operating Margin**

Historical Growth L-S CAGR +1.5%



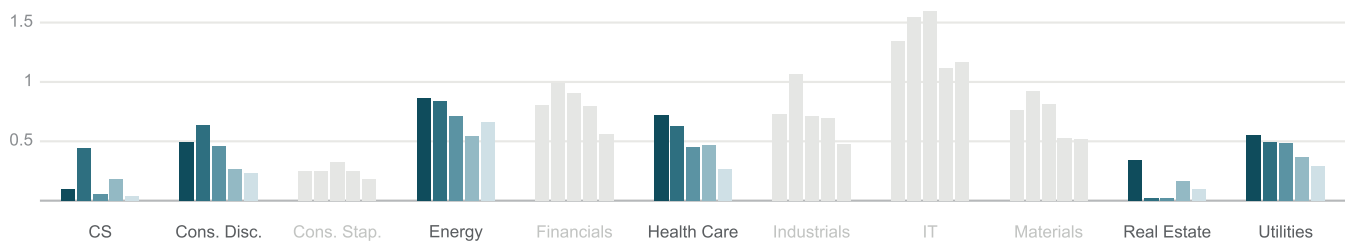
253 ● Adj 50 Day Volume Signal

Volatility L-S CAGR +1.5%



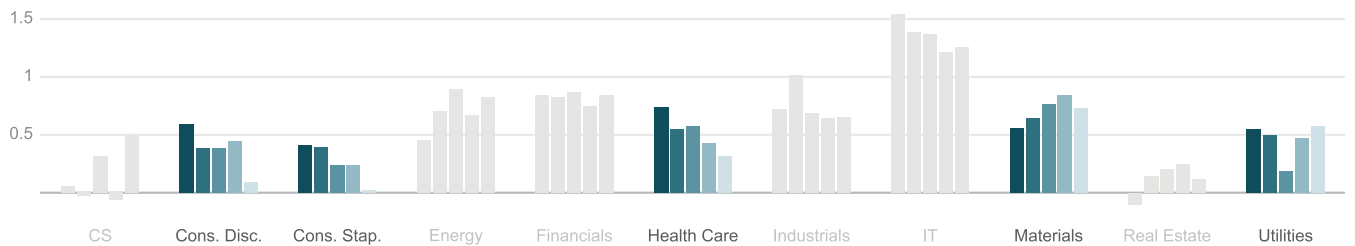
254 ● Investment Duration

Price Momentum L-S CAGR +1.5%



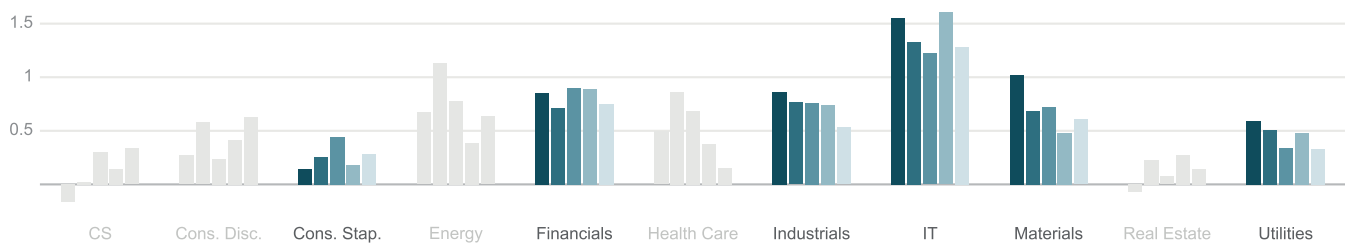
255 ● EBIT Estimate Dispersion

Analyst Expectations L-S CAGR +1.4%



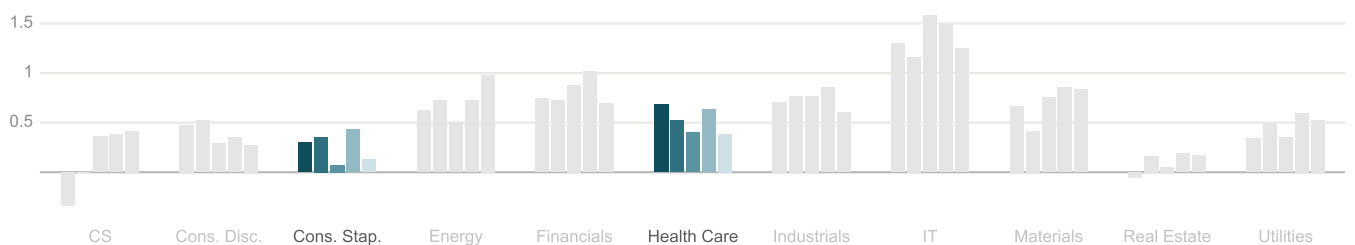
256 ● Sales Acceleration

Historical Growth L-S CAGR +0.9%



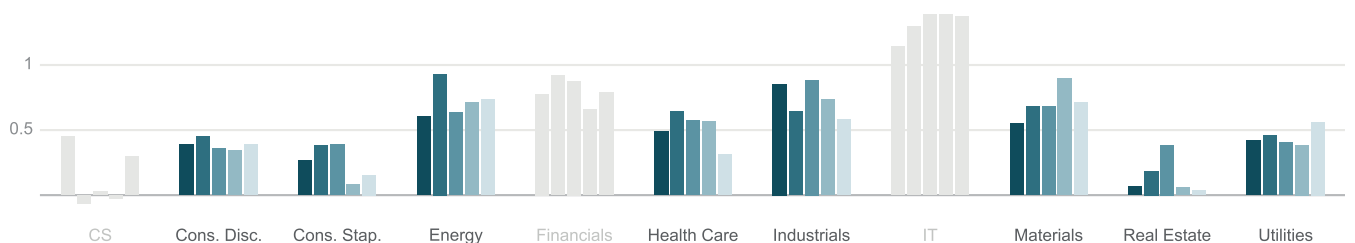
257 ● Revenue Estimate Dispersion

Analyst Expectations L-S CAGR +0.8%



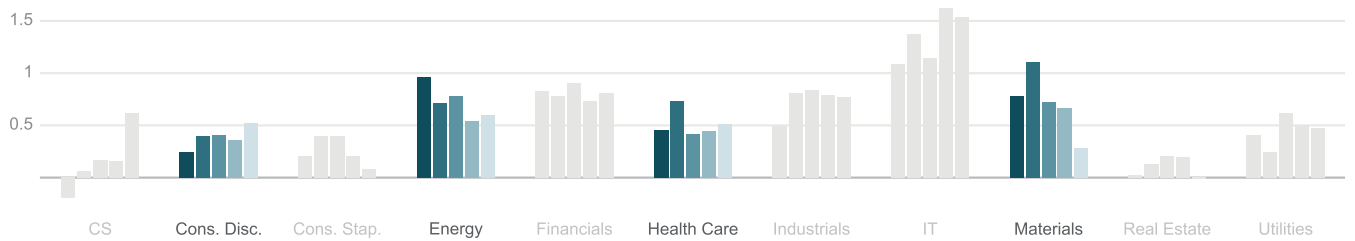
258 ● Change in TTM Depr. to Capex

Earnings Quality L-S CAGR +0.8%



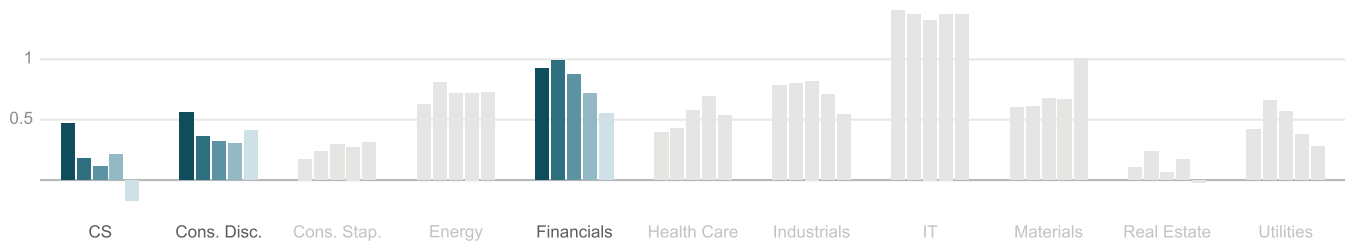
259 ● 50 Day Volume Signal

Price Momentum L-S CAGR +0.7%



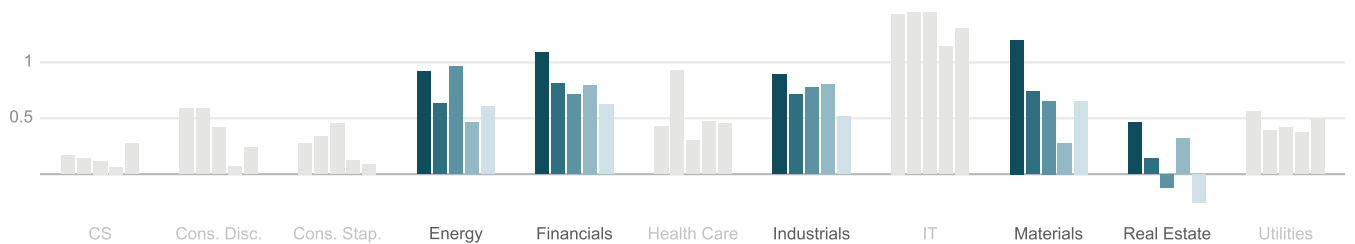
260 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

Analyst Expectations L-S CAGR +0.6%



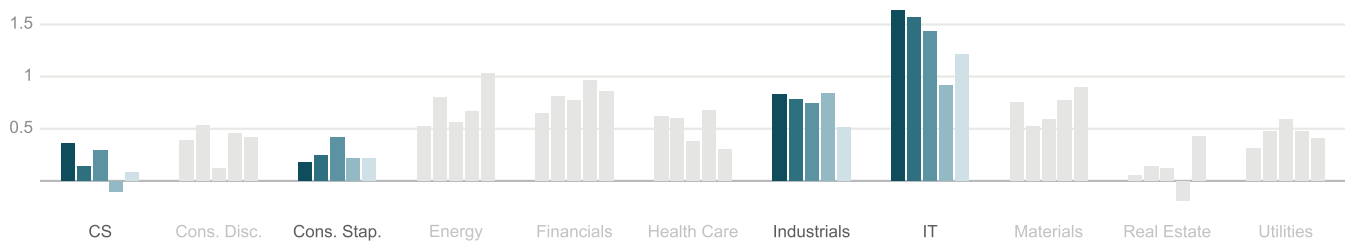
261 ● 5 Day Price Reversal

Price Momentum L-S CAGR +0.6%



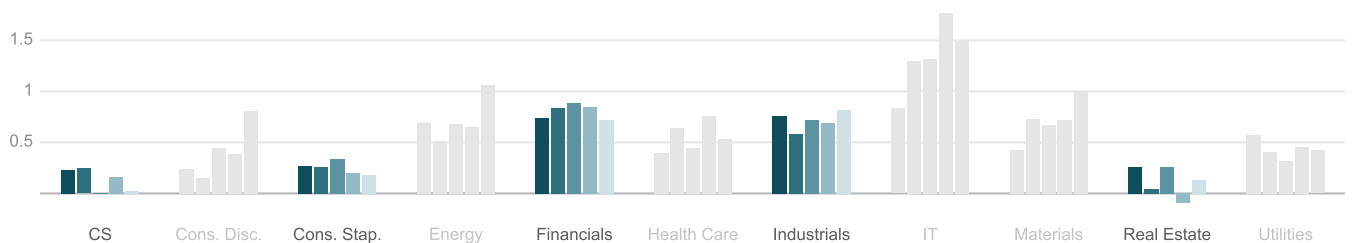
262 ● Closing Price to 52 Week High

Price Momentum L-S CAGR +0.6%



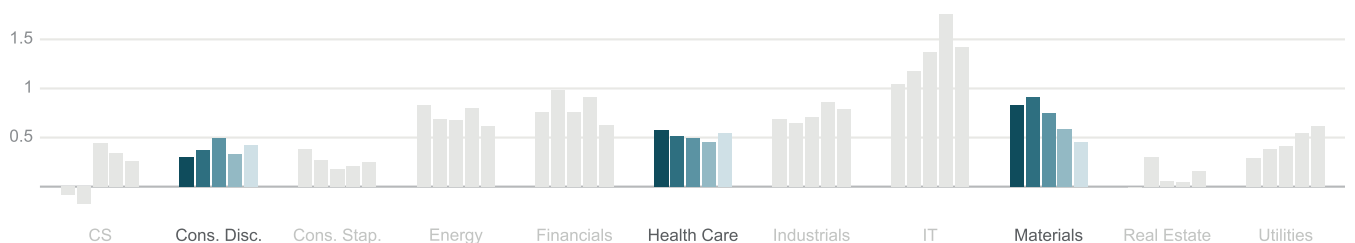
263 ● ROE 20 Qtr Standard Deviation

Earnings Quality L-S CAGR +0.5%



264 ● 1Y Chg in Share Turnover

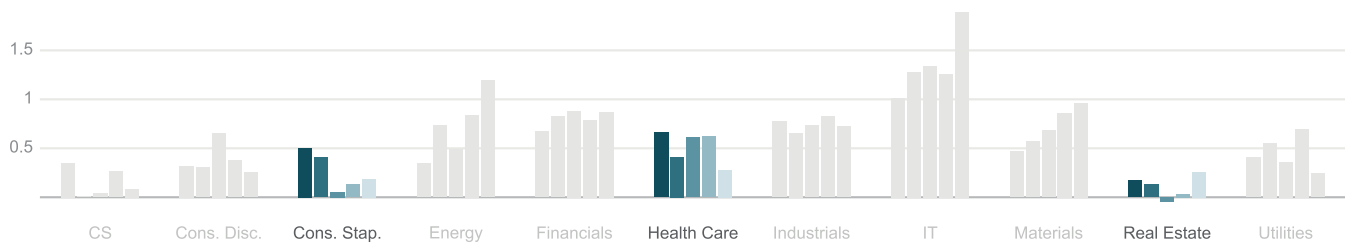
Price Momentum L-S CAGR +0.3%



265 ● 20 Day Coefficient of Variation of Volume to Price

Price Momentum

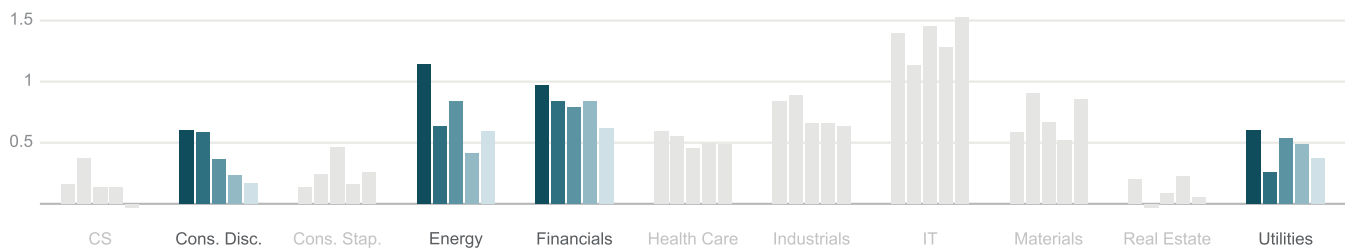
L-S CAGR -0.0%



266 ● 6M Avg Chg 1M Recommendation

Analyst Expectations

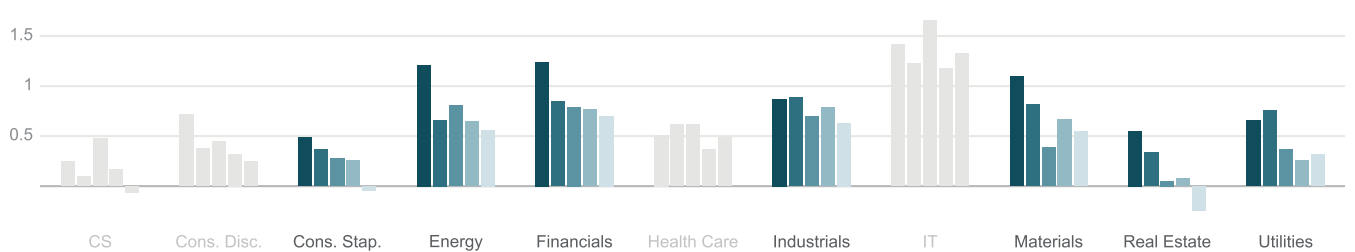
L-S CAGR -0.2%



267 ● 5-Day Money Flow/Volume

Price Momentum

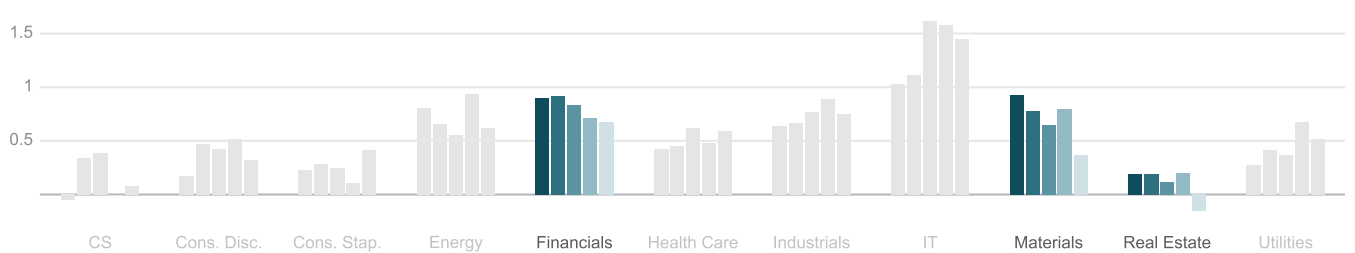
L-S CAGR -0.3%



268 ● 5 Day Volume Signal

Price Momentum

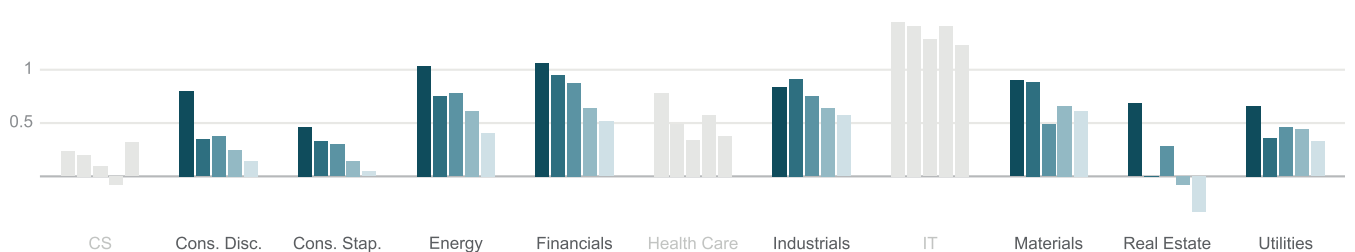
L-S CAGR -0.4%



269 ● 20-Day Lane's Stochastic Indicator

Price Momentum

L-S CAGR -0.6%



270 ● 4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation

L-S CAGR -0.9%

